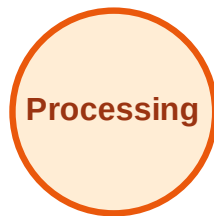


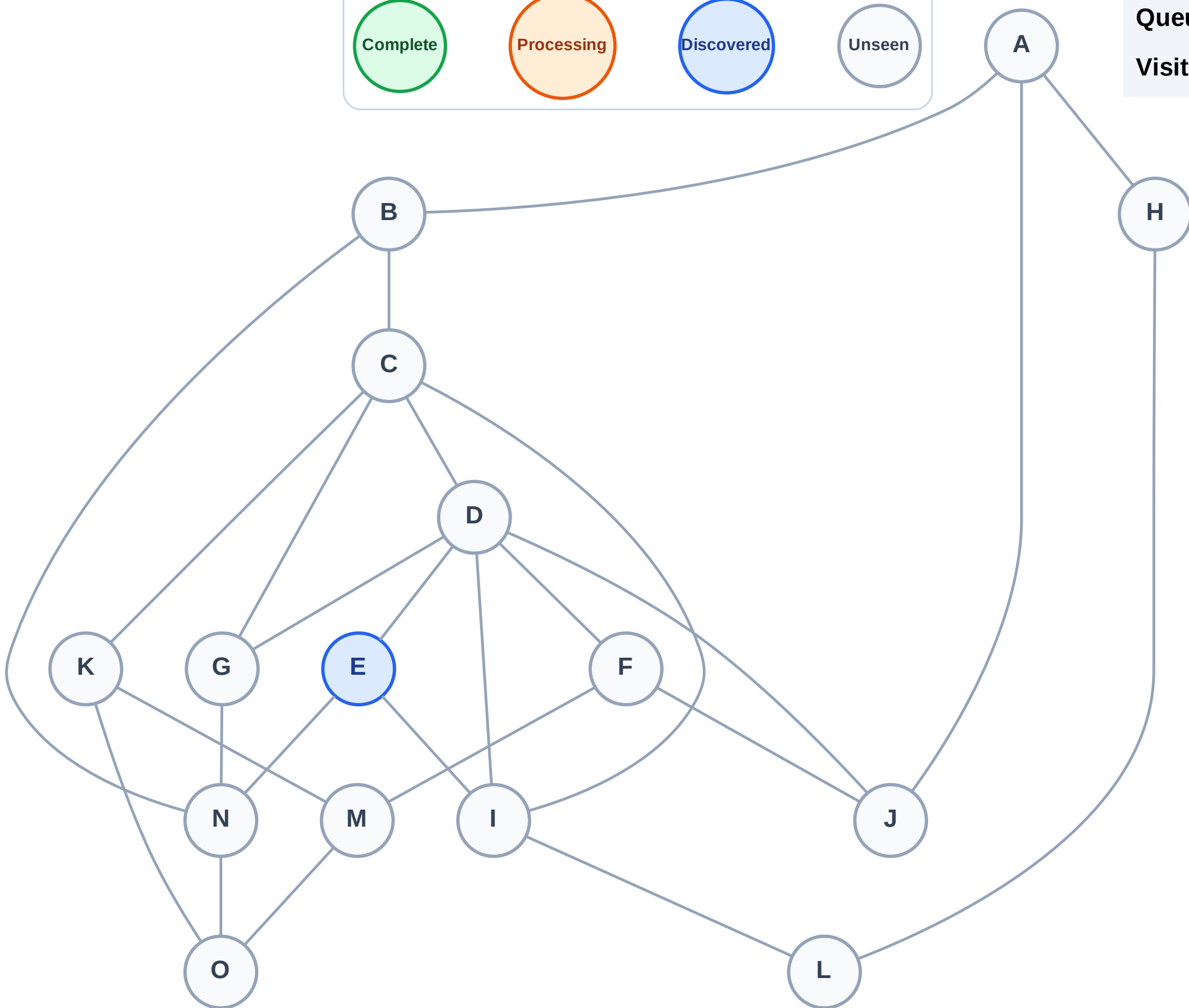
BFS Traversal

Step 1: Start at E

Legend



Queue: E
Visited: (none)



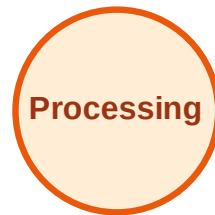
BFS Traversal

Step 2: Process node E

Legend



Complete



Processing

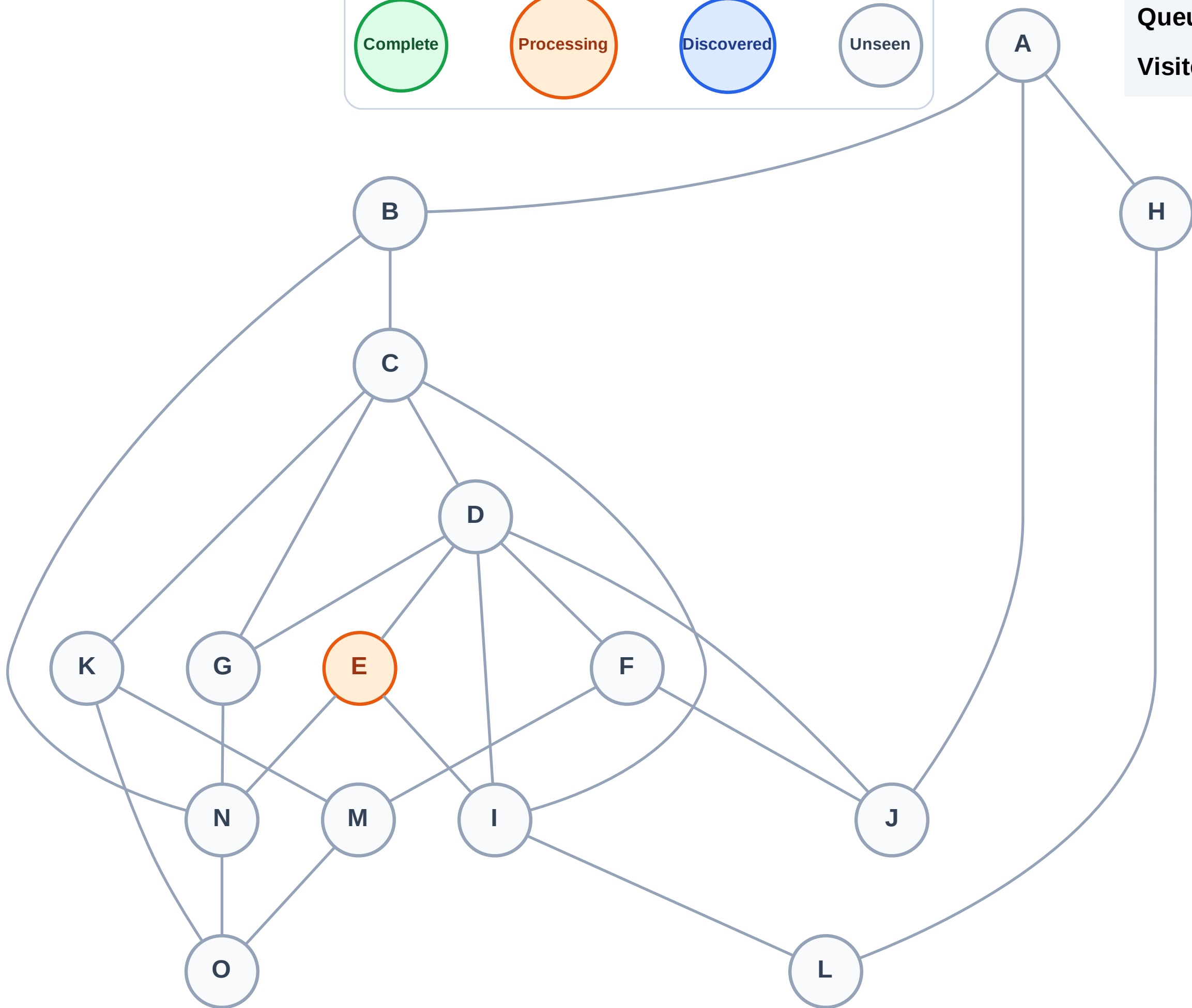


Discovered



Unseen

Queue: (empty)
Visited: (none)



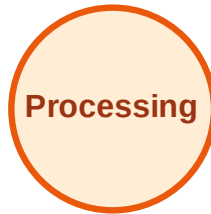
BFS Traversal

Step 3: Discover D, I, N from E

Legend



Complete



Processing

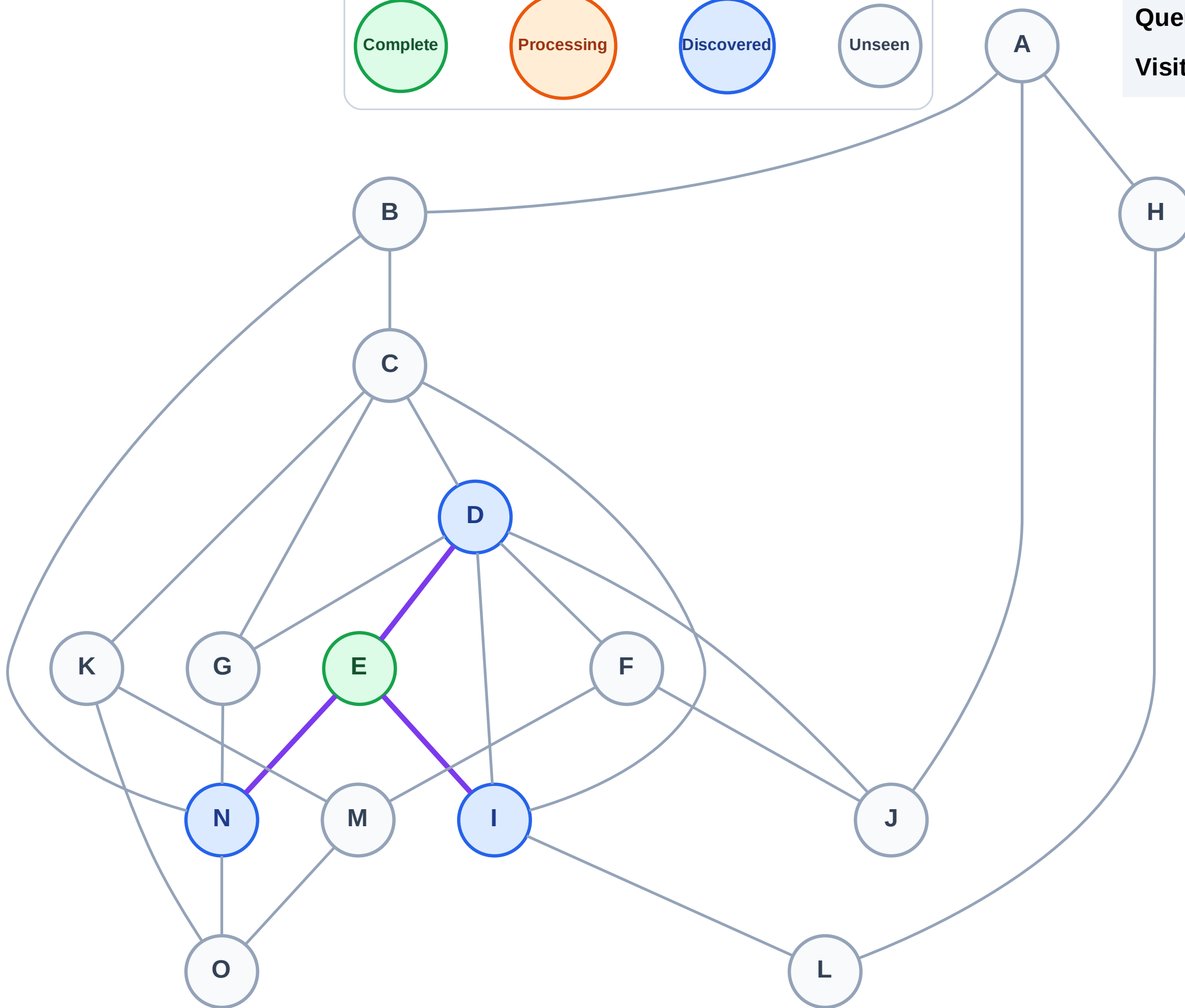


Discovered



Unseen

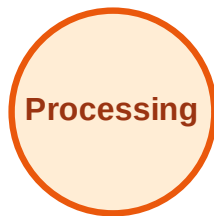
Queue: D → I → N
Visited: E



BFS Traversal

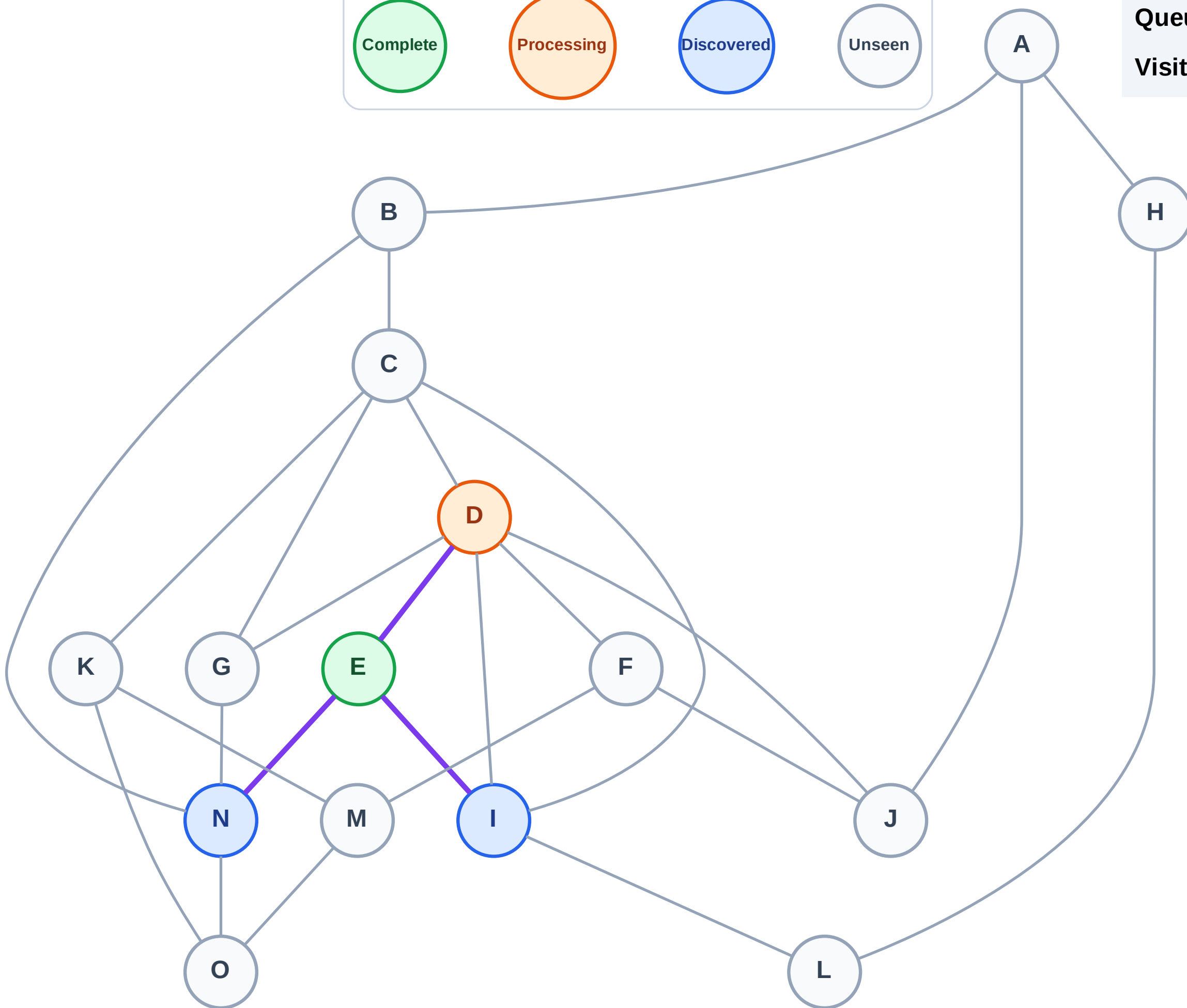
Step 4: Process node D

Legend



Queue: I → N

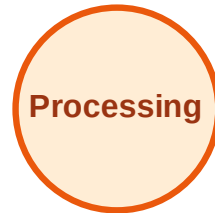
Visited: E



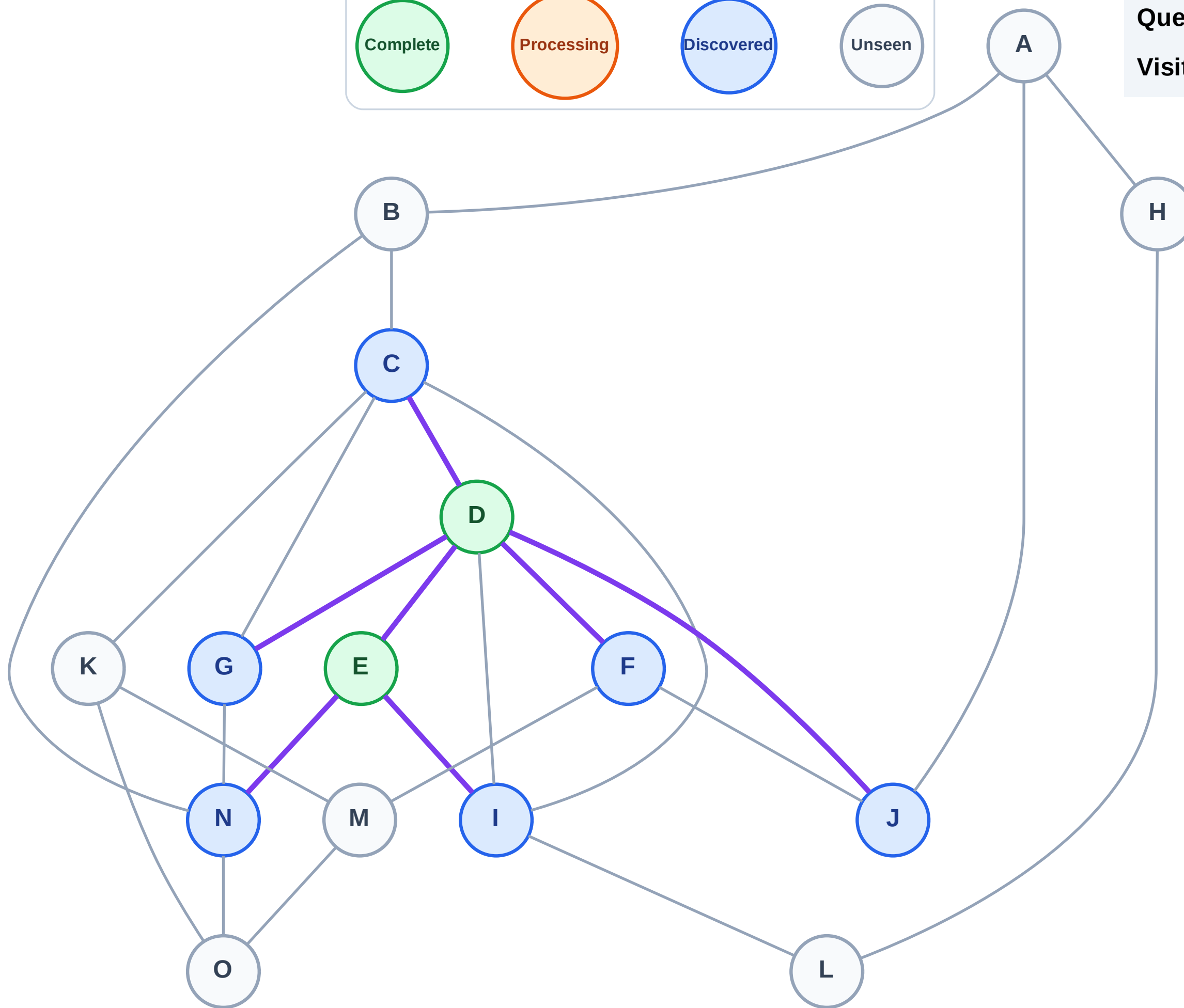
BFS Traversal

Step 5: Discover C, F, G, J from D

Legend



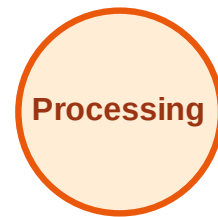
Queue: I → N → C → F → G → J
Visited: D, E



BFS Traversal

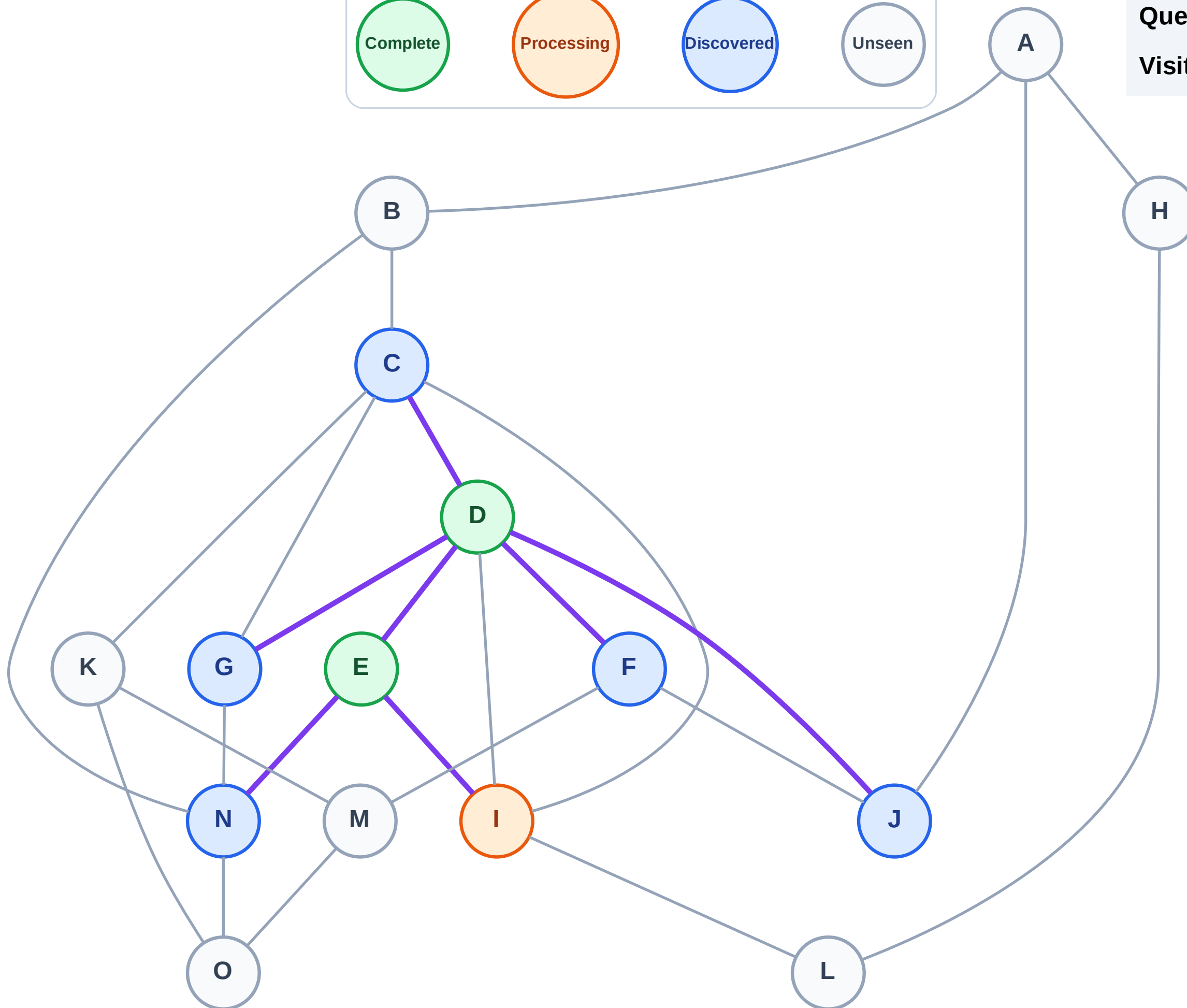
Step 6: Process node I

Legend



Queue: N → C → F → G → J

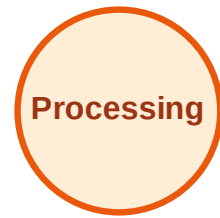
Visited: D, E



BFS Traversal

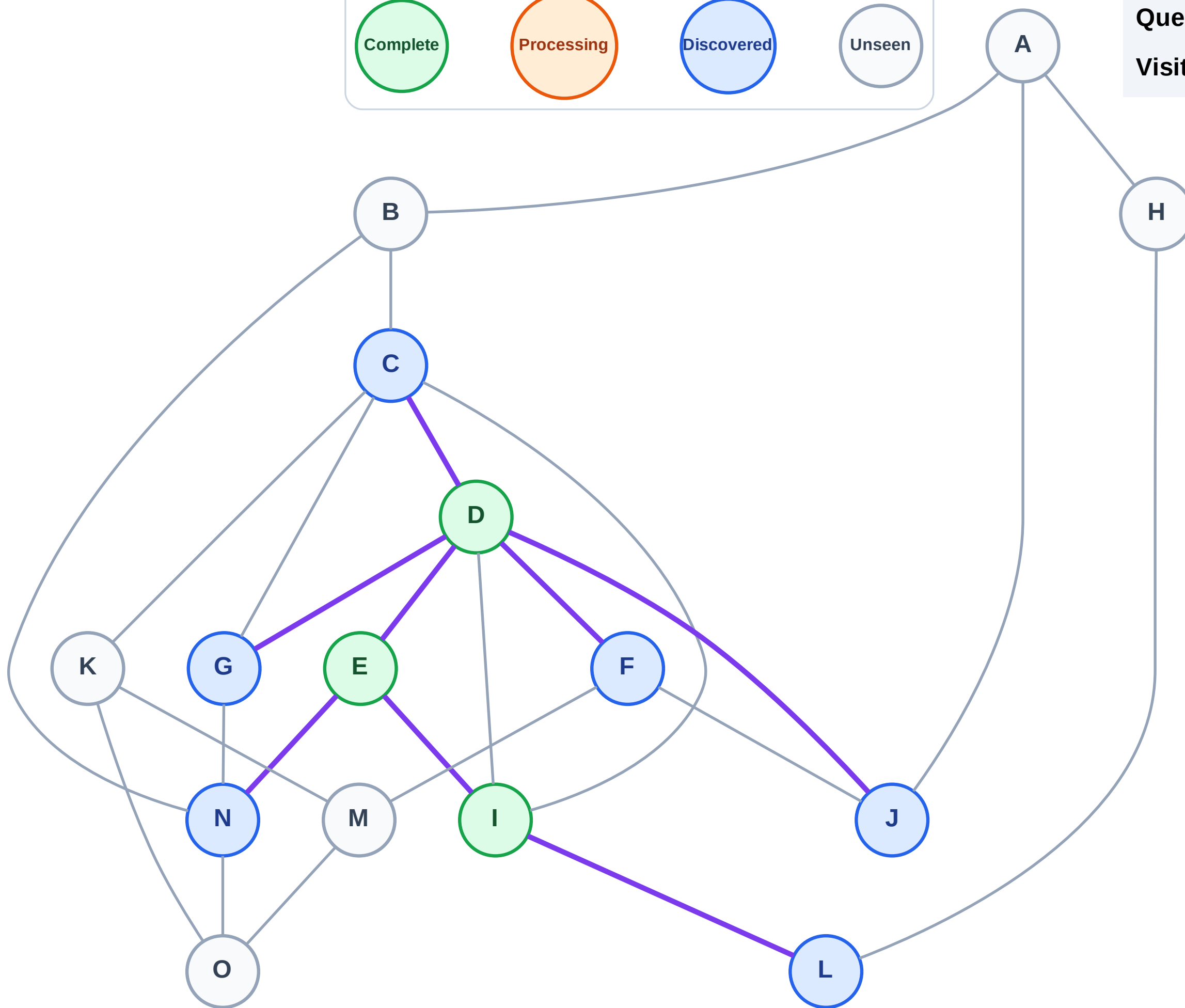
Step 7: Discover L from I

Legend



Queue: N → C → F → G → J → L

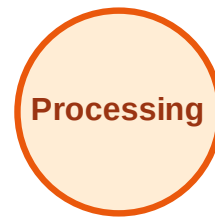
Visited: D, E, I



BFS Traversal

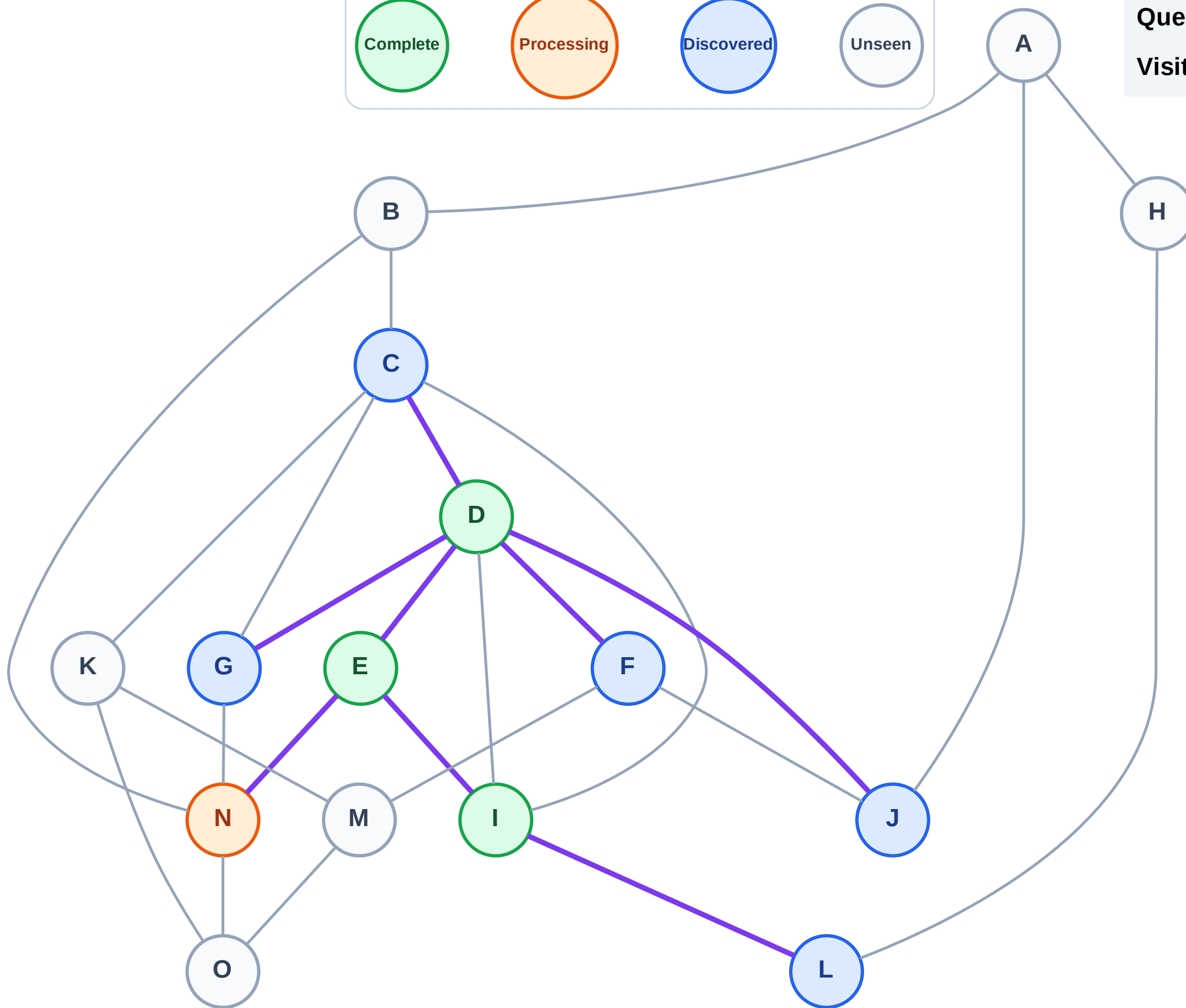
Step 8: Process node N

Legend



Queue: C → F → G → J → L

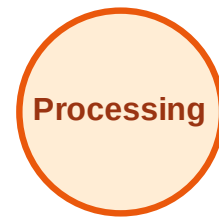
Visited: D, E, I



BFS Traversal

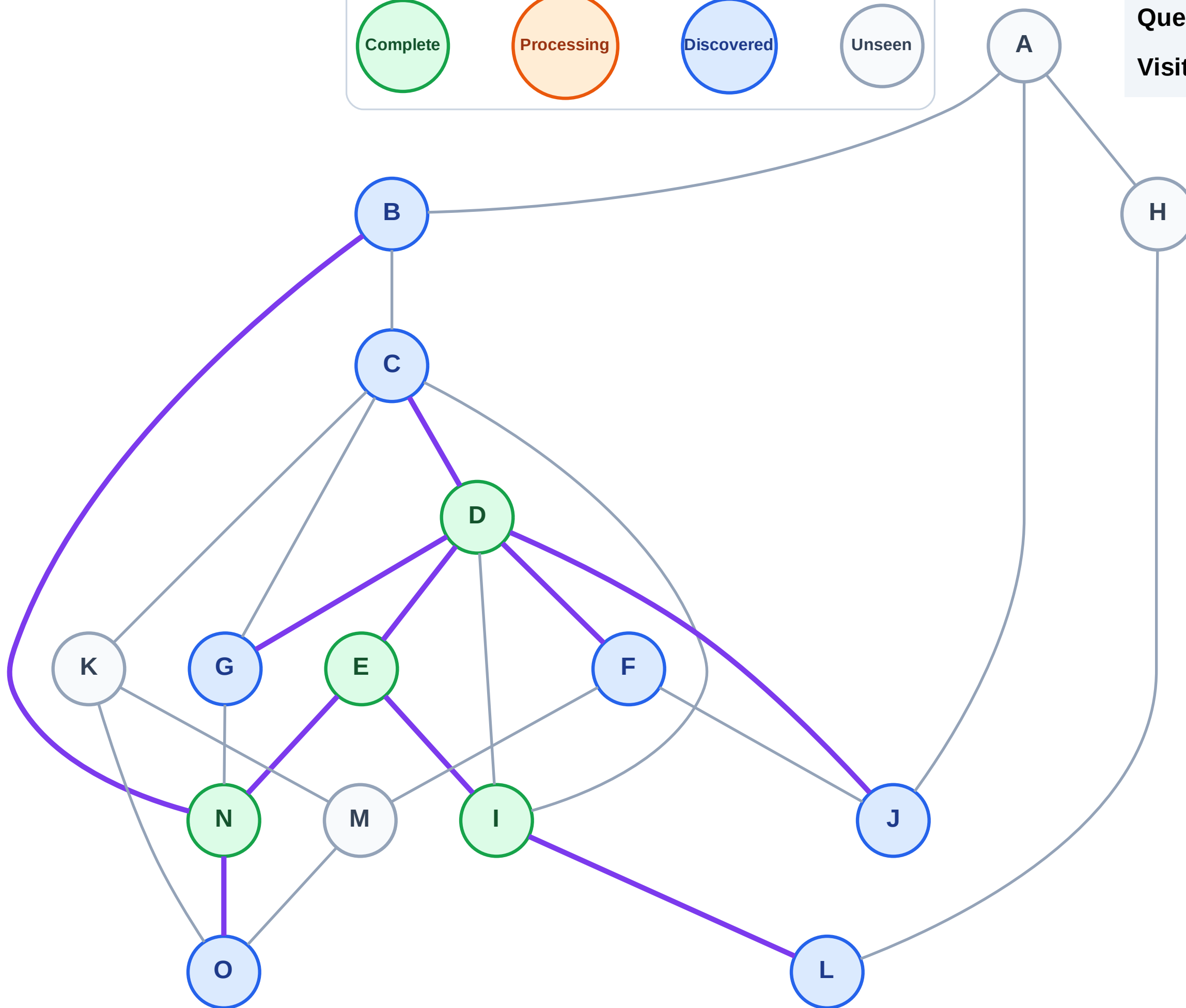
Step 9: Discover B, O from N

Legend



Queue: C → F → G → J → L → B → O

Visited: D, E, I, N



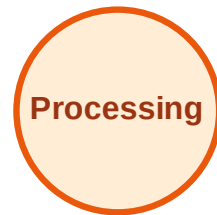
BFS Traversal

Step 10: Process node C

Legend



Complete



Processing



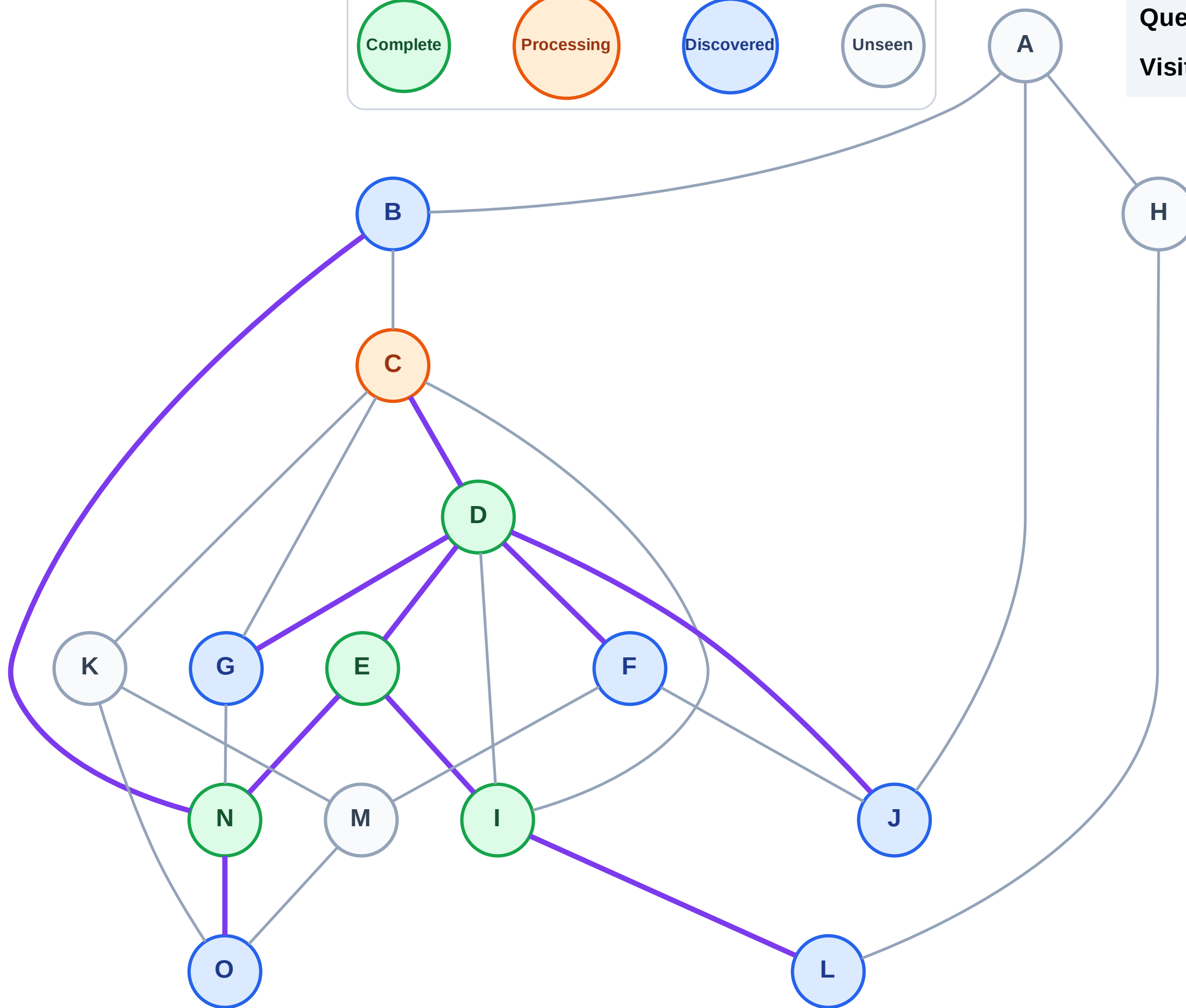
Discovered



Unseen

Queue: F → G → J → L → B → O

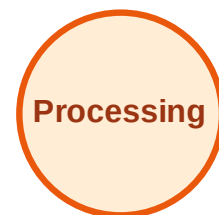
Visited: D, E, I, N



BFS Traversal

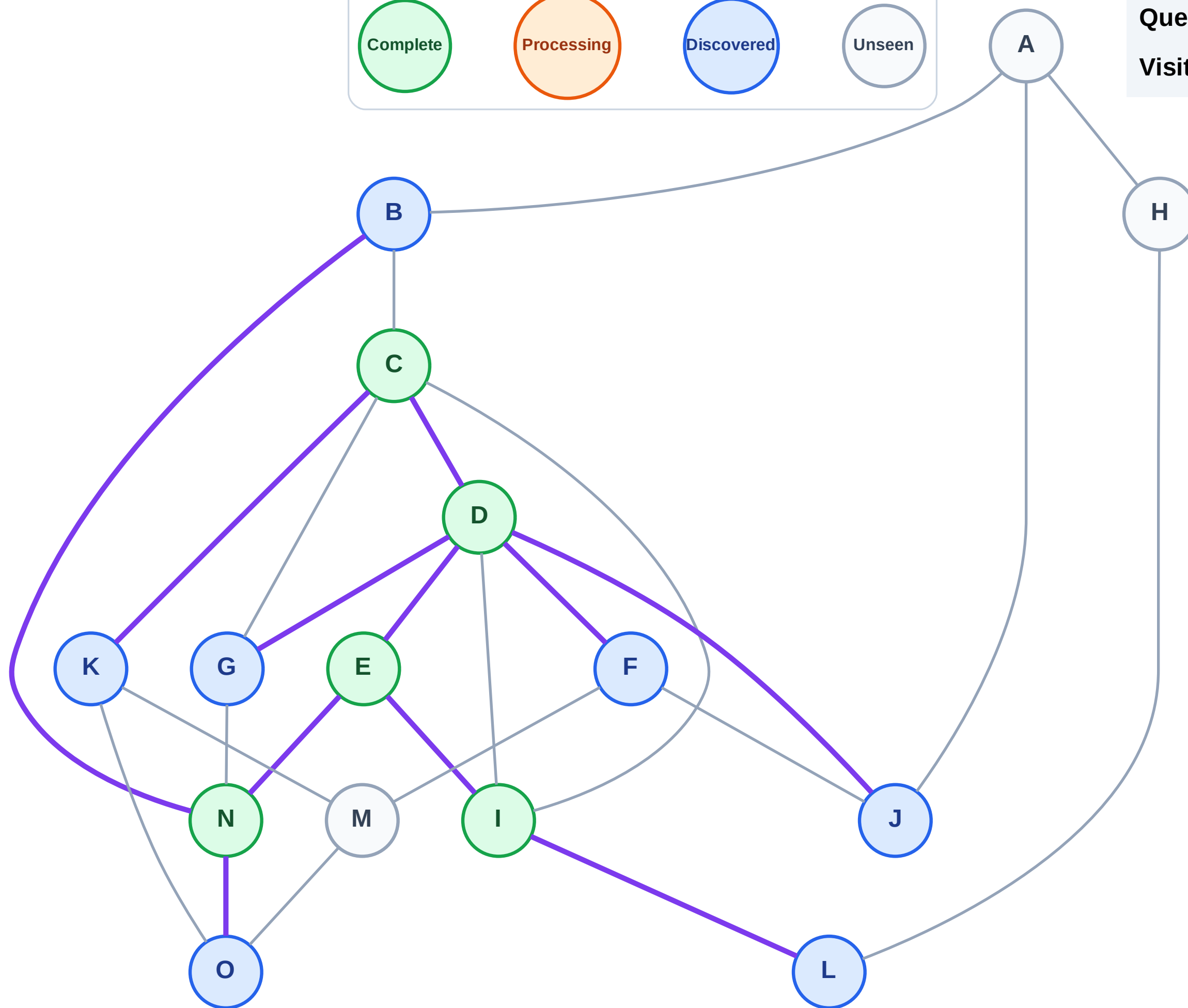
Step 11: Discover K from C

Legend



Queue: F → G → J → L → B → O → K

Visited: C, D, E, I, N



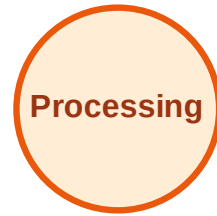
BFS Traversal

Step 12: Process node F

Legend



Complete



Processing

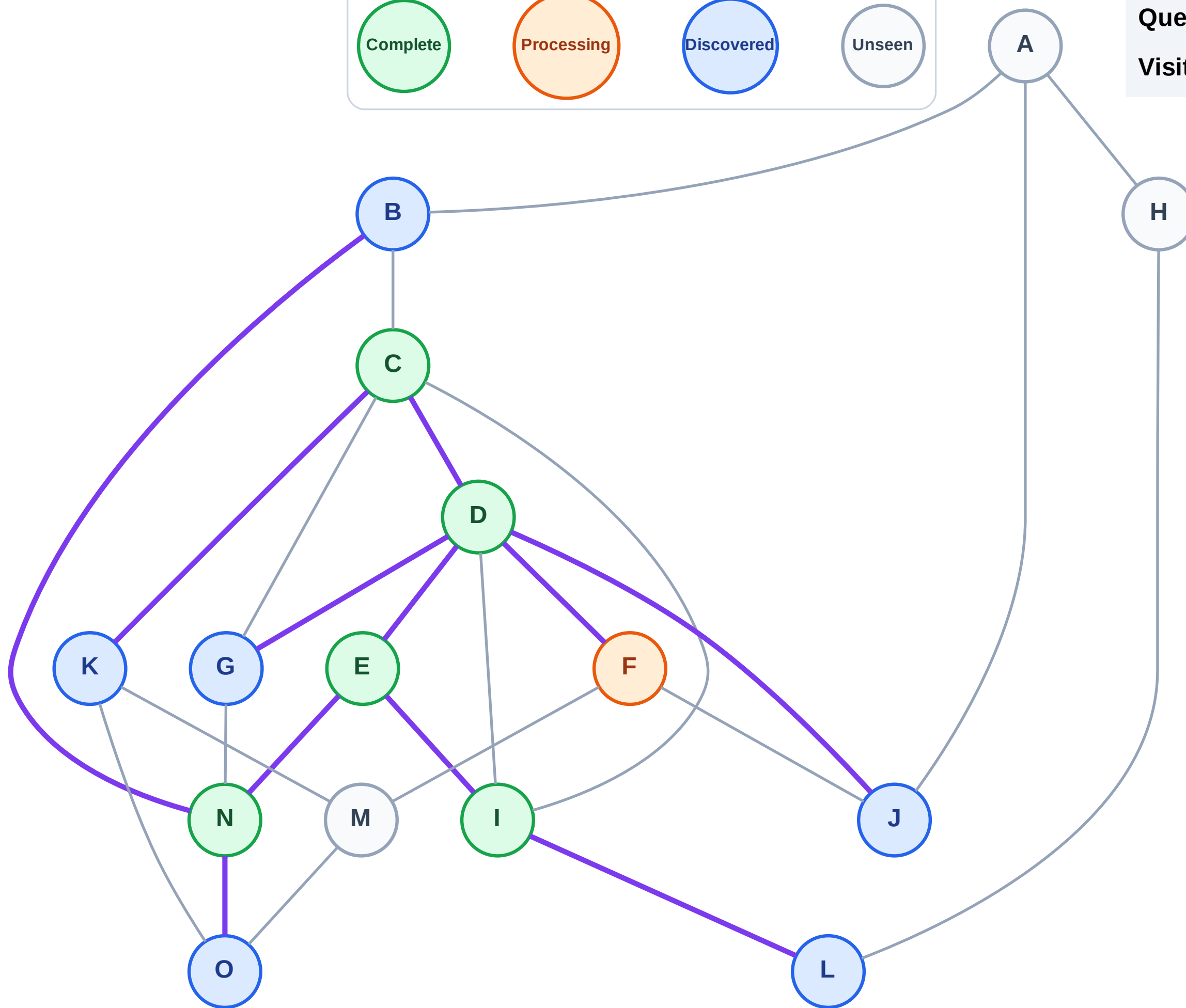


Discovered



Unseen

Queue: G → J → L → B → O → K
Visited: C, D, E, I, N



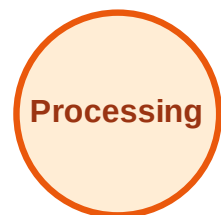
BFS Traversal

Step 13: Discover M from F

Legend



Complete



Processing



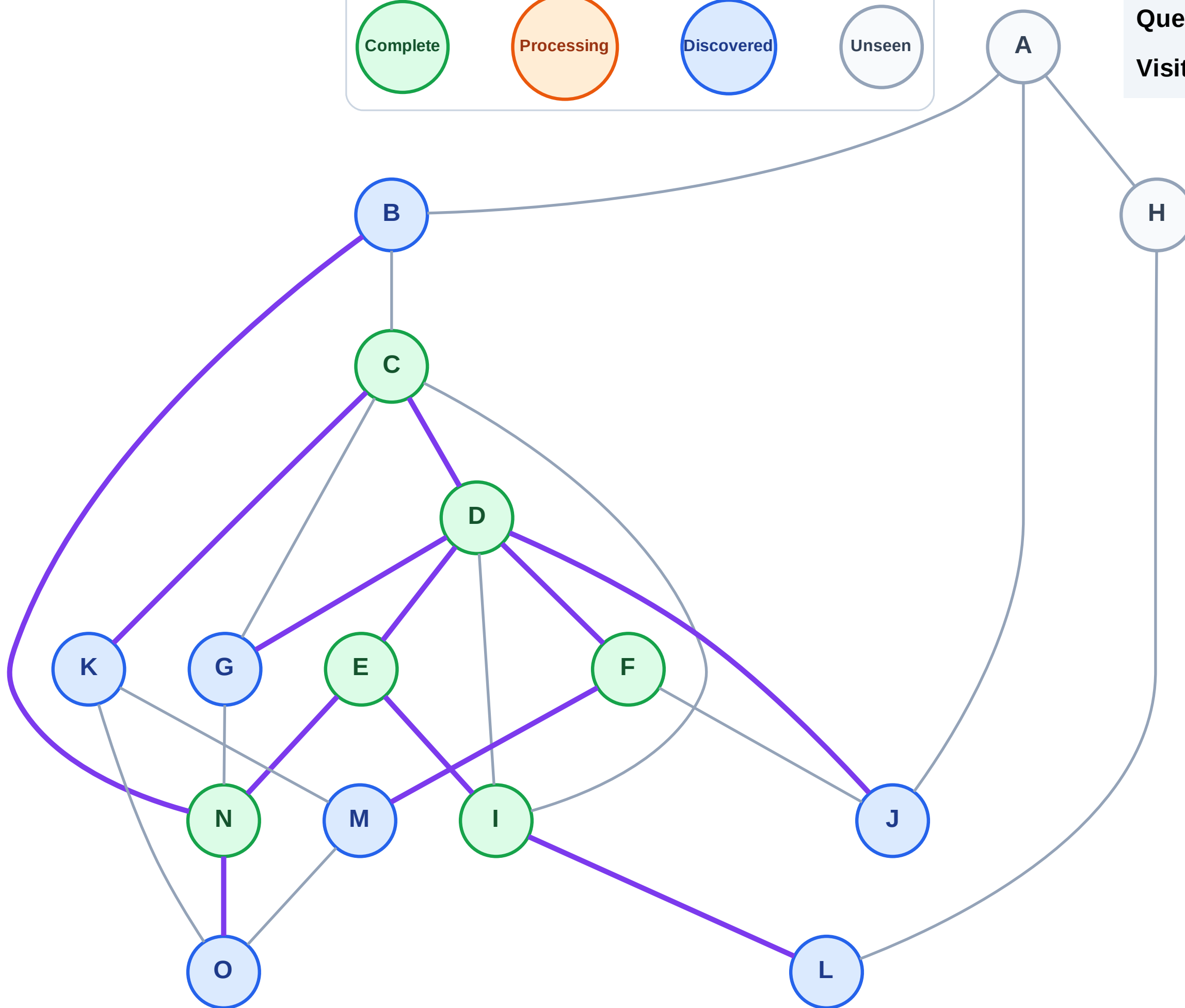
Discovered



Unseen

Queue: G → J → L → B → O → K → M

Visited: C, D, E, F, I, N



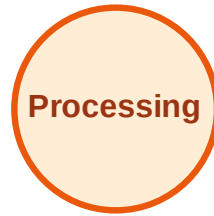
BFS Traversal

Step 14: Process node G

Legend



Complete



Processing

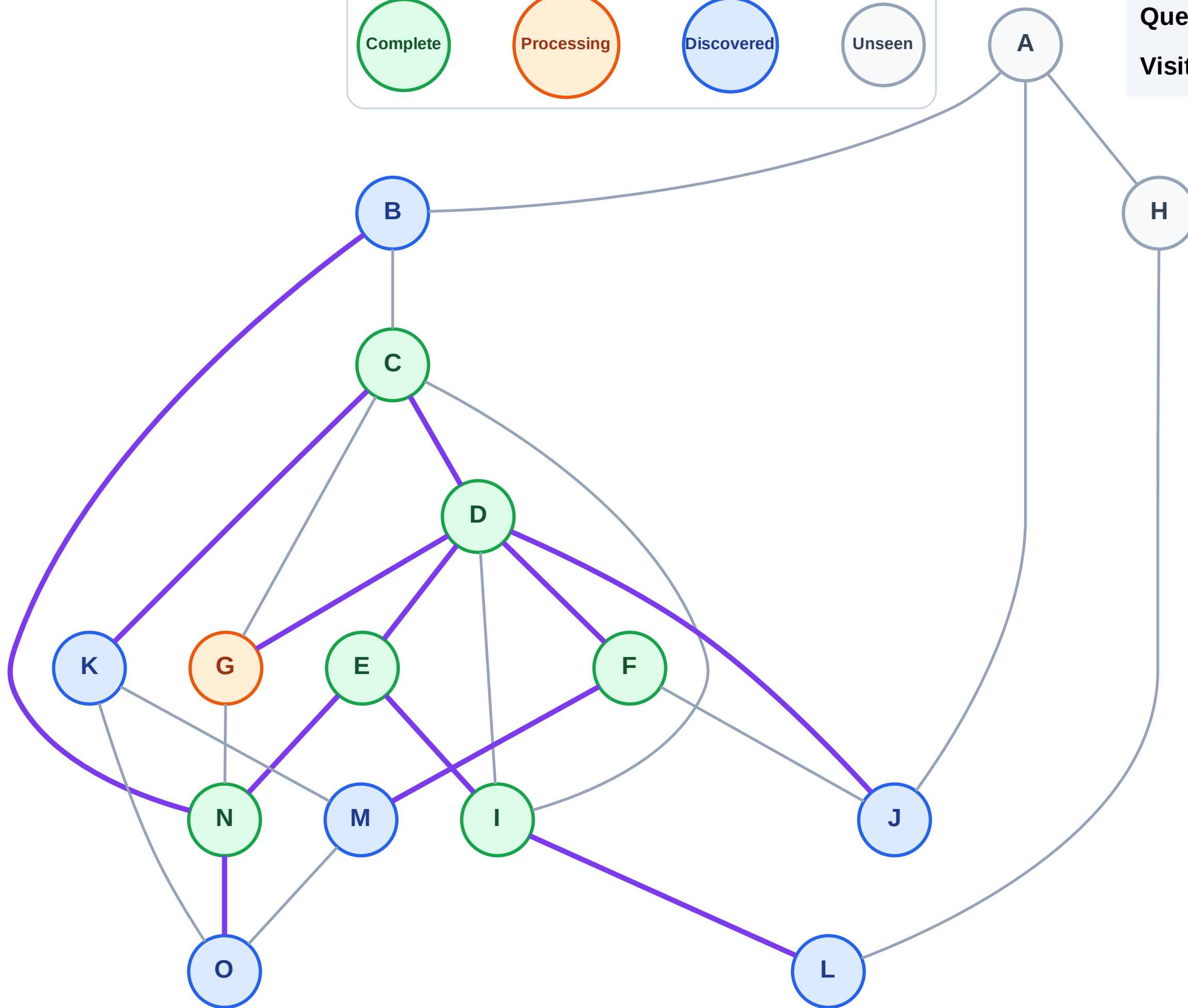


Discovered



Unseen

Queue: J → L → B → O → K → M
Visited: C, D, E, F, I, N



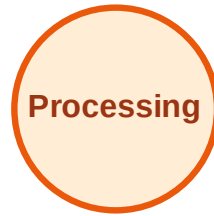
BFS Traversal

Step 15: No new neighbors from G

Legend



Complete



Processing

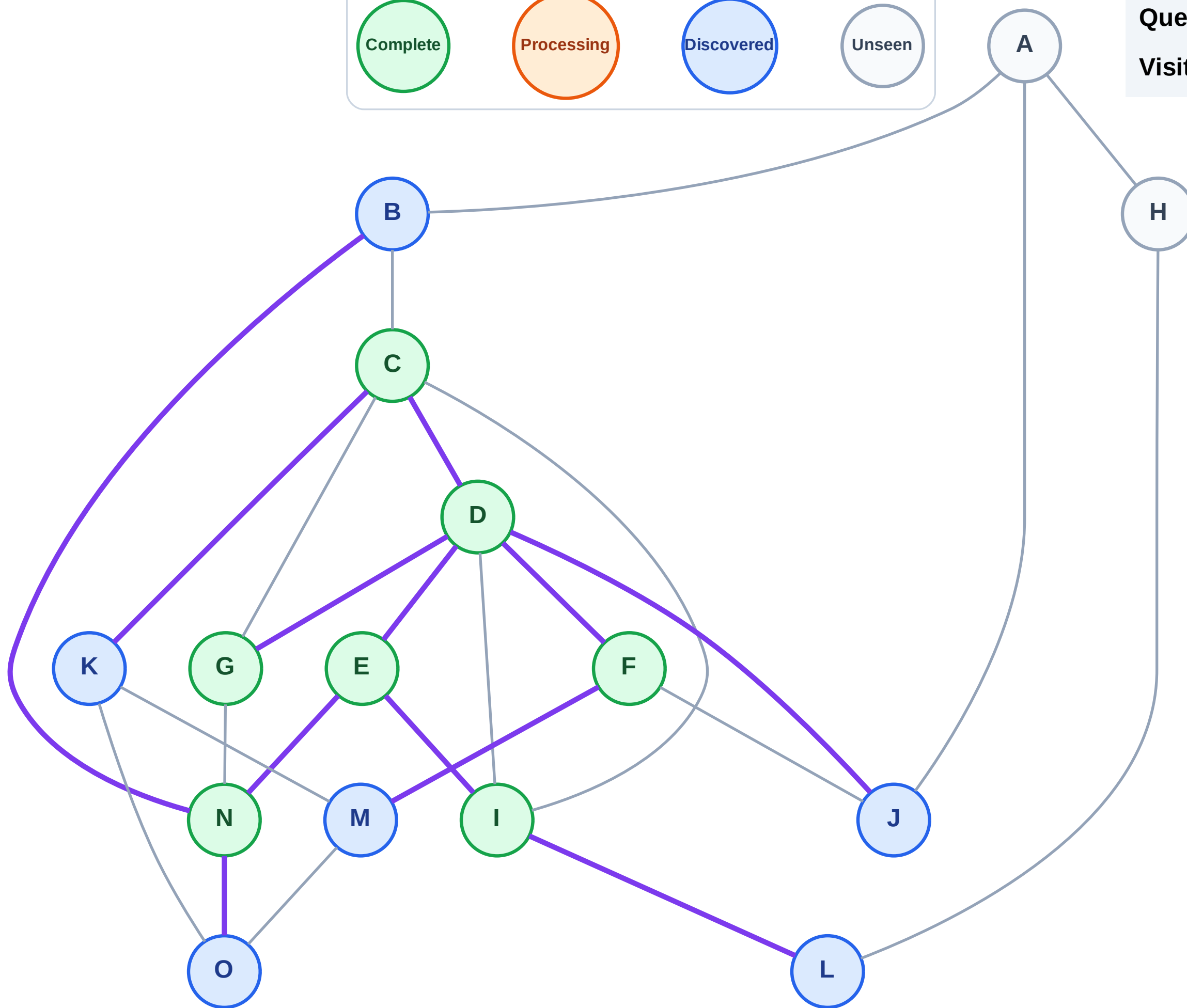


Discovered



Unseen

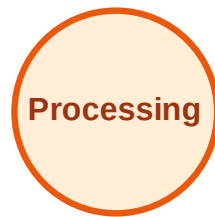
Queue: J → L → B → O → K → M
Visited: C, D, E, F, G, I, N



BFS Traversal

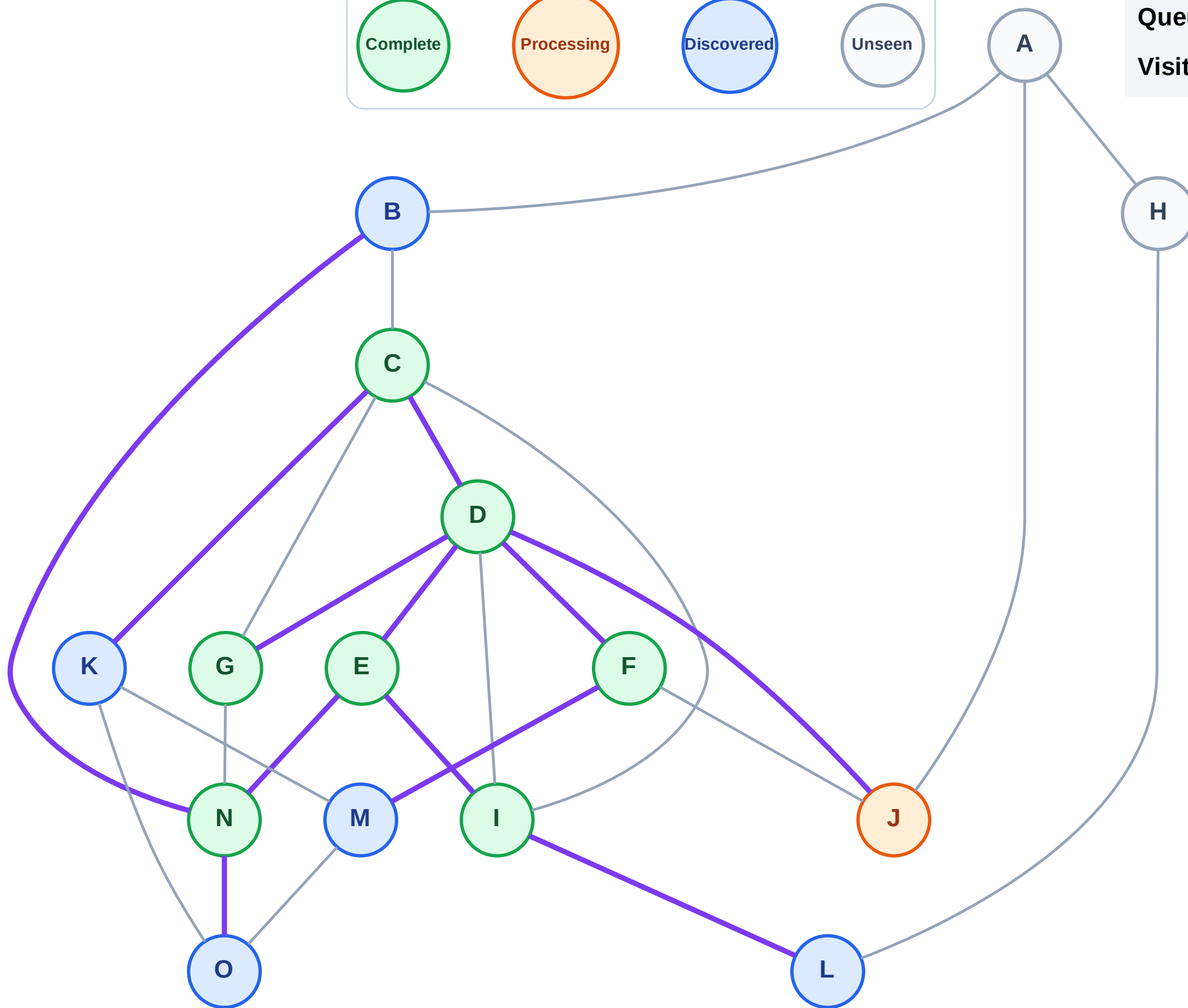
Step 16: Process node J

Legend



Queue: L → B → O → K → M

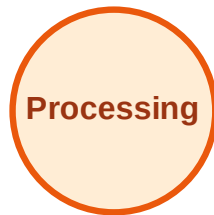
Visited: C, D, E, F, G, I, N



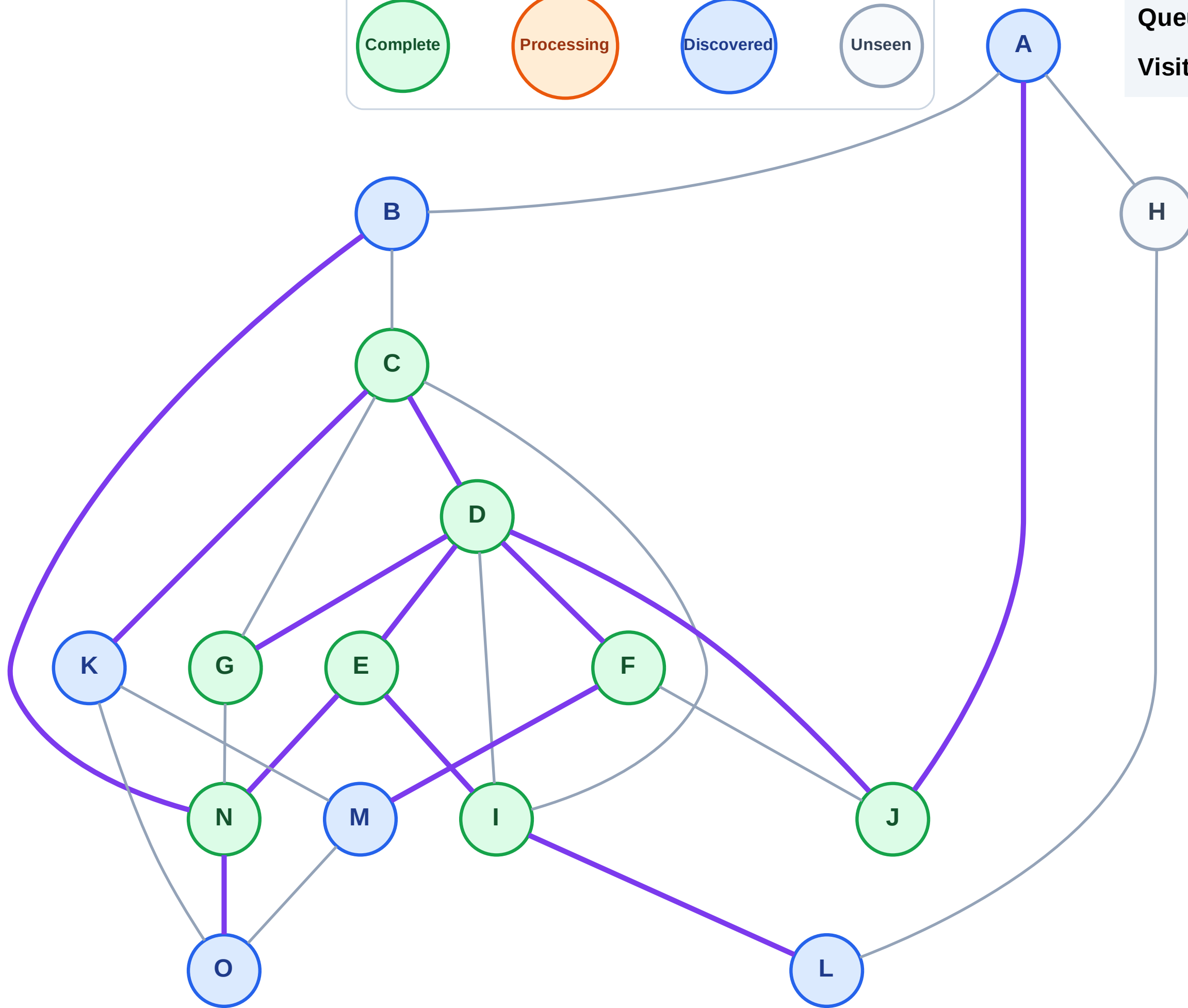
BFS Traversal

Step 17: Discover A from J

Legend



Queue: L → B → O → K → M → A
Visited: C, D, E, F, G, I, J, N



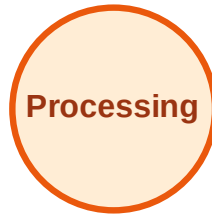
BFS Traversal

Step 18: Process node L

Legend



Complete



Processing

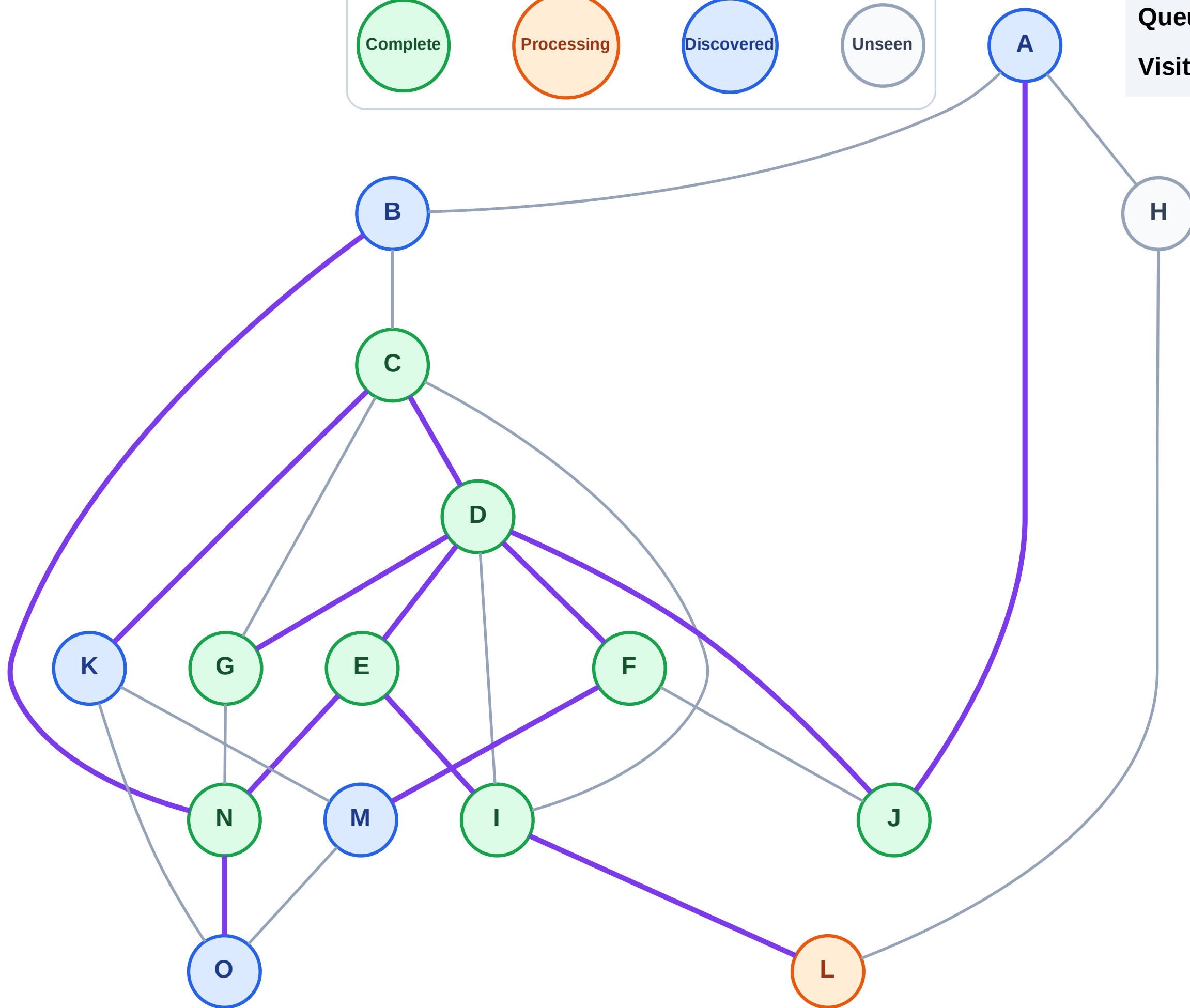


Discovered



Unseen

Queue: B → O → K → M → A
Visited: C, D, E, F, G, I, J, N



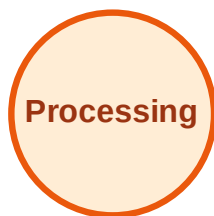
BFS Traversal

Step 19: Discover H from L

Legend



Complete



Processing



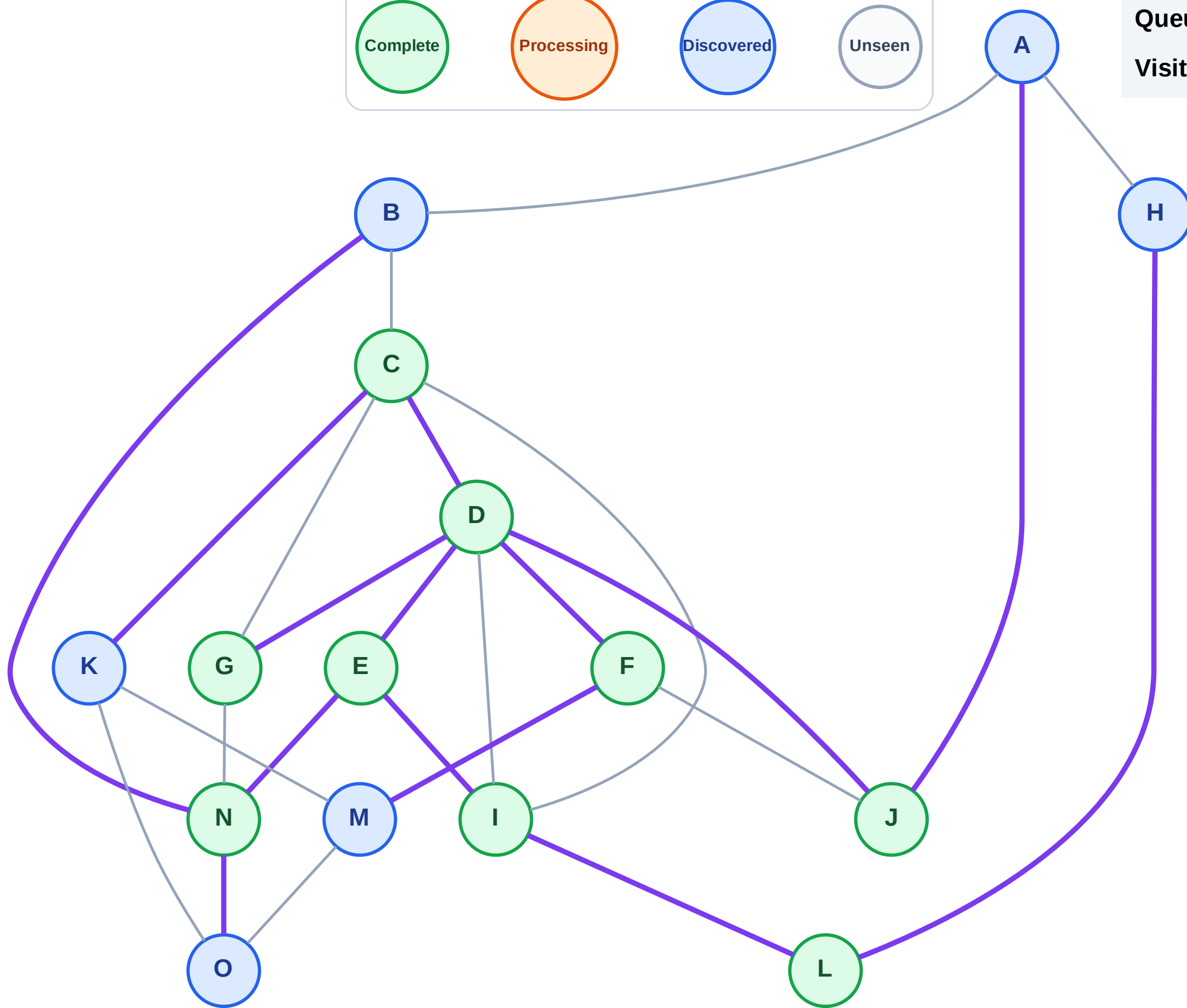
Discovered



Unseen

Queue: B → O → K → M → A → H

Visited: C, D, E, F, G, I, J, L, N



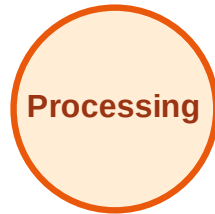
BFS Traversal

Step 20: Process node B

Legend



Complete



Processing



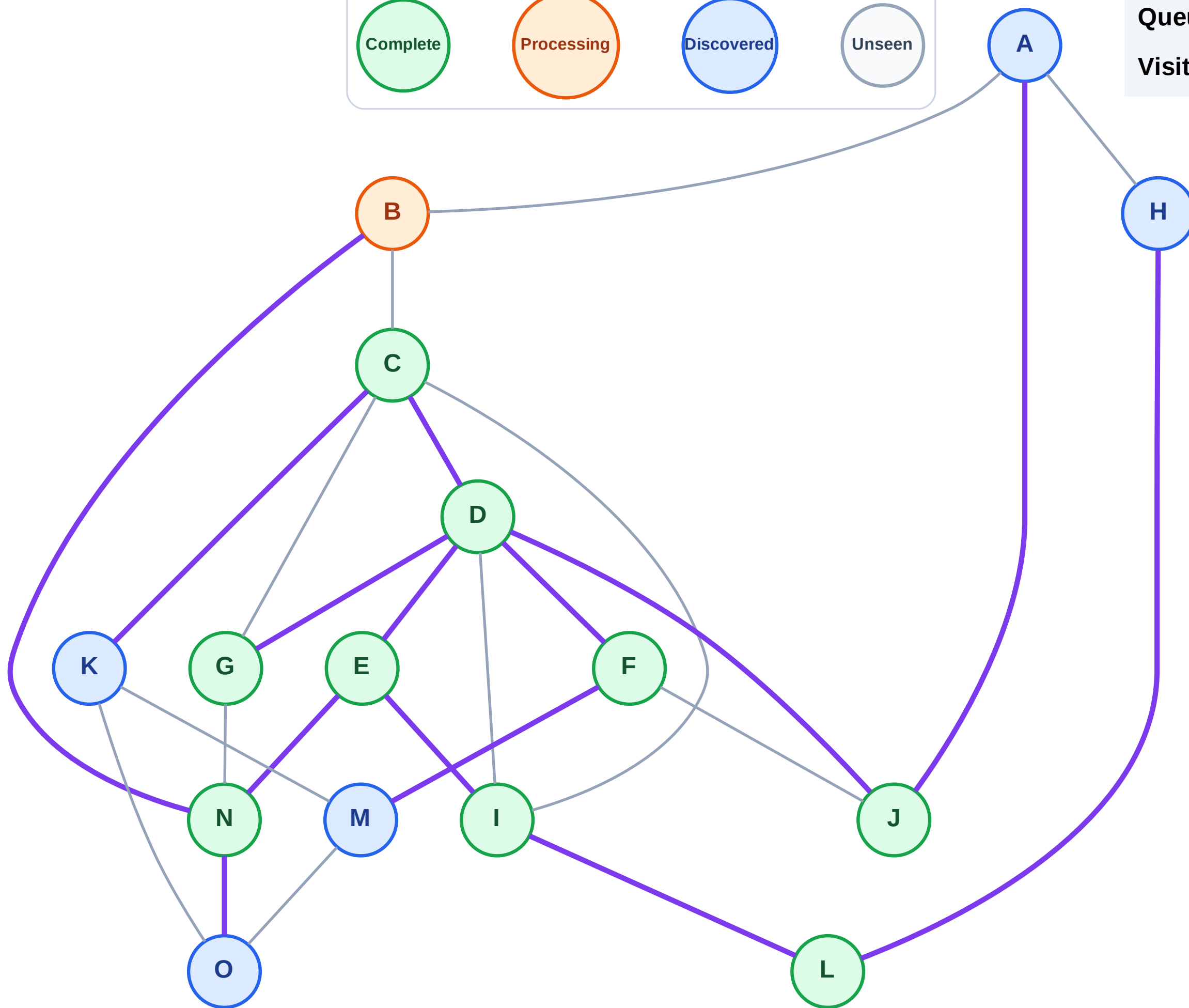
Discovered



Unseen

Queue: O → K → M → A → H

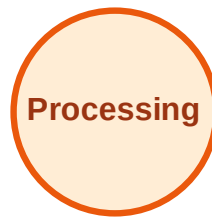
Visited: C, D, E, F, G, I, J, L, N



BFS Traversal

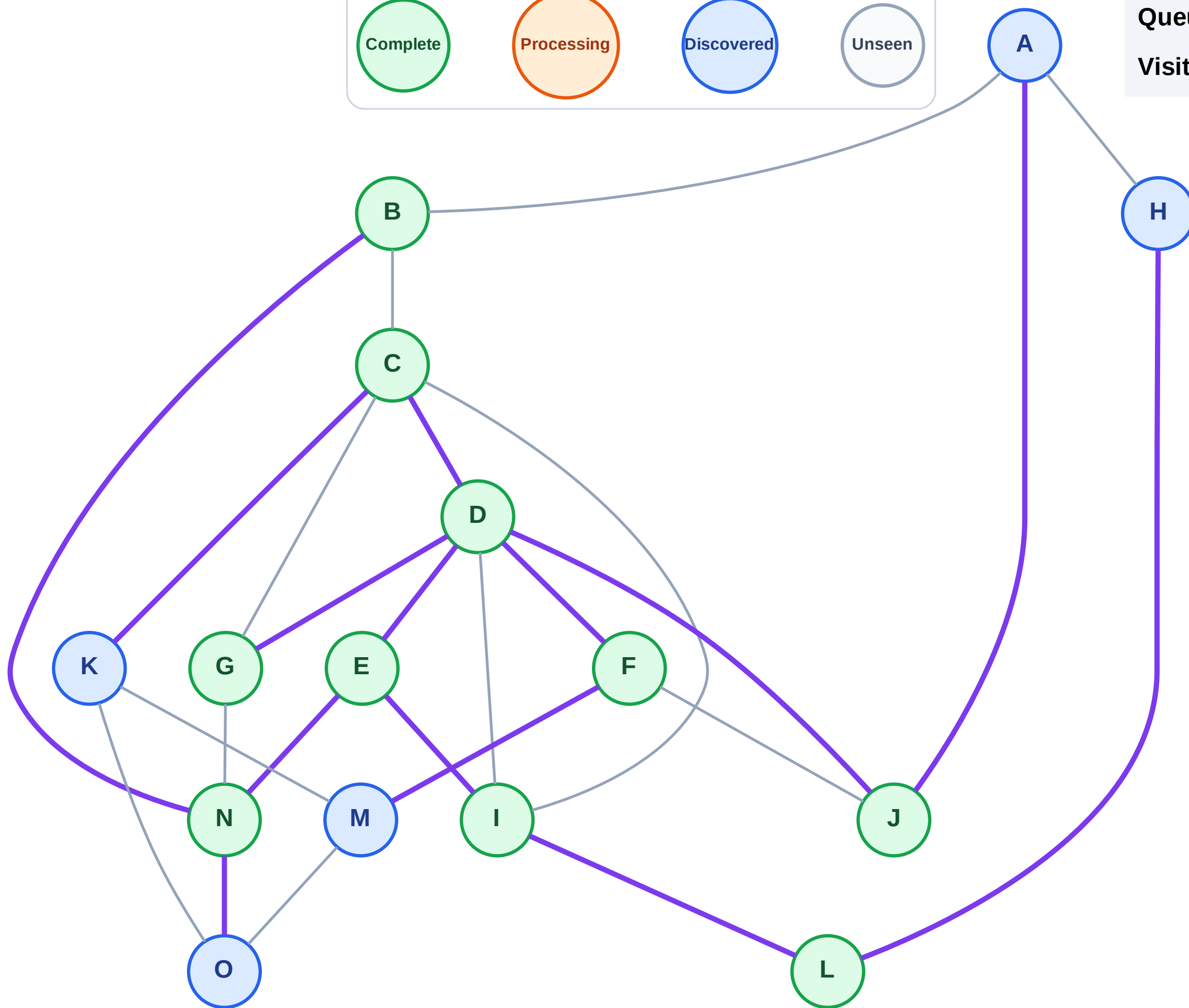
Step 21: No new neighbors from B

Legend



Queue: O → K → M → A → H

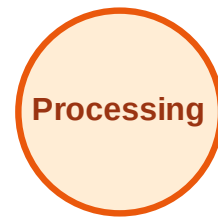
Visited: B, C, D, E, F, G, I, J, L, N



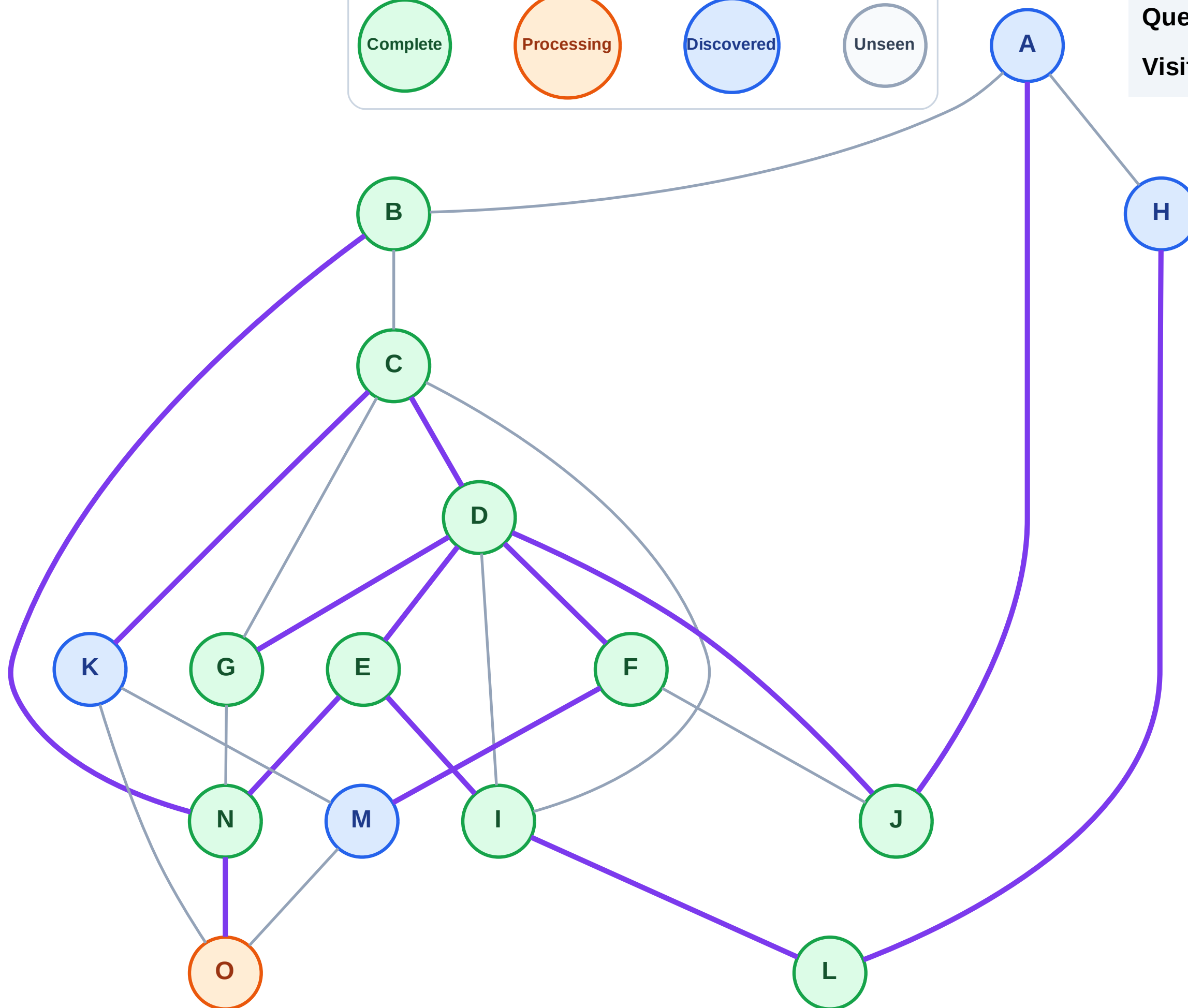
BFS Traversal

Step 22: Process node O

Legend



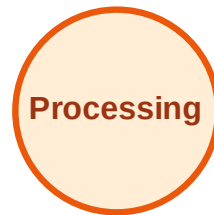
Queue: K → M → A → H
Visited: B, C, D, E, F, G, I, J, L, N



BFS Traversal

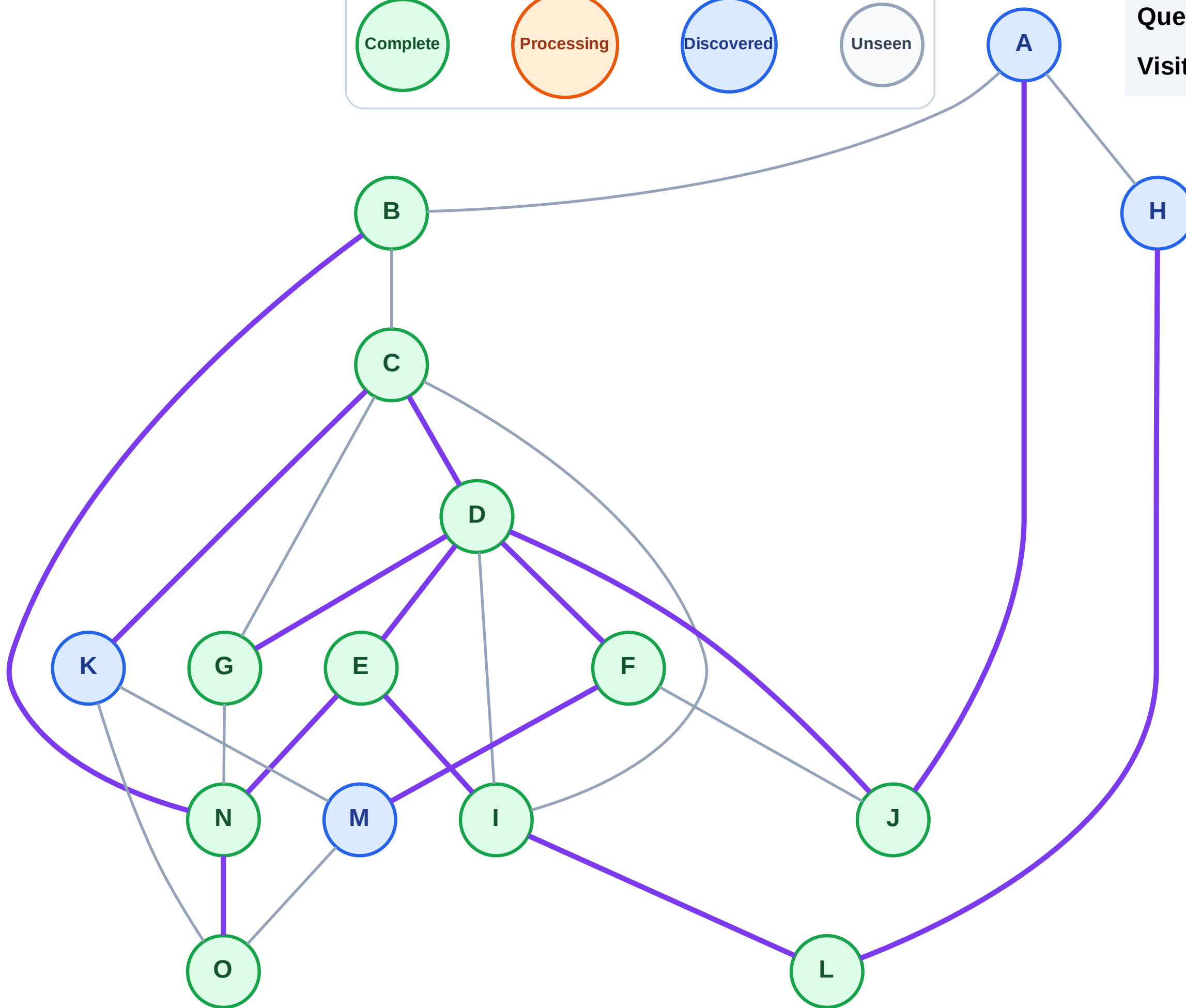
Step 23: No new neighbors from O

Legend



Queue: K → M → A → H

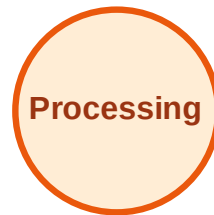
Visited: B, C, D, E, F, G, I, J, L, N, O



BFS Traversal

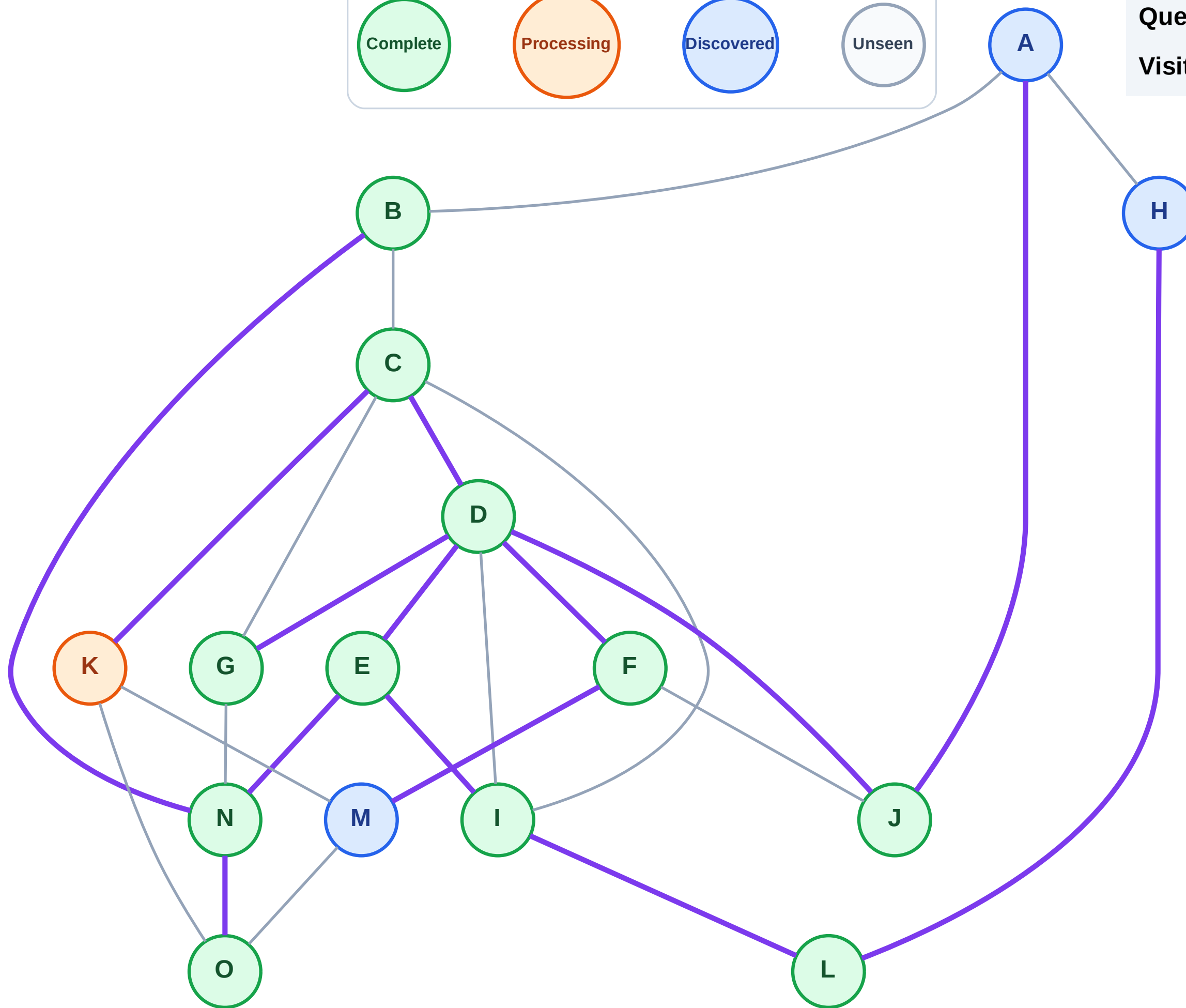
Step 24: Process node K

Legend



Queue: M → A → H

Visited: B, C, D, E, F, G, I, J, L, N, O



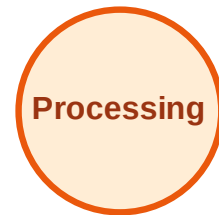
BFS Traversal

Step 25: No new neighbors from K

Legend



Complete



Processing



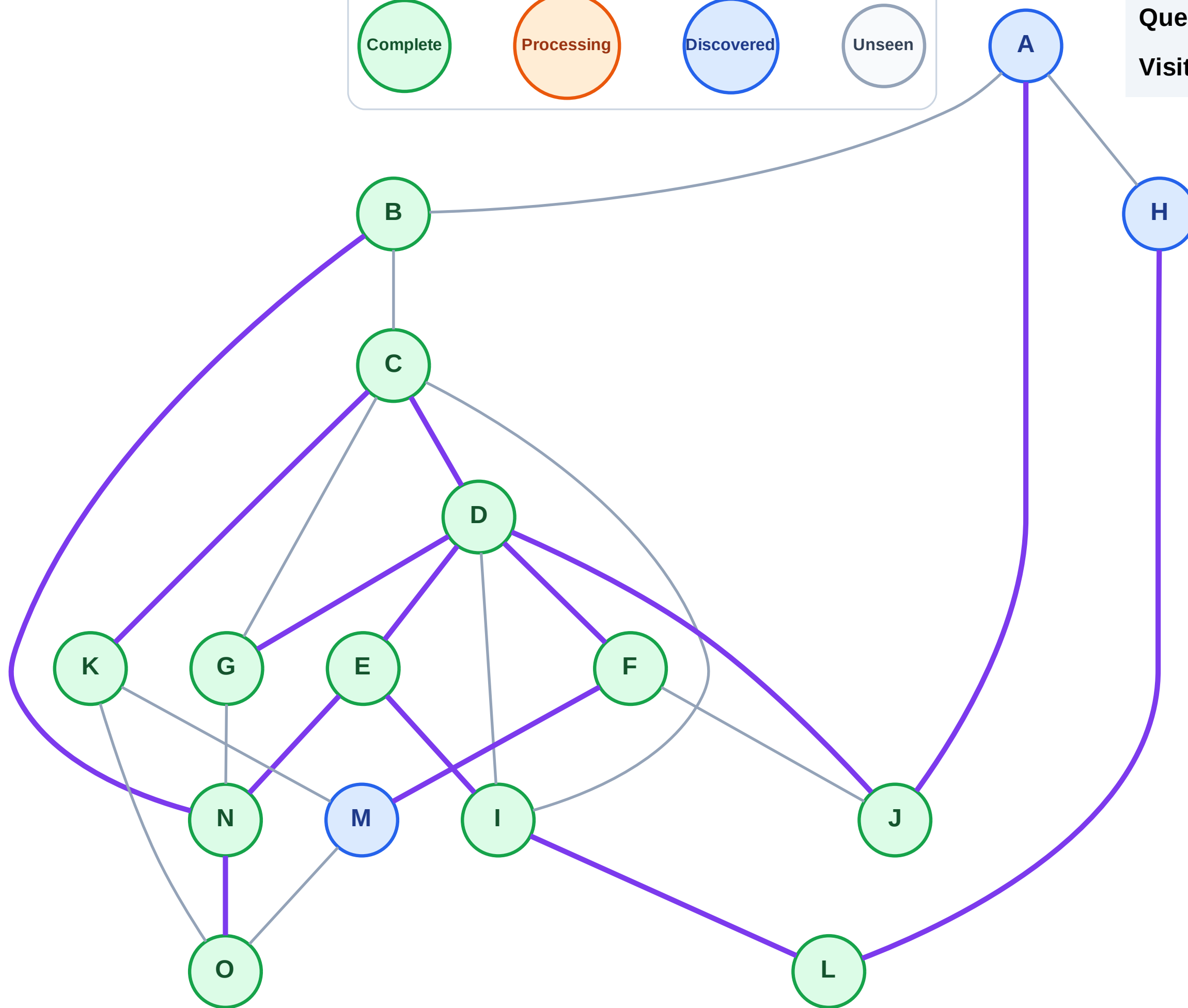
Discovered



Unseen

Queue: M → A → H

Visited: B, C, D, E, F, G, I, J, K, L, N, O



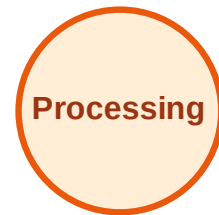
BFS Traversal

Step 26: Process node M

Legend



Complete



Processing



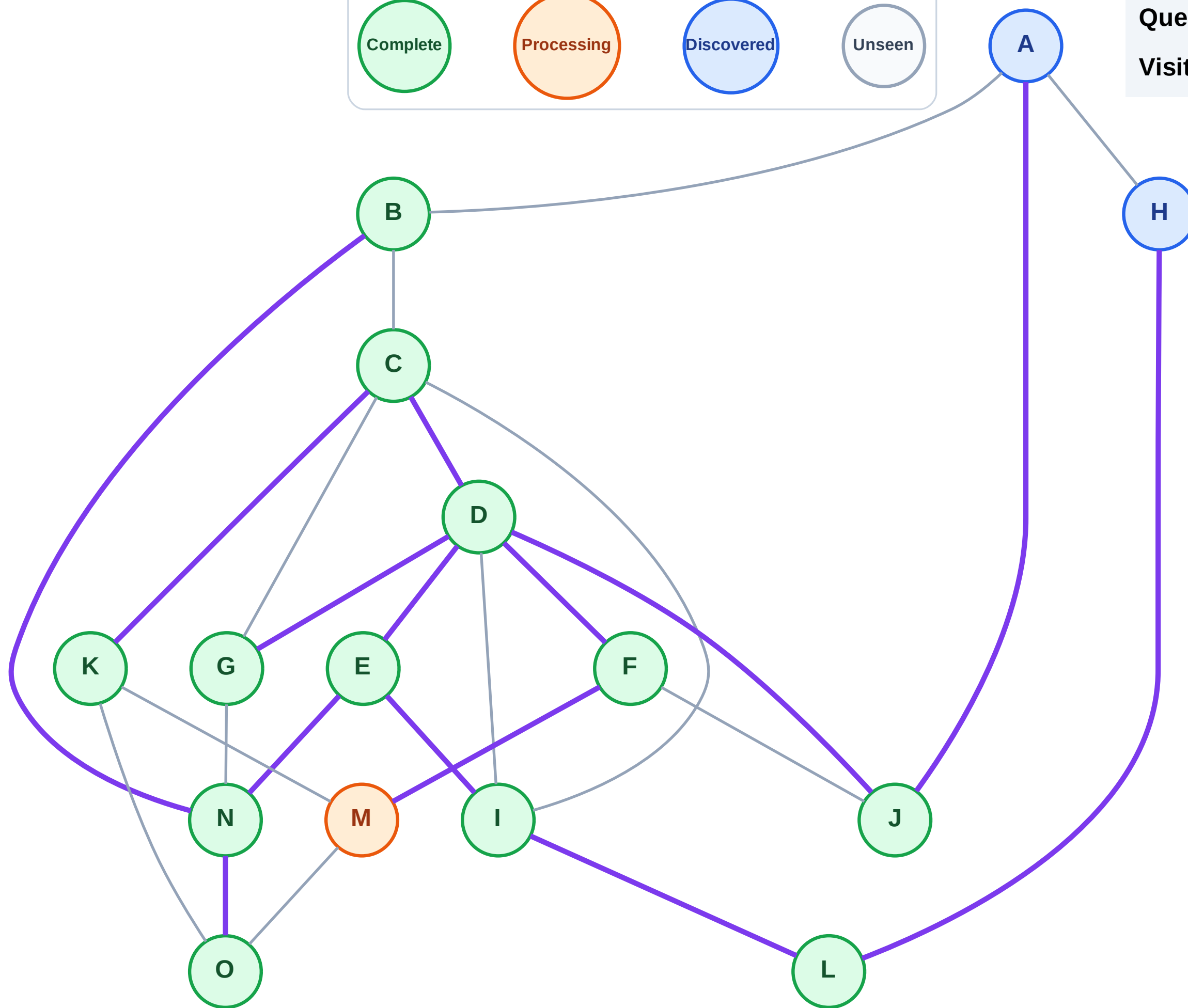
Discovered



Unseen

Queue: A → H

Visited: B, C, D, E, F, G, I, J, K, L, N, O

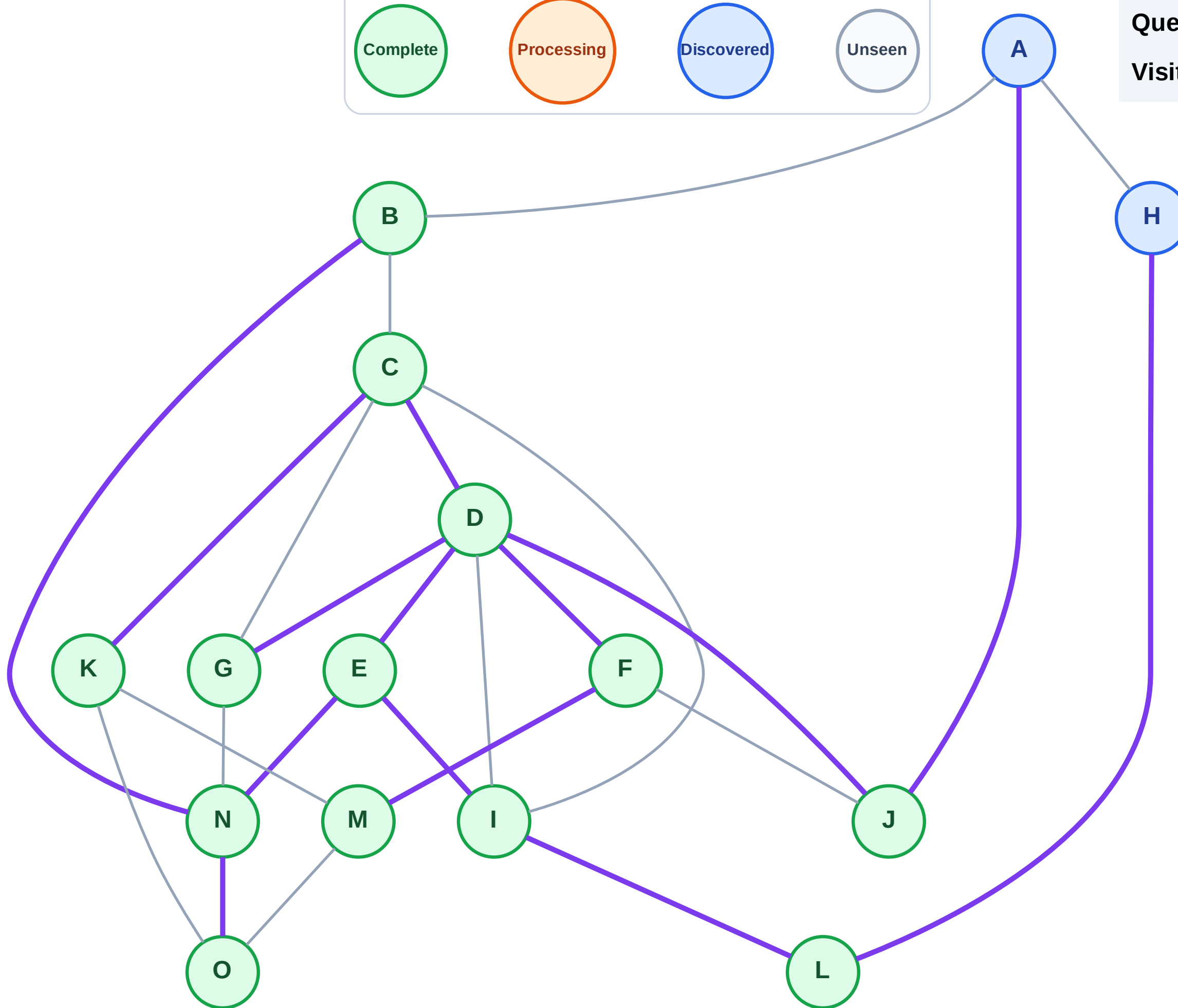


Step 27: No new neighbors from M

Legend

Unseen

Visited: B, C, D, E, F, G, I, J, K, L, M, N, O



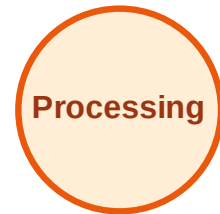
BFS Traversal

Step 28: Process node A

Legend



Complete



Processing



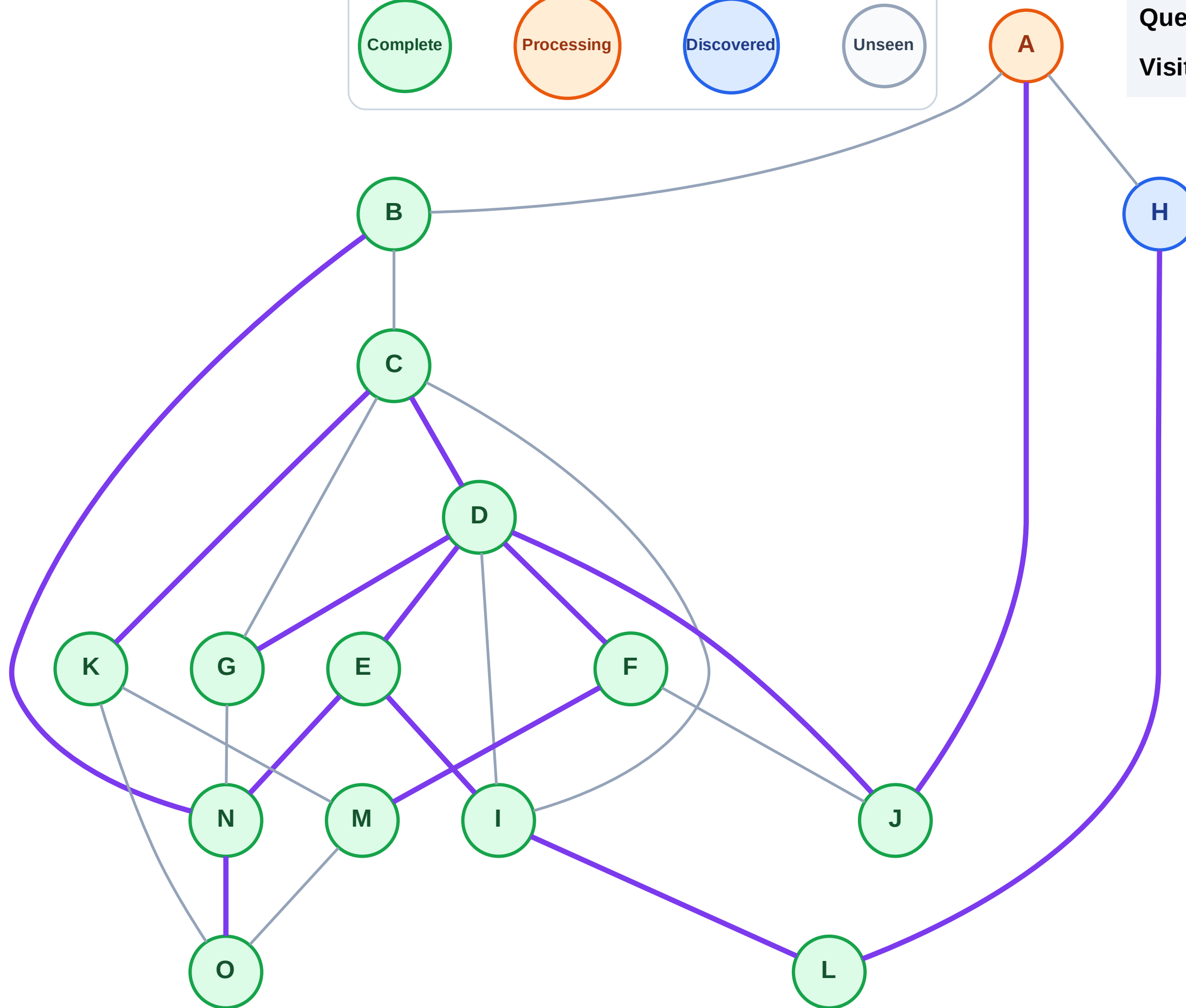
Discovered



Unseen

Queue: H

Visited: B, C, D, E, F, G, I, J, K, L, M, N, O



BFS Traversal

Step 29: No new neighbors from A

Legend

Complete

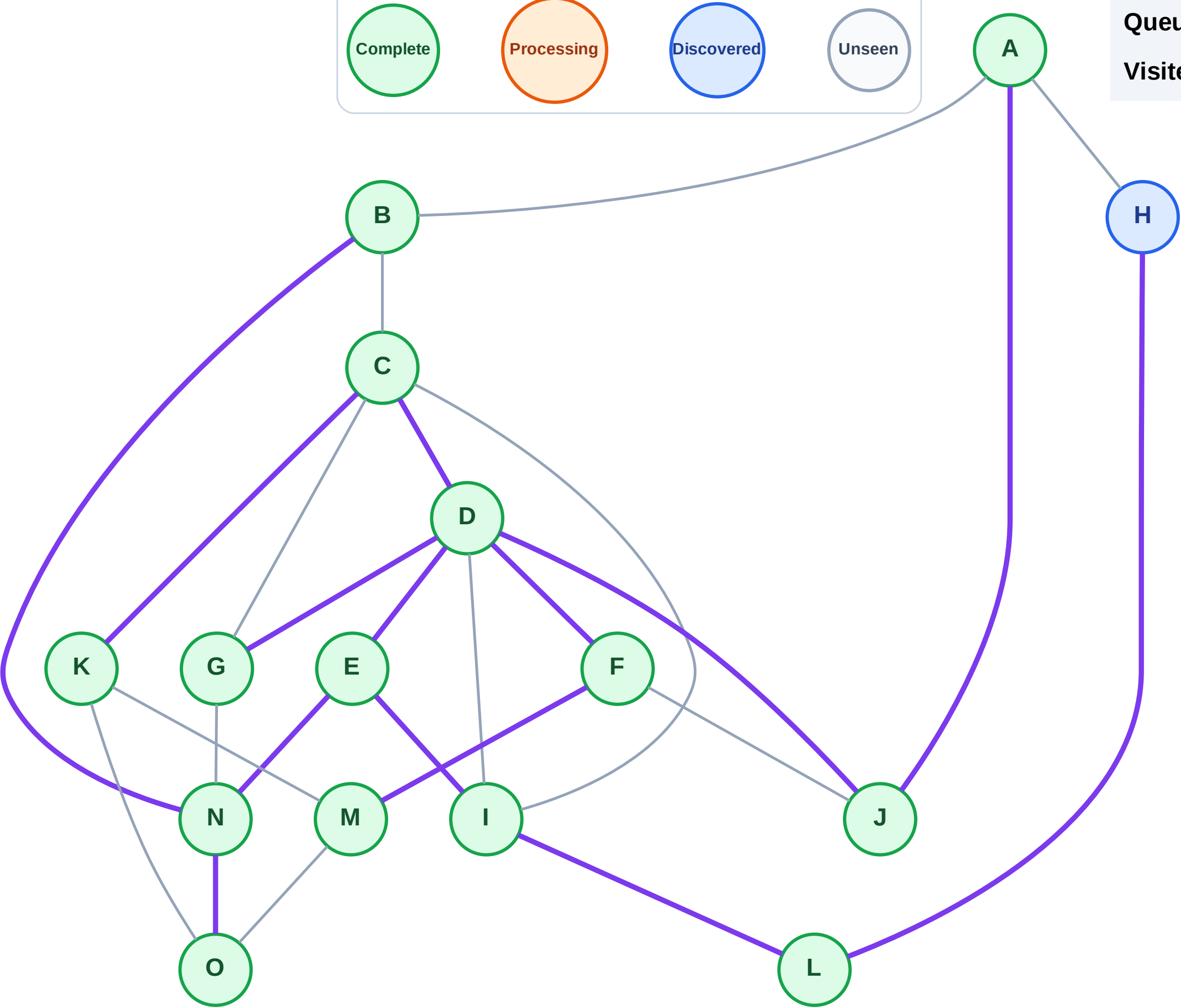
Processing

Discovered

Unseen

Queue: H

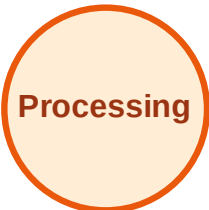
Visited: A, B, C, D, E, F, G, I, J, K, L, M, N, O



BFS Traversal

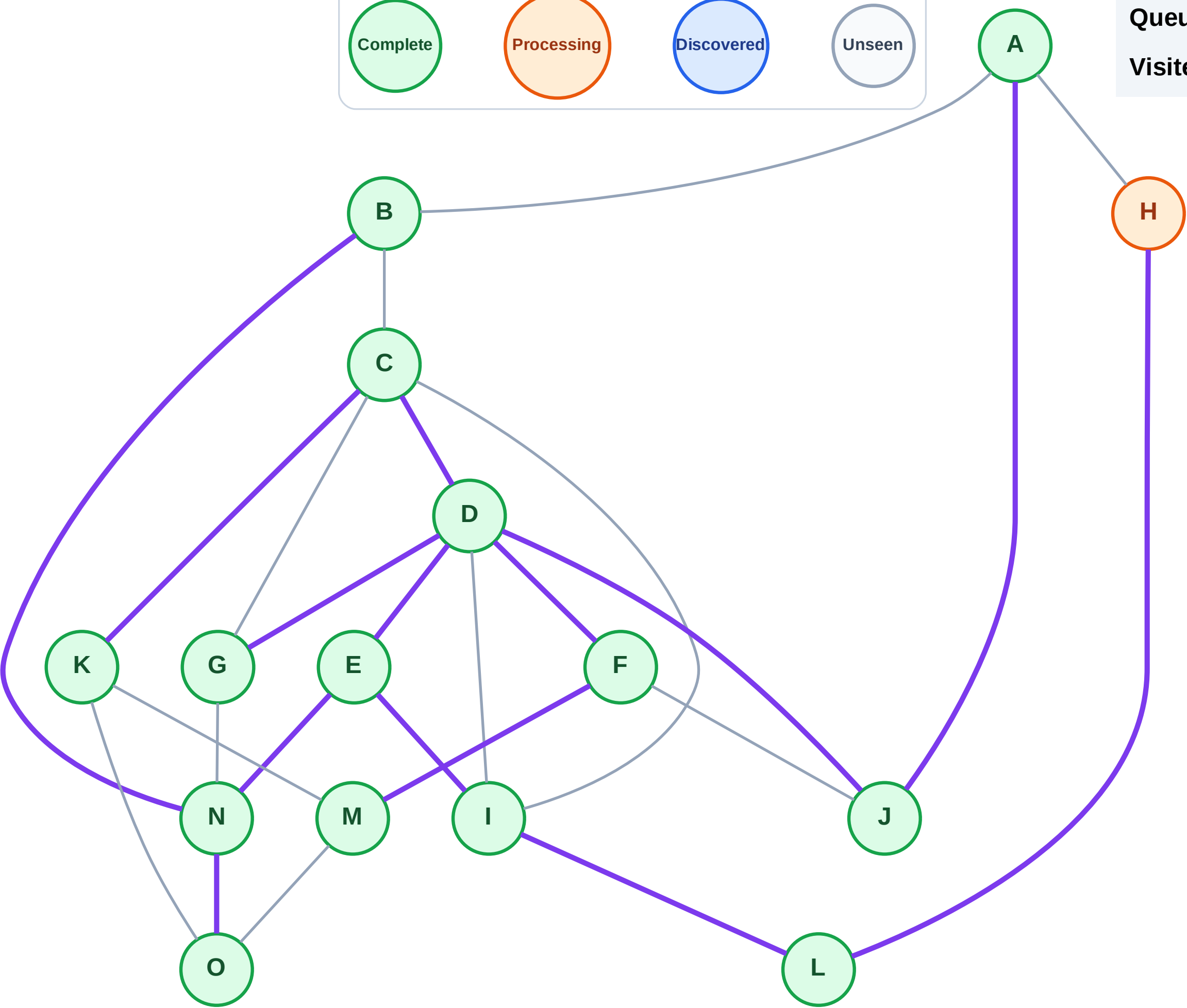
Step 30: Process node H

Legend



Queue: (empty)

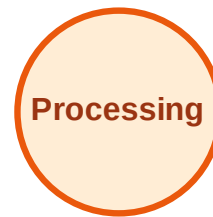
Visited: A, B, C, D, E, F, G, I, J, K, L, M, N, O



BFS Traversal

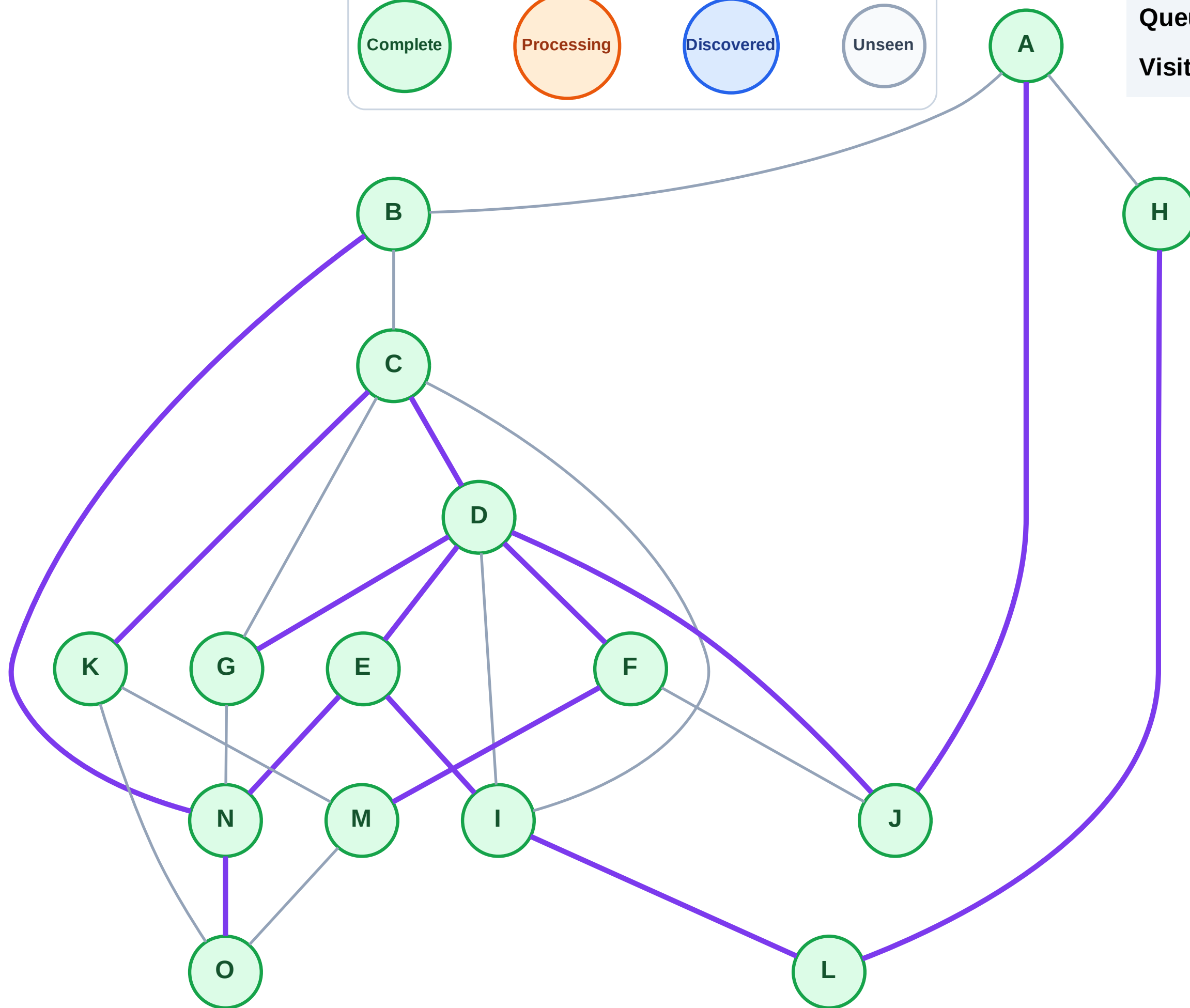
Step 31: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



DFS Traversal

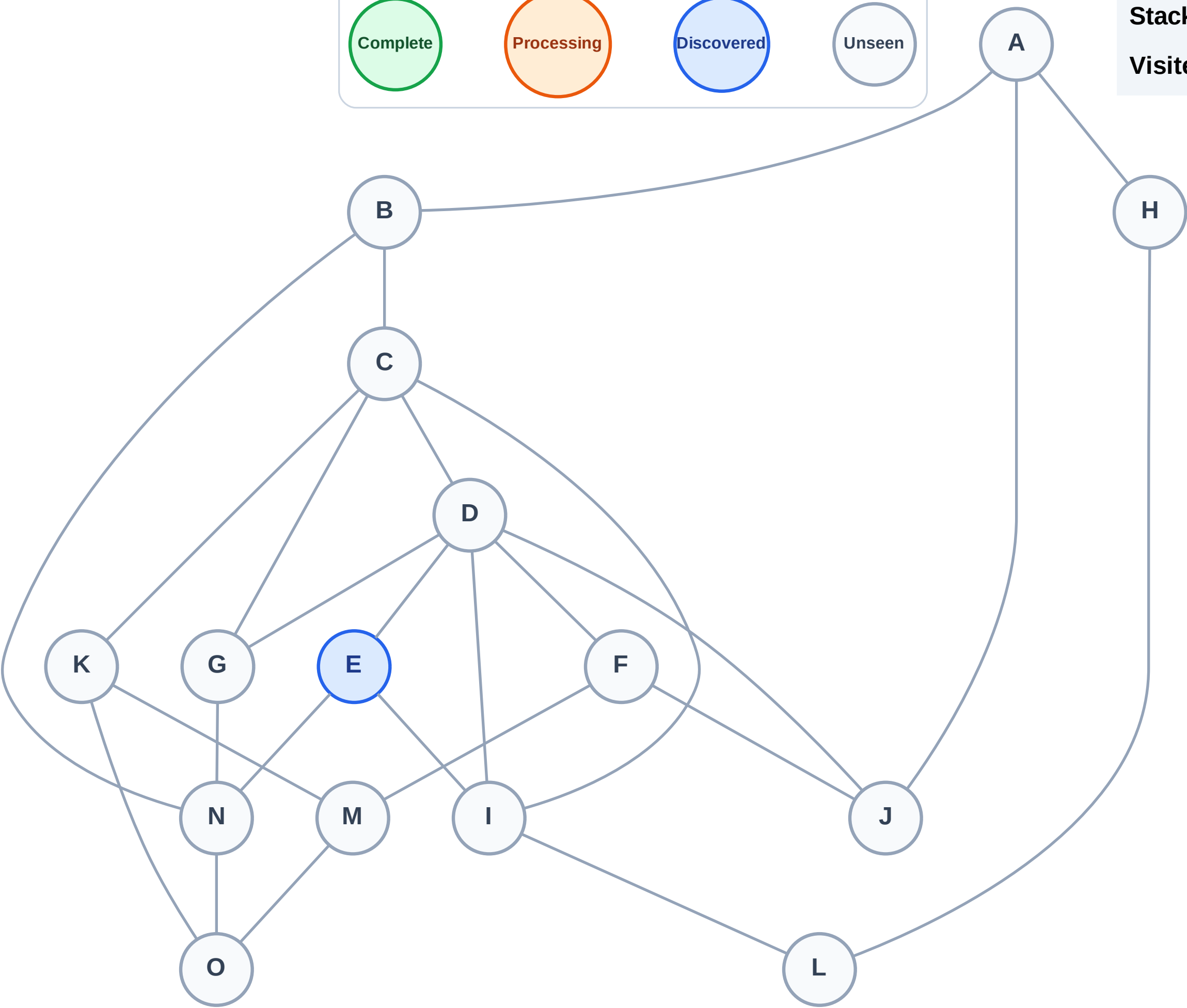
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



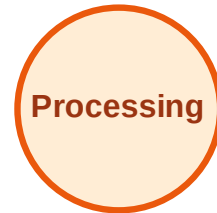
DFS Traversal

Step 2: Process node E

Legend



Complete



Processing



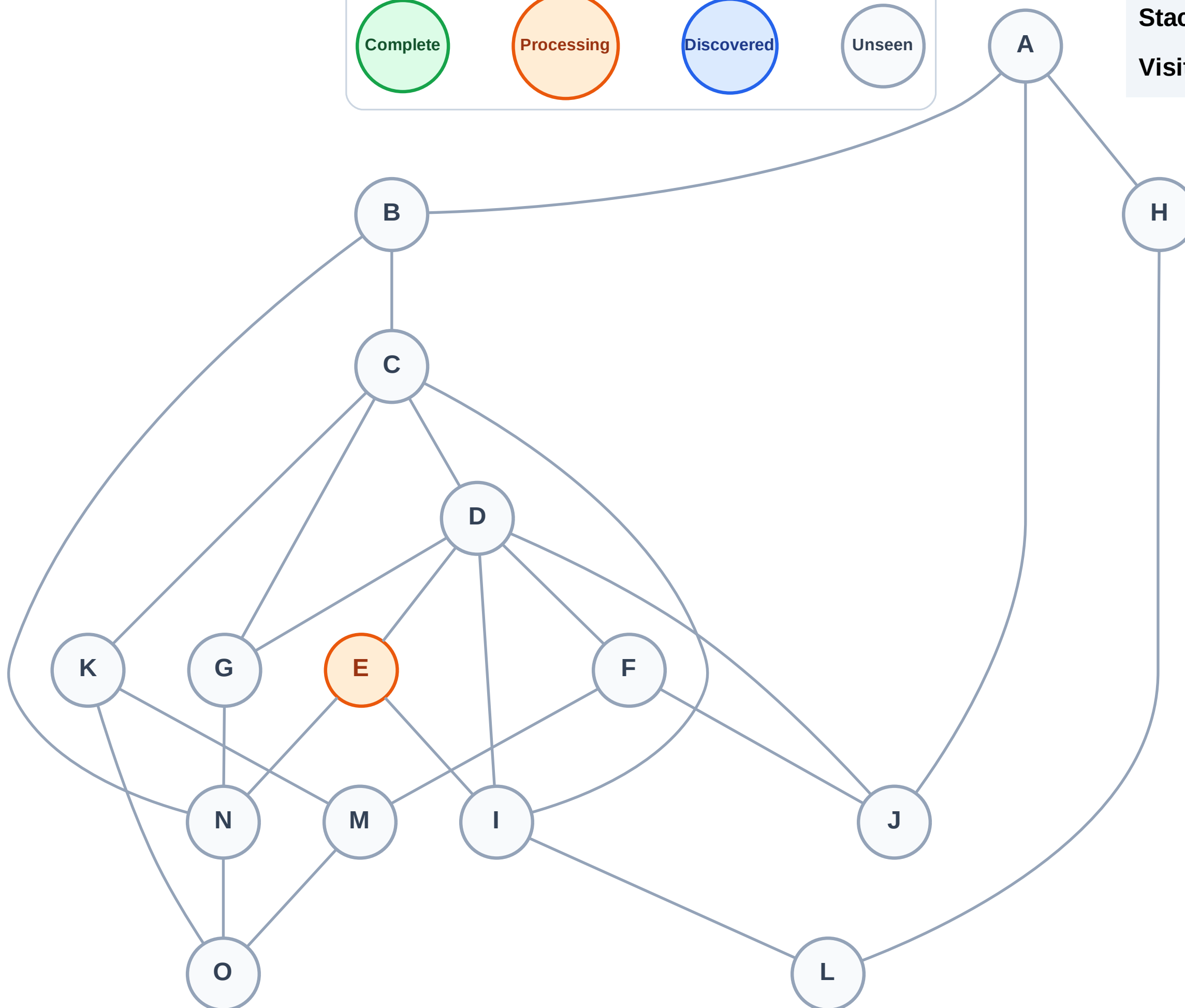
Discovered



Unseen

Stack (top at right): (empty)

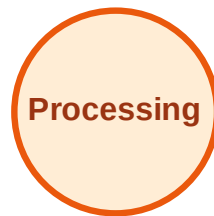
Visited: (none)



DFS Traversal

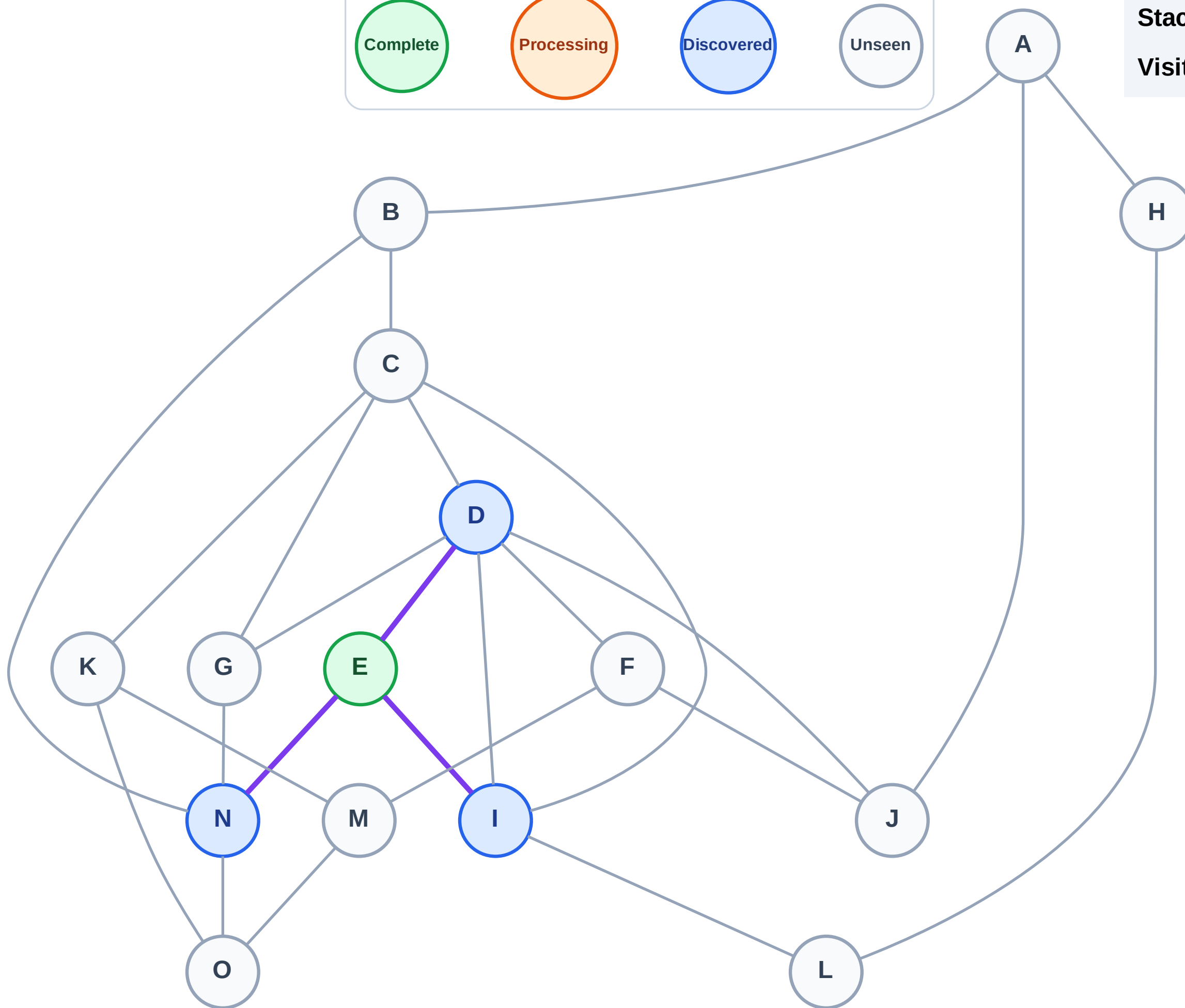
Step 3: Discover N, I, D from E

Legend



Stack (top at right): N | I | D

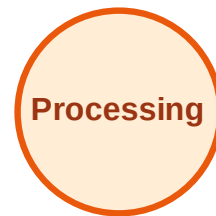
Visited: E



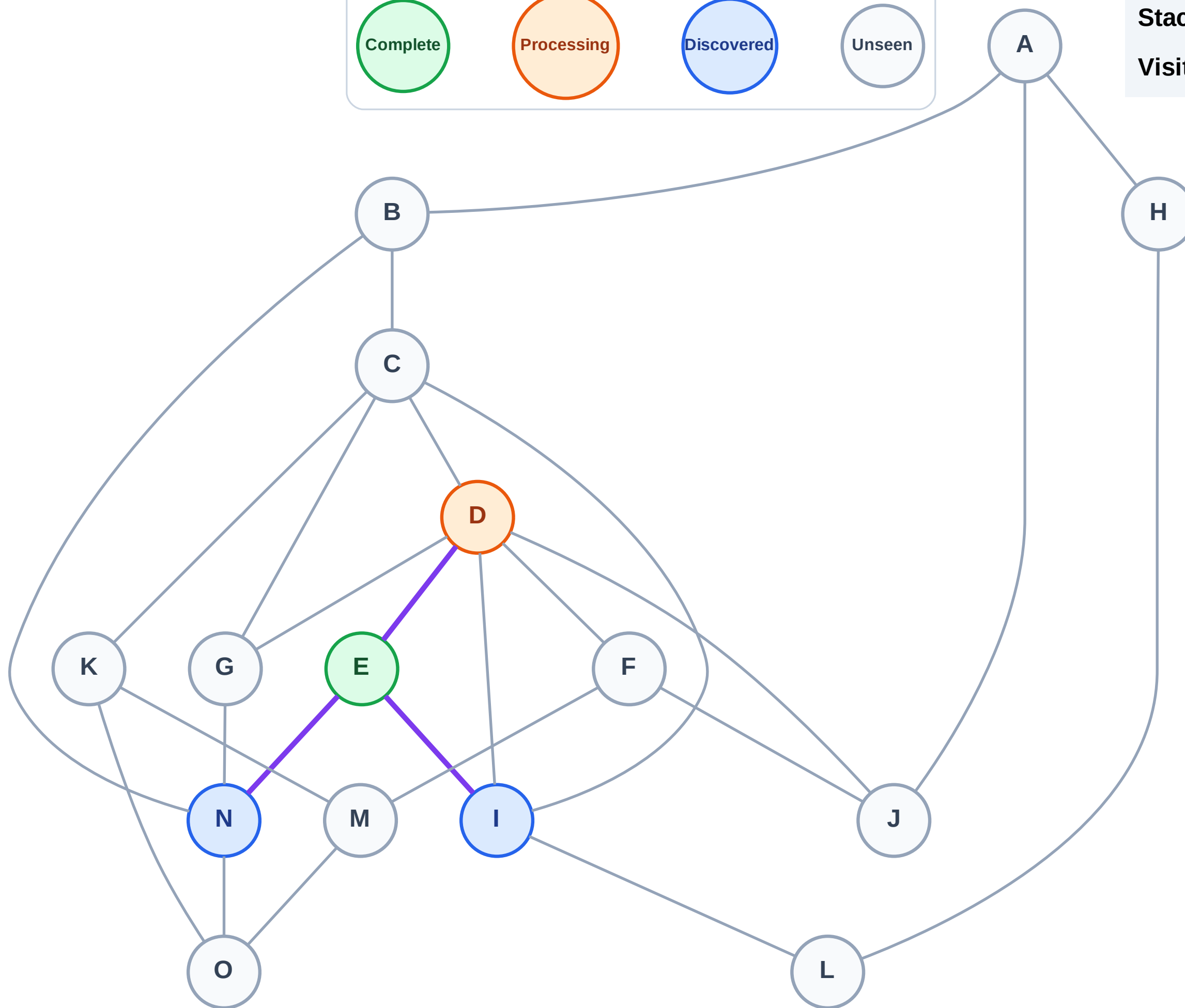
DFS Traversal

Step 4: Process node D

Legend



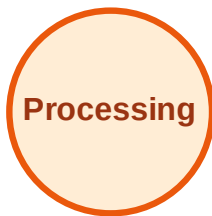
Stack (top at right): N | I
Visited: E



DFS Traversal

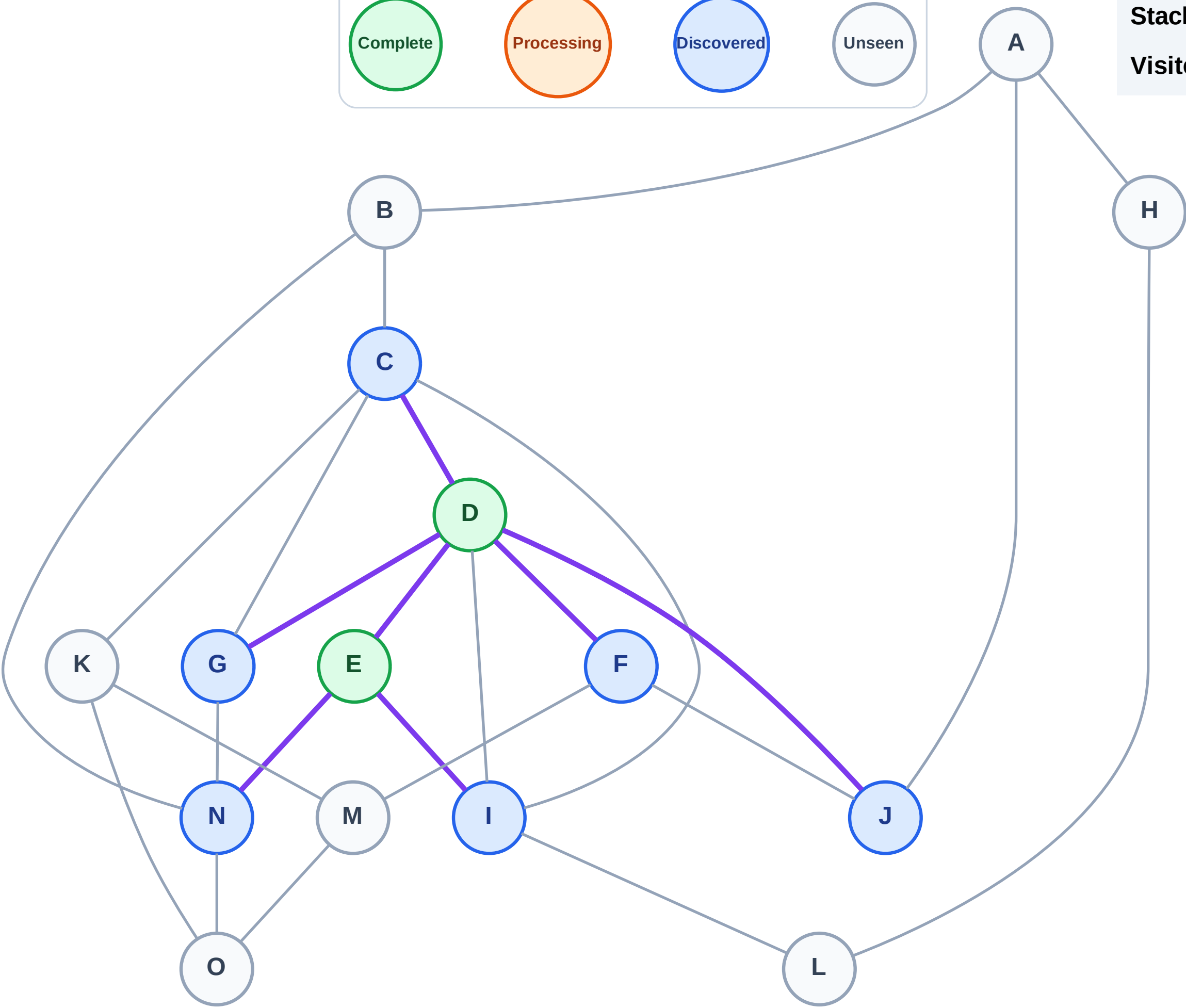
Step 5: Discover J, G, F, C from D

Legend



Stack (top at right): N | I | J | G | F | C

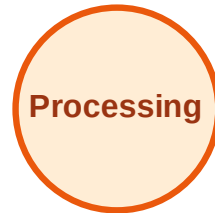
Visited: D, E



DFS Traversal

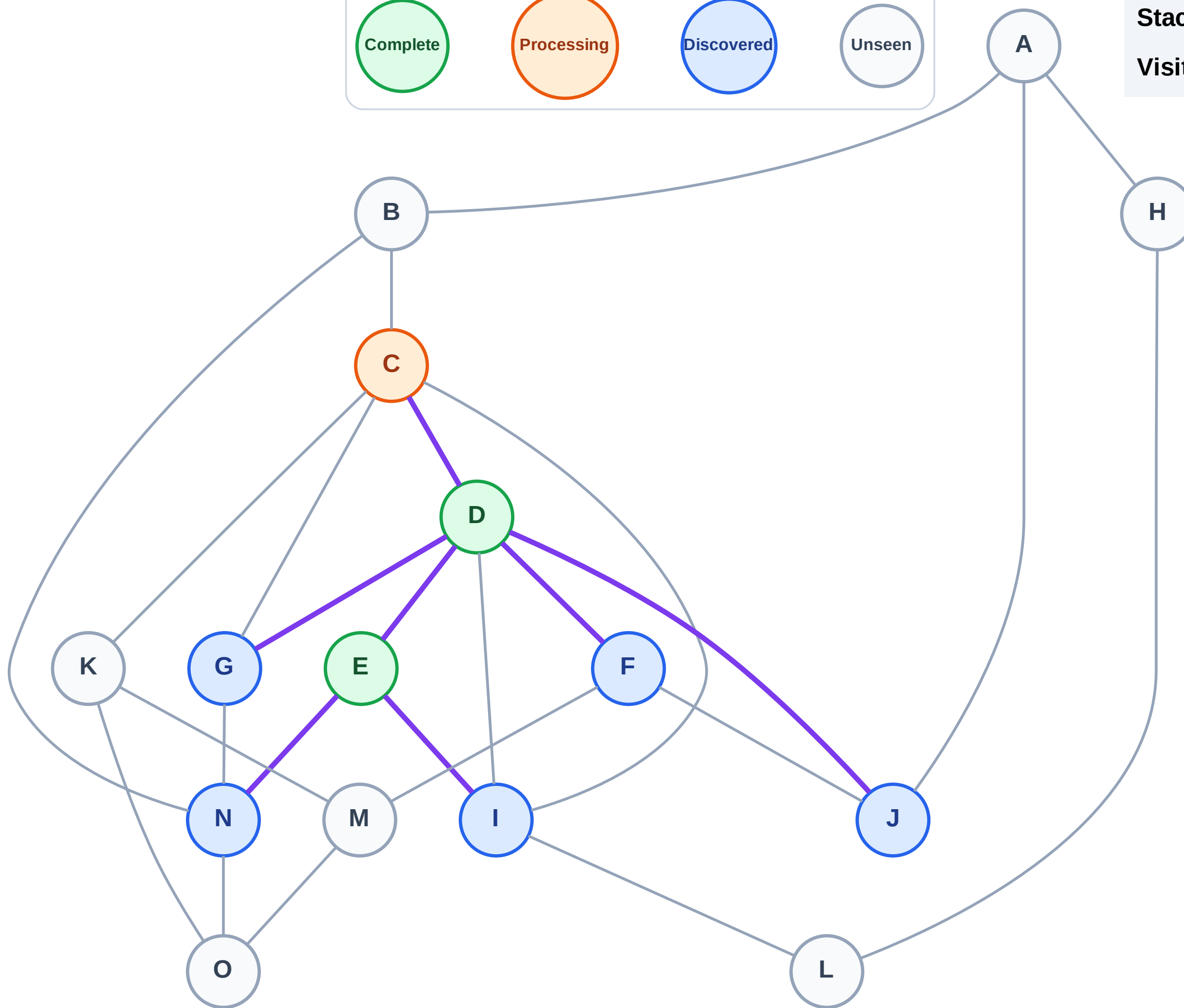
Step 6: Process node C

Legend



Stack (top at right): N | I | J | G | F

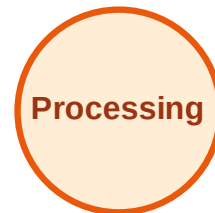
Visited: D, E



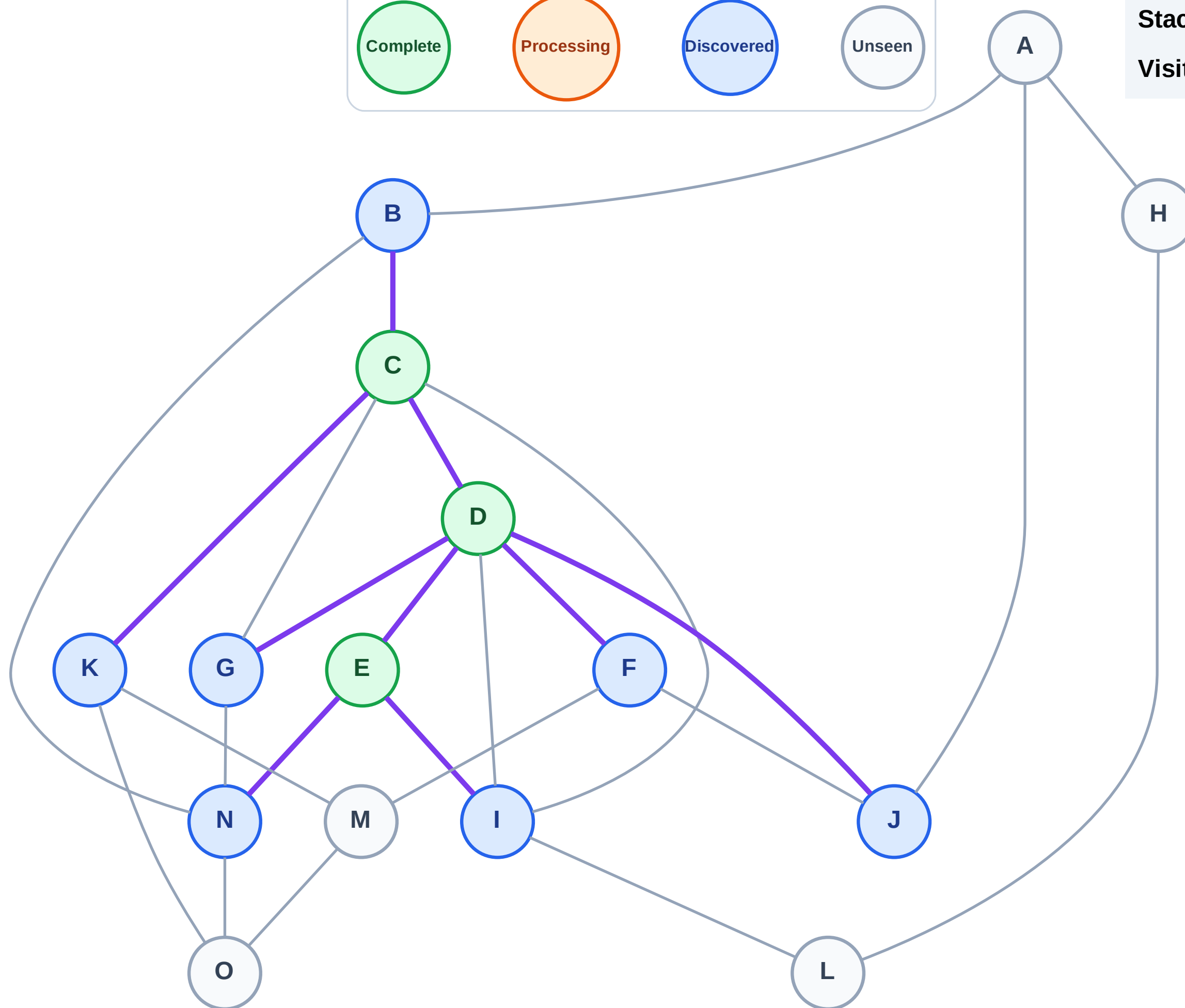
DFS Traversal

Step 7: Discover K, B from C

Legend



Stack (top at right): N | I | J | G | F | K | B
Visited: C, D, E



DFS Traversal

Step 8: Process node B

Legend

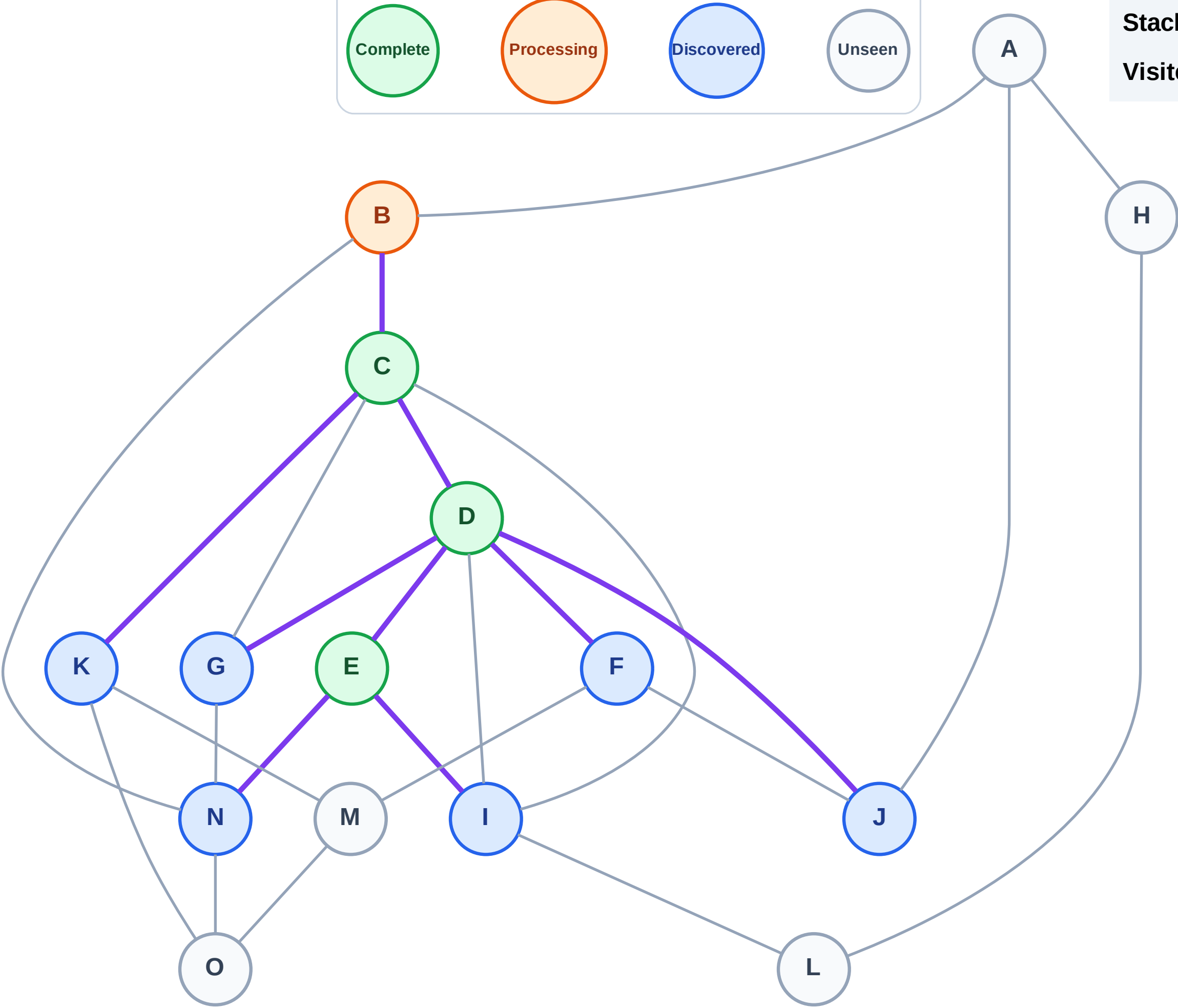
Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | J | G | F | K
Visited: C, D, E



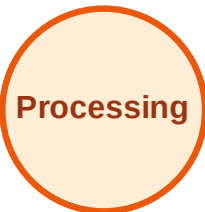
Step 9: Discover A from B



DFS Traversal

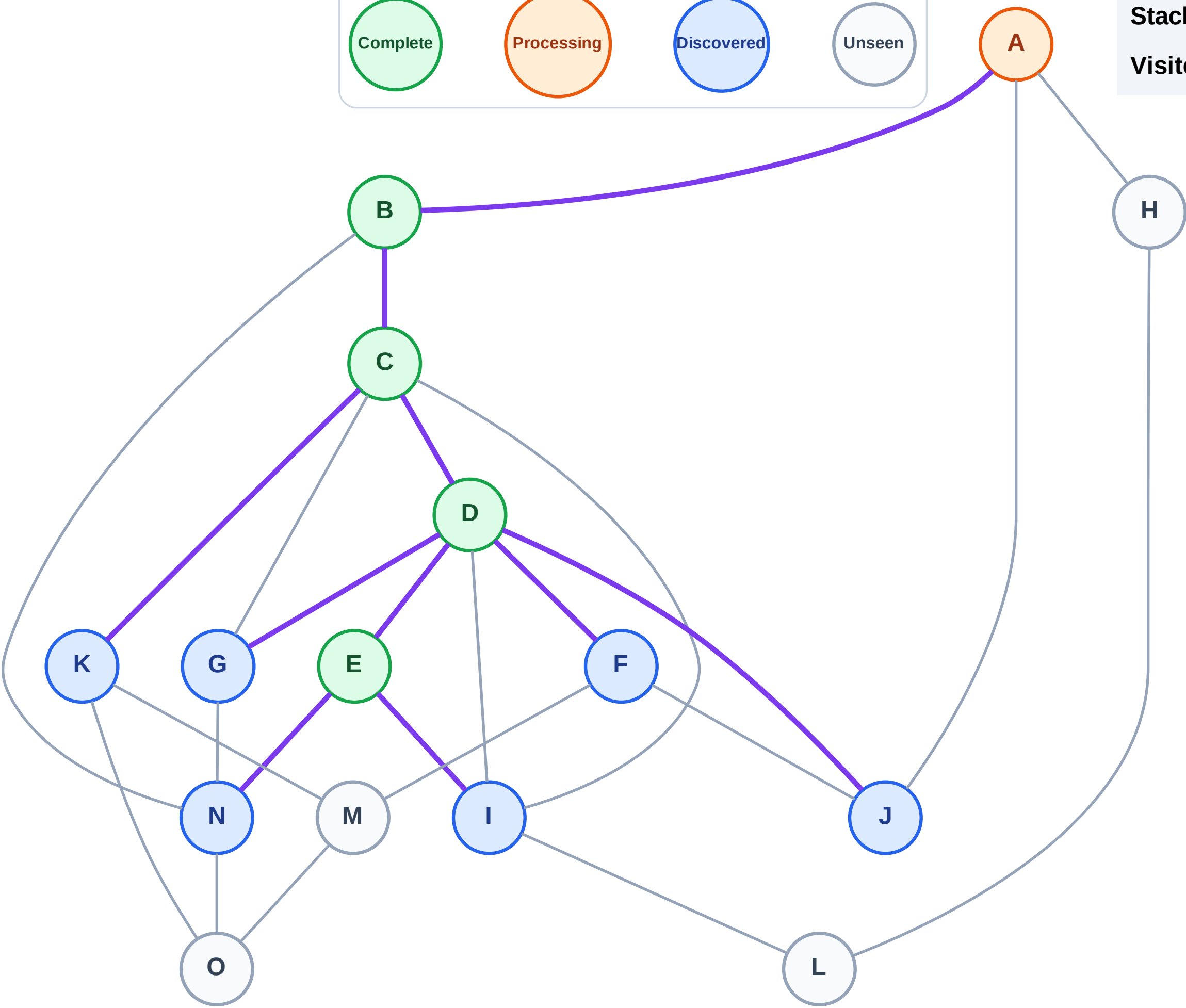
Step 10: Process node A

Legend



Stack (top at right): N | I | J | G | F | K

Visited: B, C, D, E



DFS Traversal

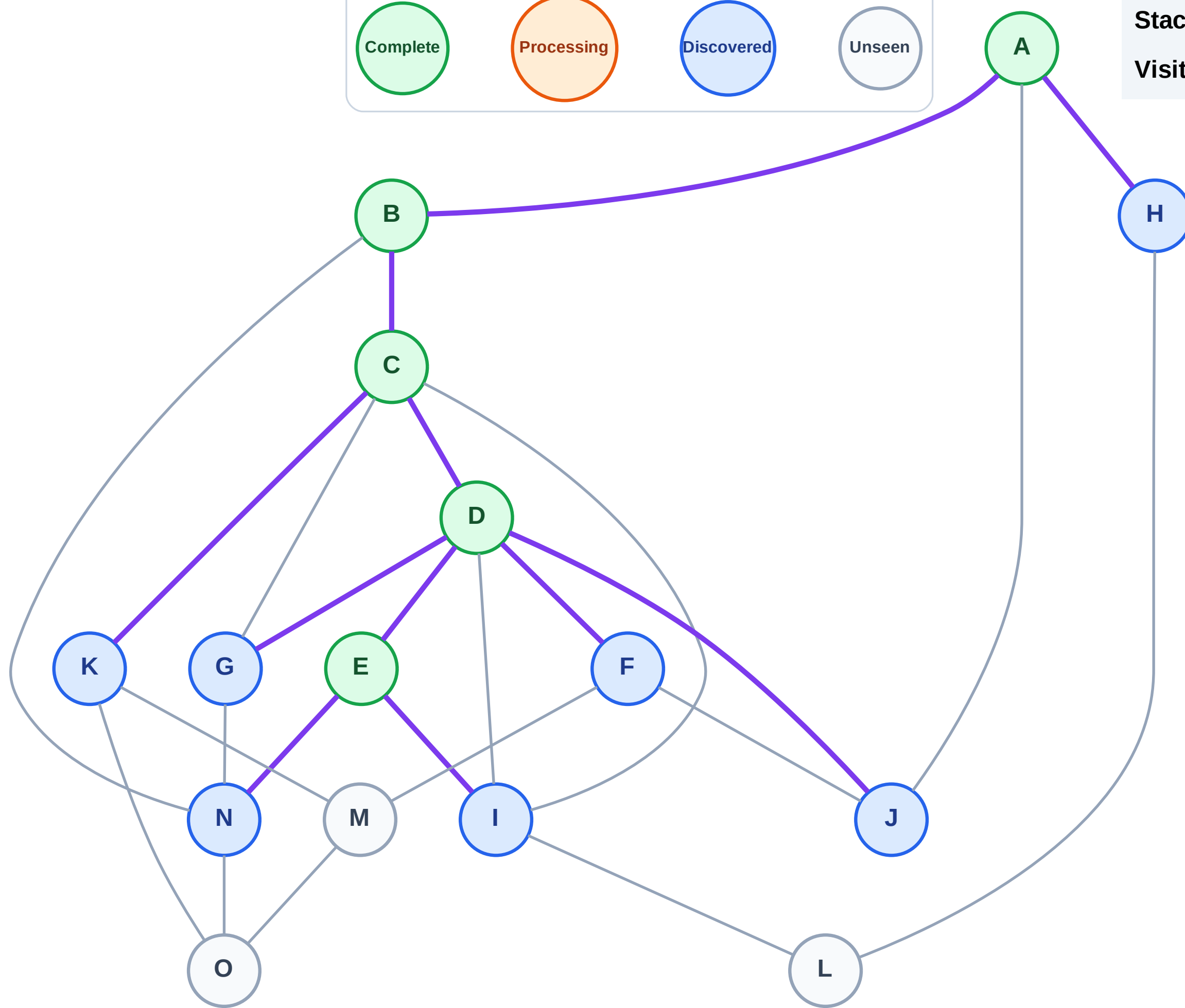
Step 11: Discover H from A

Legend



Stack (top at right): N | I | J | G | F | K | H

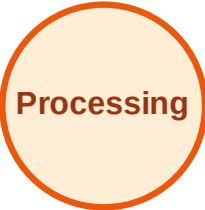
Visited: A, B, C, D, E



DFS Traversal

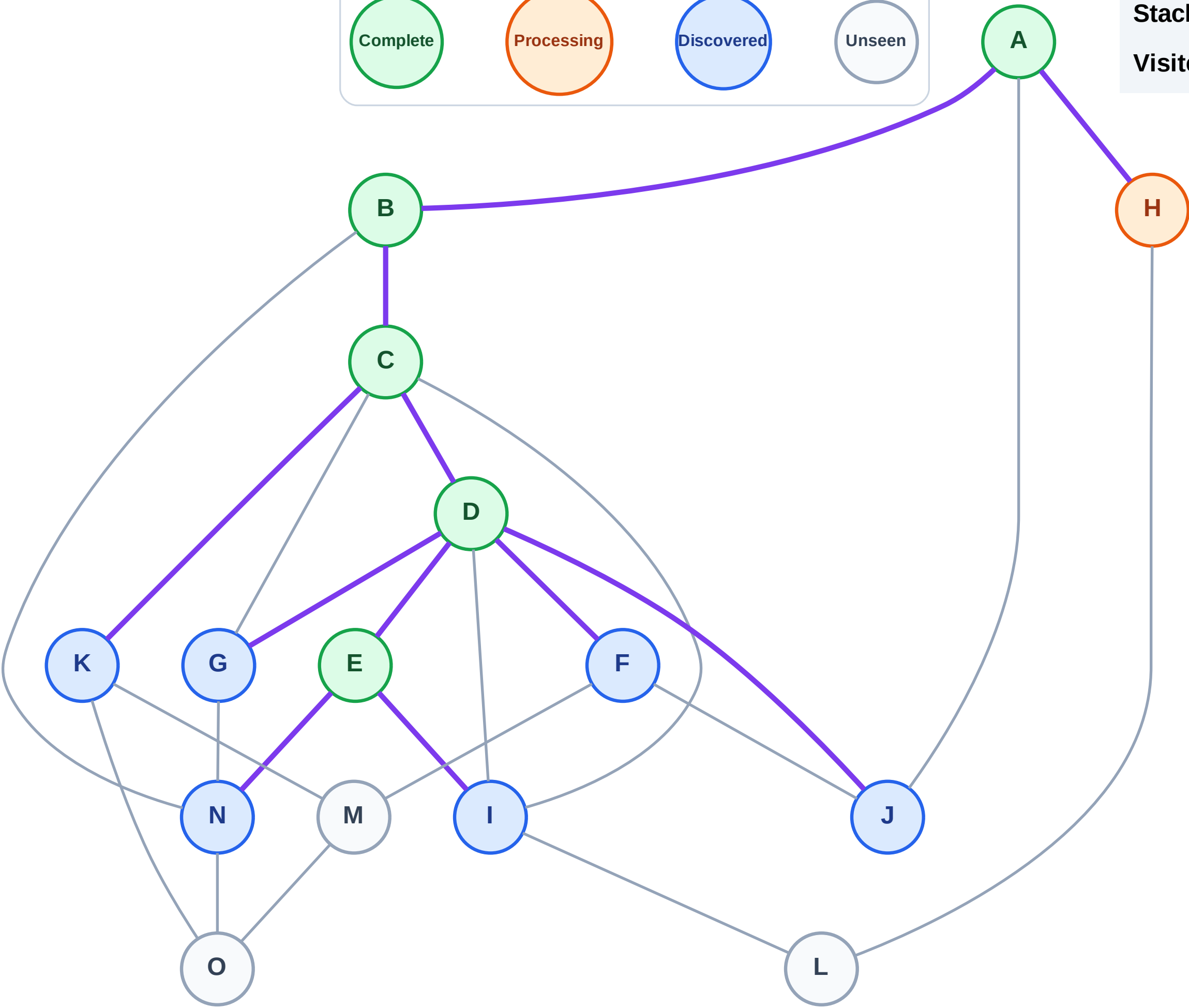
Step 12: Process node H

Legend



Stack (top at right): N | I | J | G | F | K

Visited: A, B, C, D, E



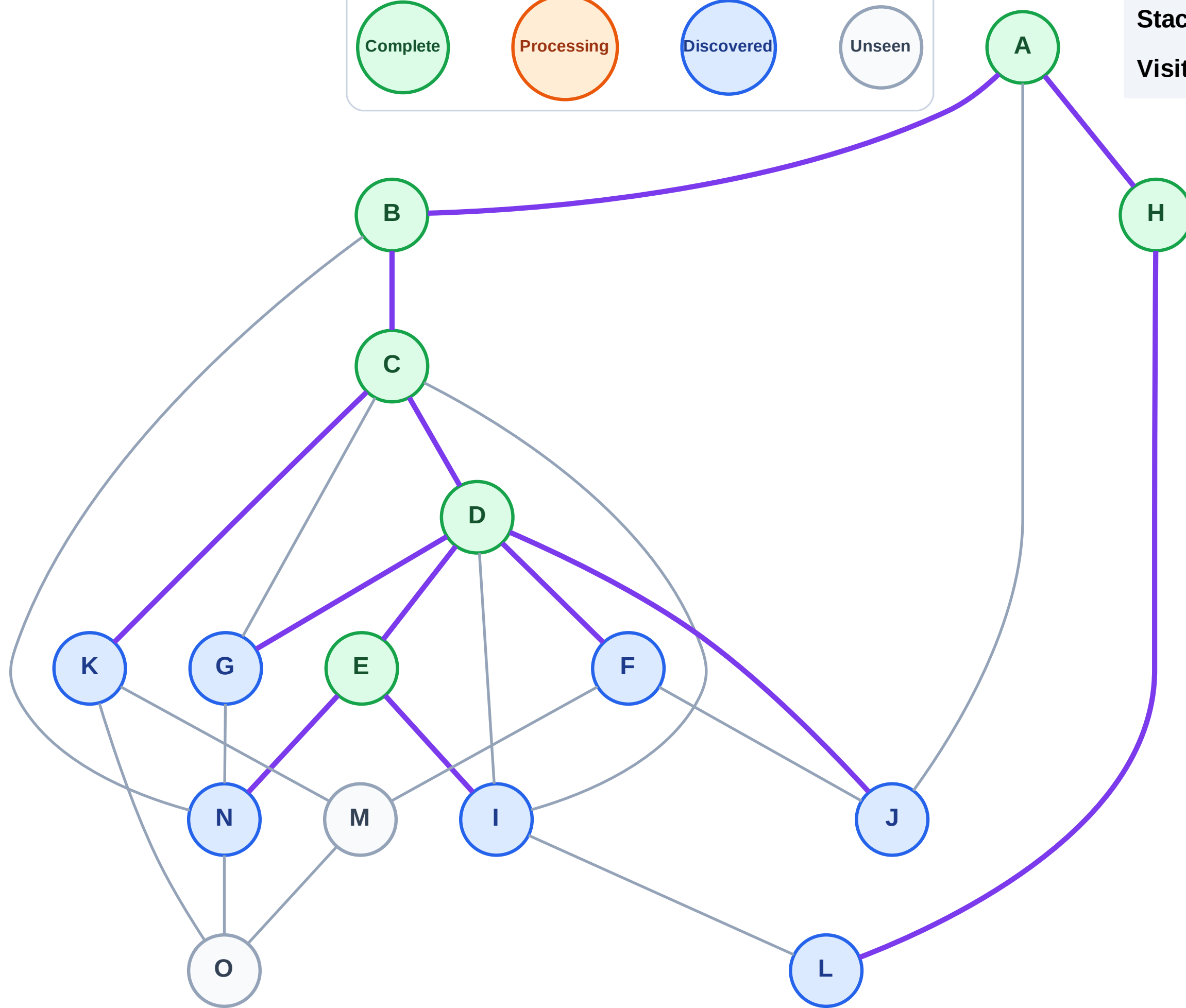
DFS Traversal

Step 13: Discover L from H

Legend



Stack (top at right): N | I | J | G | F | K | L
Visited: A, B, C, D, E, H



DFS Traversal

Step 14: Process node L

Legend

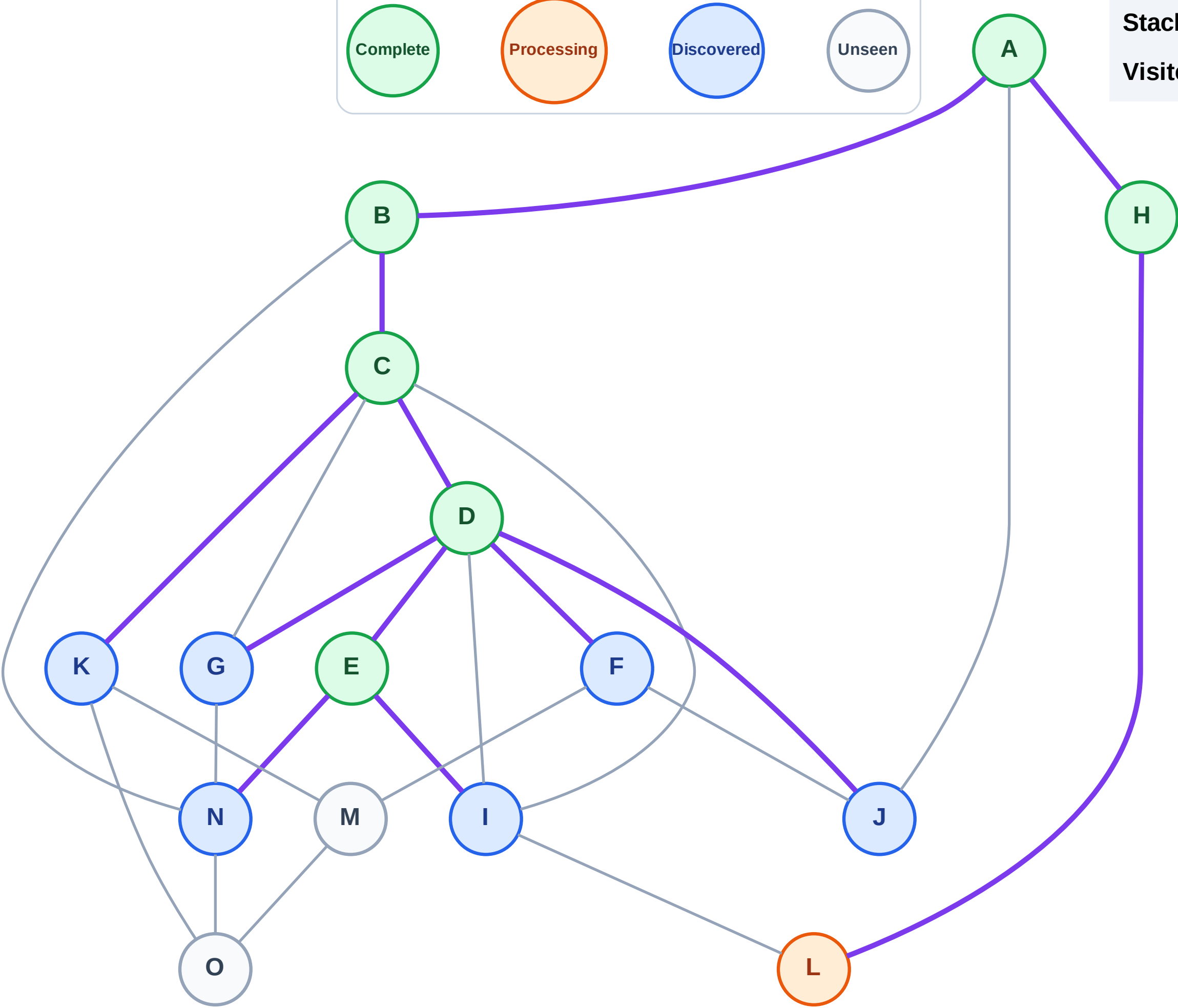
Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | J | G | F | K
Visited: A, B, C, D, E, H



DFS Traversal

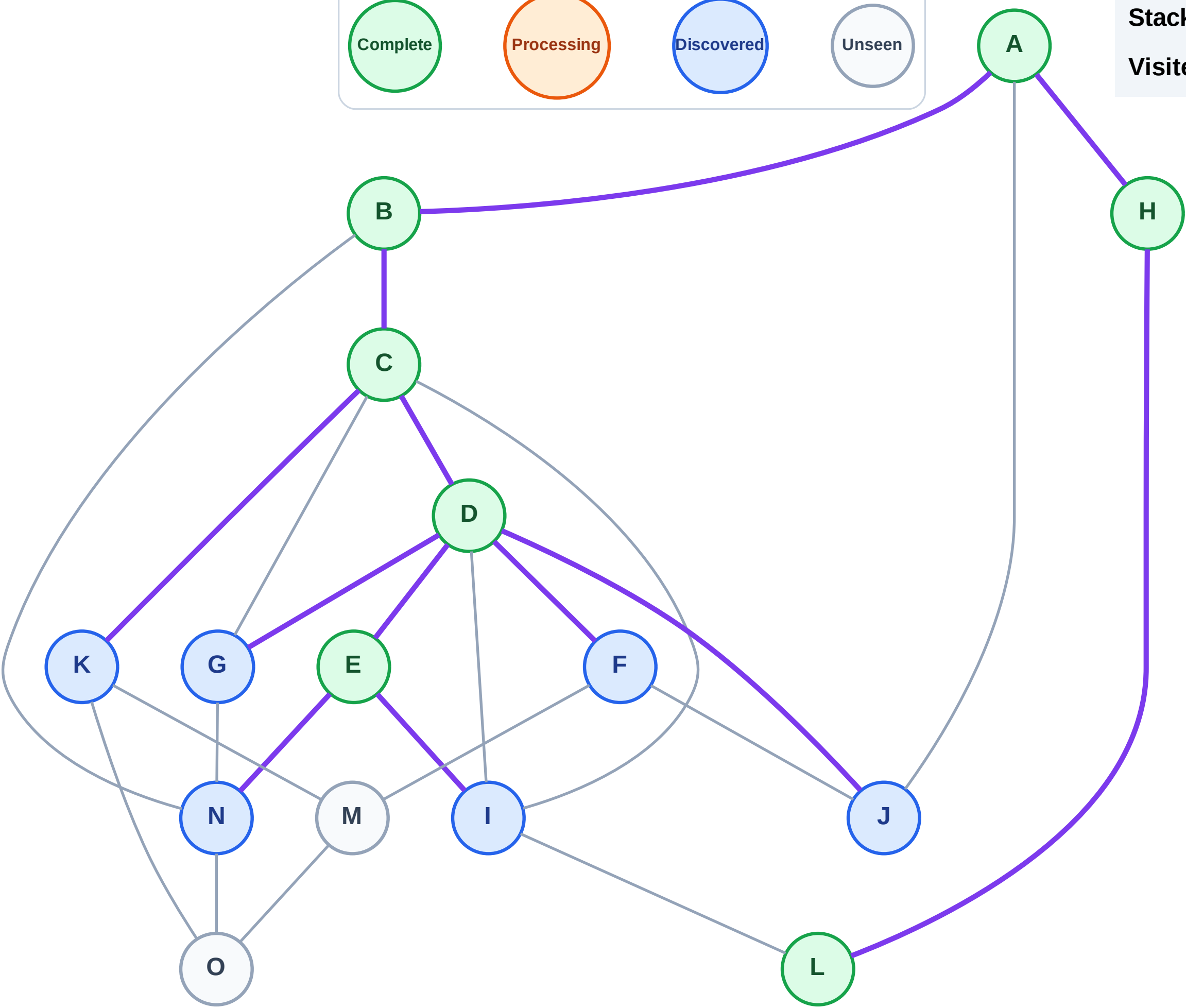
Step 15: No new neighbors from L

Legend



Stack (top at right): N | I | J | G | F | K

Visited: A, B, C, D, E, H, L



Step 16: Process node K

Legend

Complete

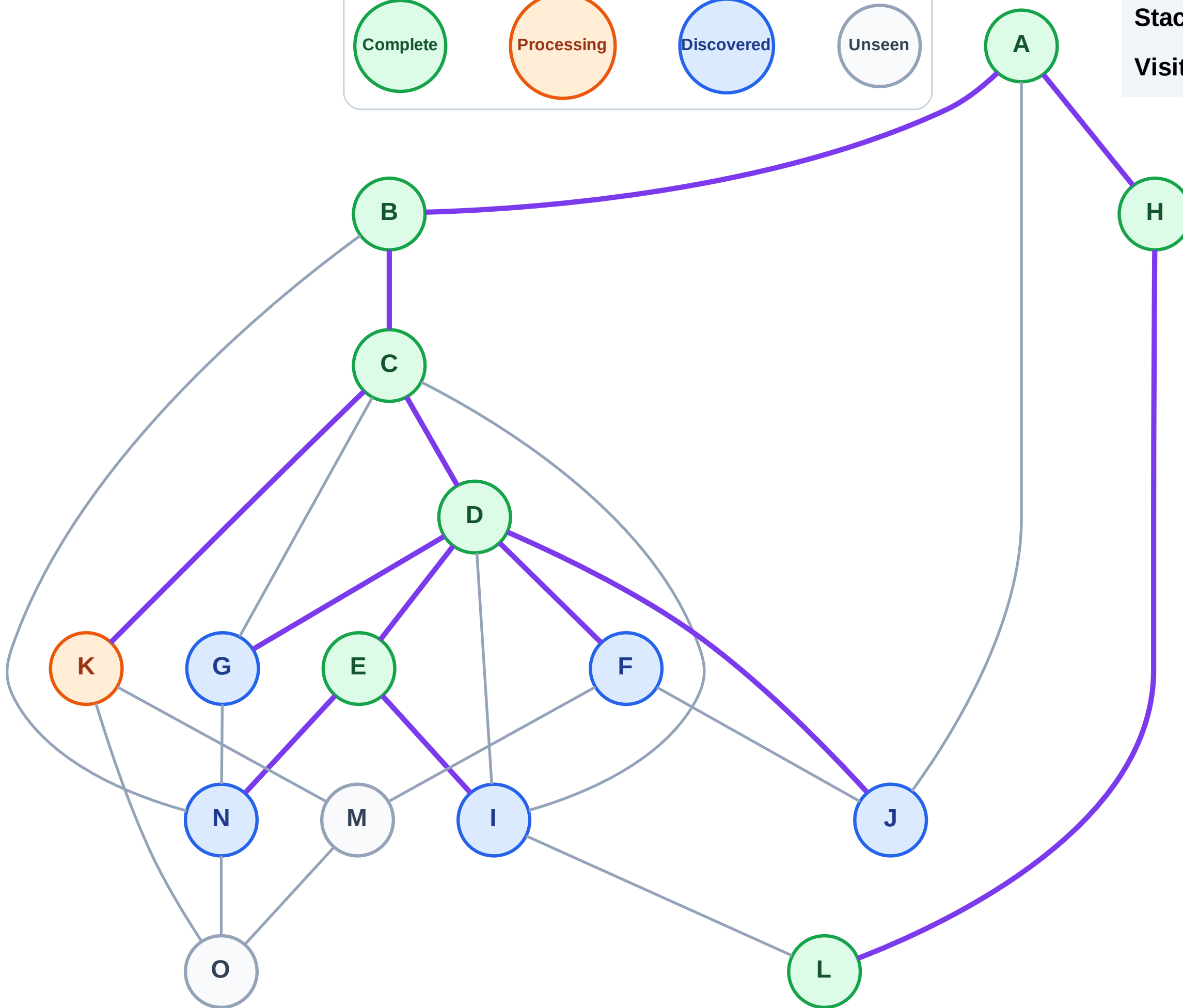
Processing

Discovered

Unseen

Stack (top at right): N | I | J | G | F

Visited: A, B, C, D, E, H, L



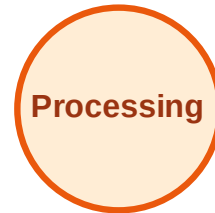
DFS Traversal

Step 17: Discover O, M from K

Legend



Complete



Processing



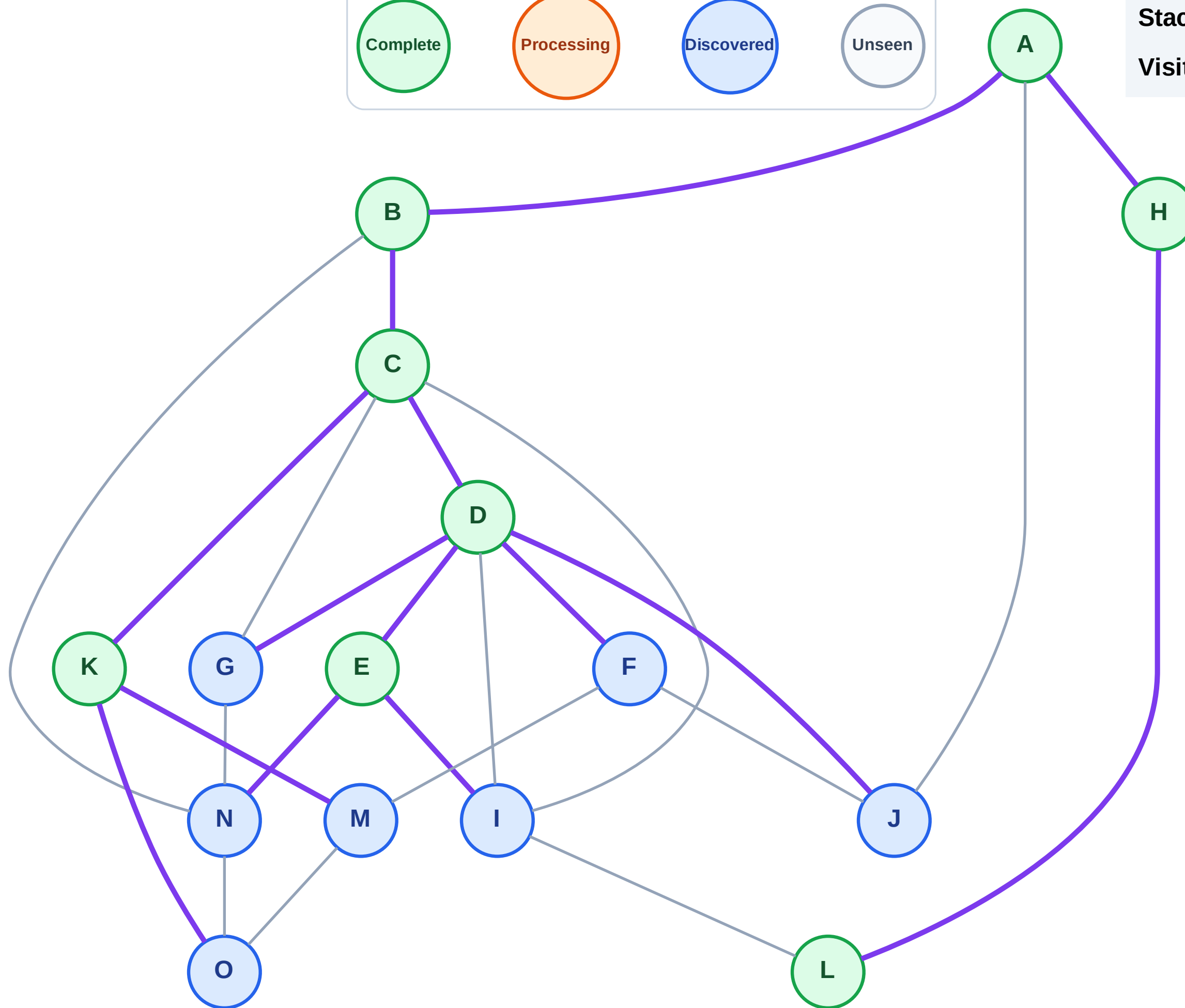
Discovered



Unseen

Stack (top at right): N | I | J | G | F | O | M

Visited: A, B, C, D, E, H, K, L



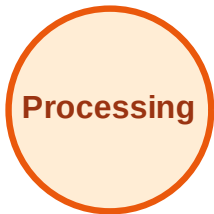
DFS Traversal

Step 18: Process node M

Legend



Complete



Processing



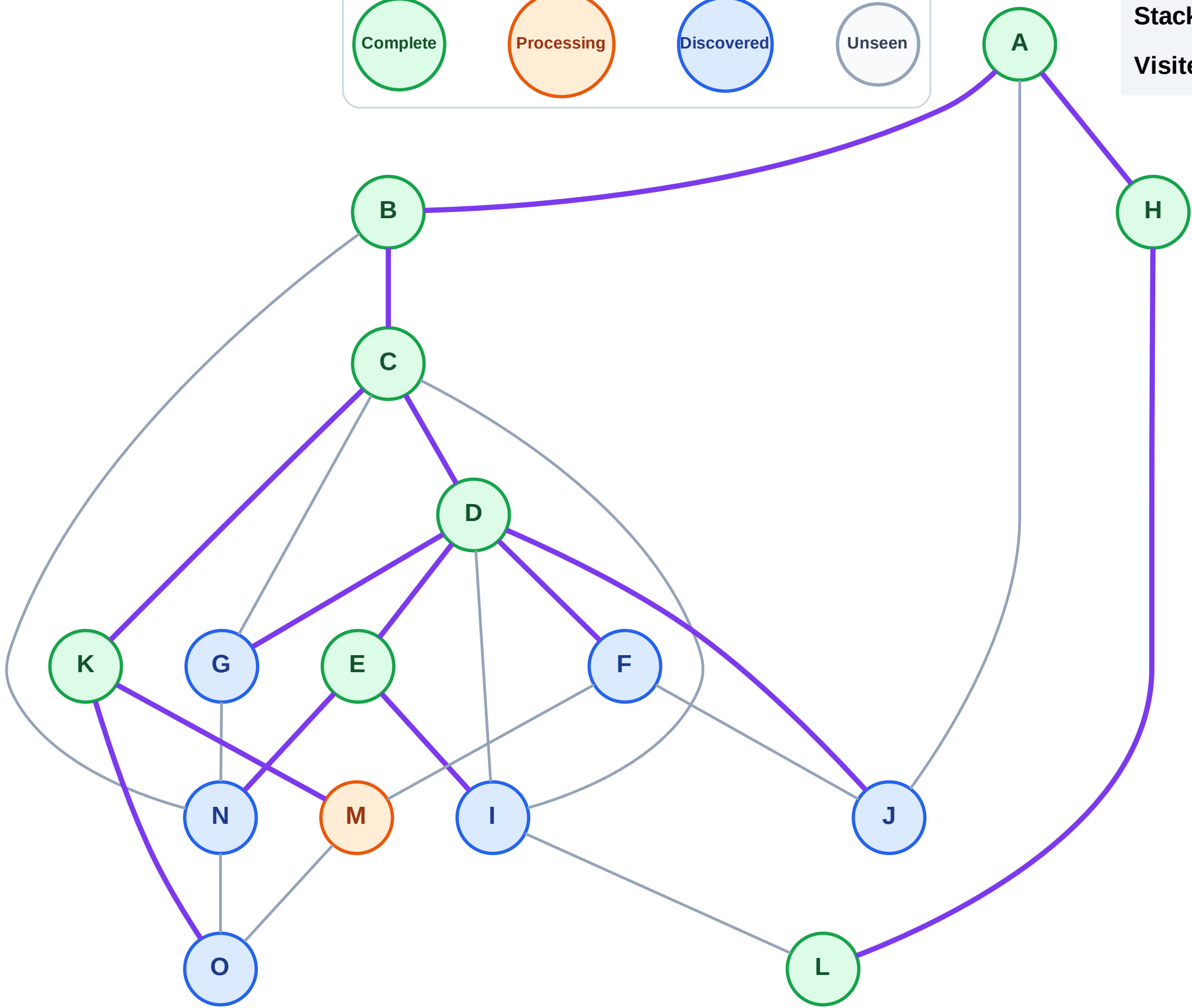
Discovered



Unseen

Stack (top at right): N | I | J | G | F | O

Visited: A, B, C, D, E, H, K, L



DFS Traversal

Step 19: No new neighbors from M

Legend

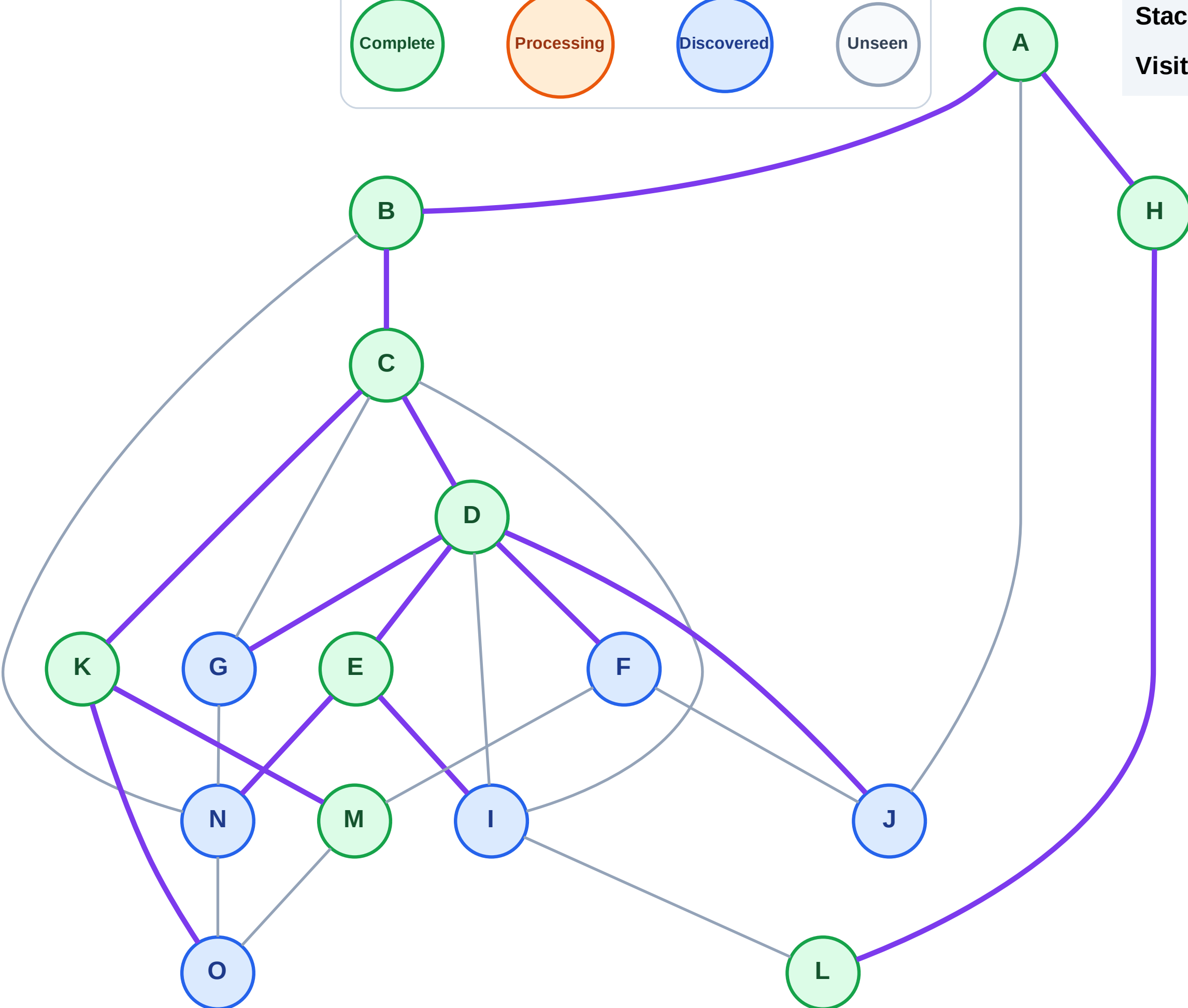
Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | J | G | F | O
Visited: A, B, C, D, E, H, K, L, M



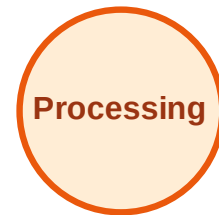
DFS Traversal

Step 20: Process node O

Legend



Complete



Processing



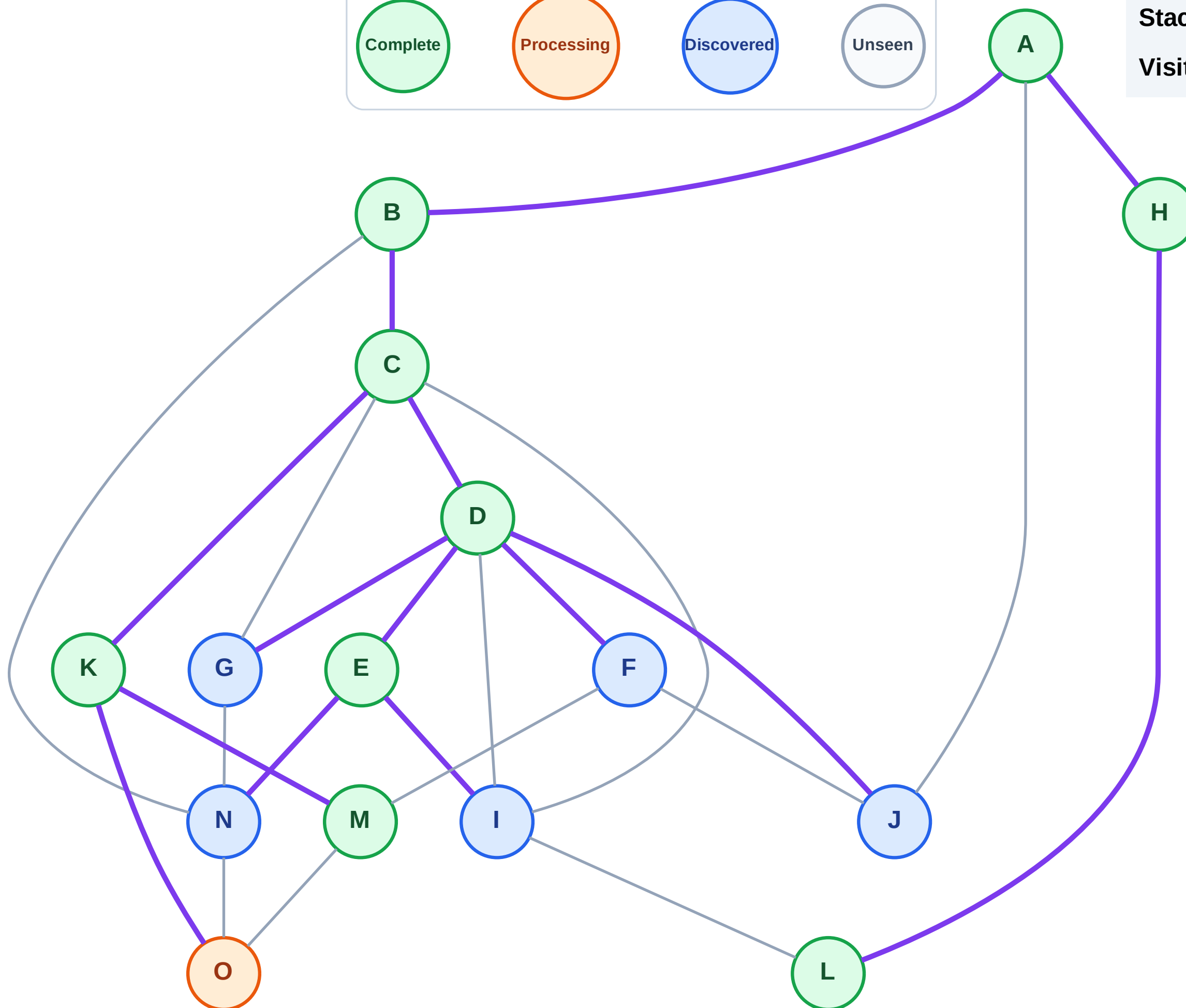
Discovered



Unseen

Stack (top at right): N | I | J | G | F

Visited: A, B, C, D, E, H, K, L, M



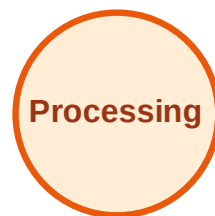
DFS Traversal

Step 21: No new neighbors from O

Legend



Complete



Processing



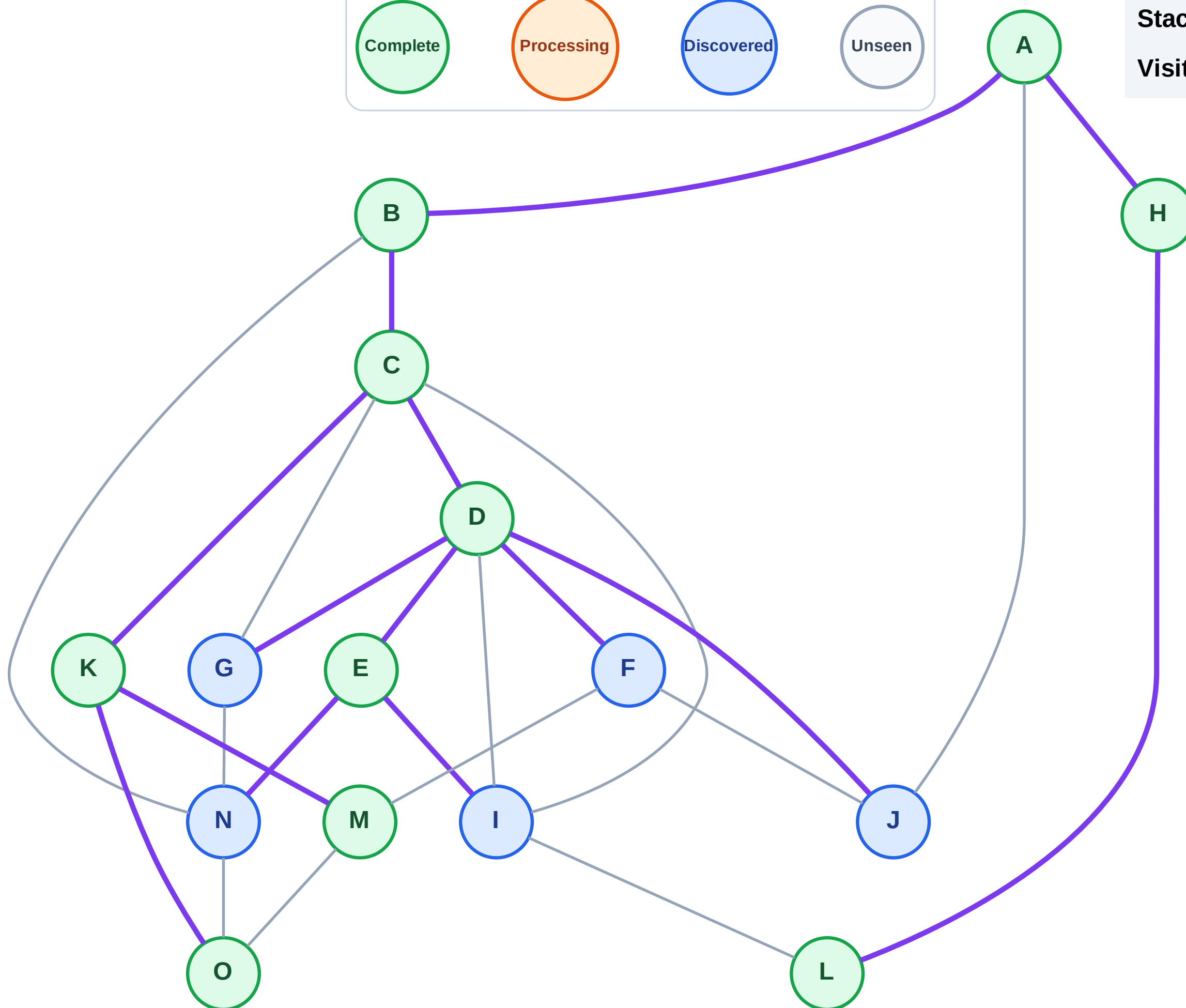
Discovered



Unseen

Stack (top at right): N | I | J | G | F

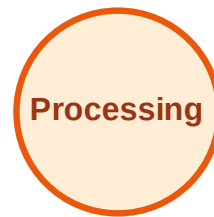
Visited: A, B, C, D, E, H, K, L, M, O



DFS Traversal

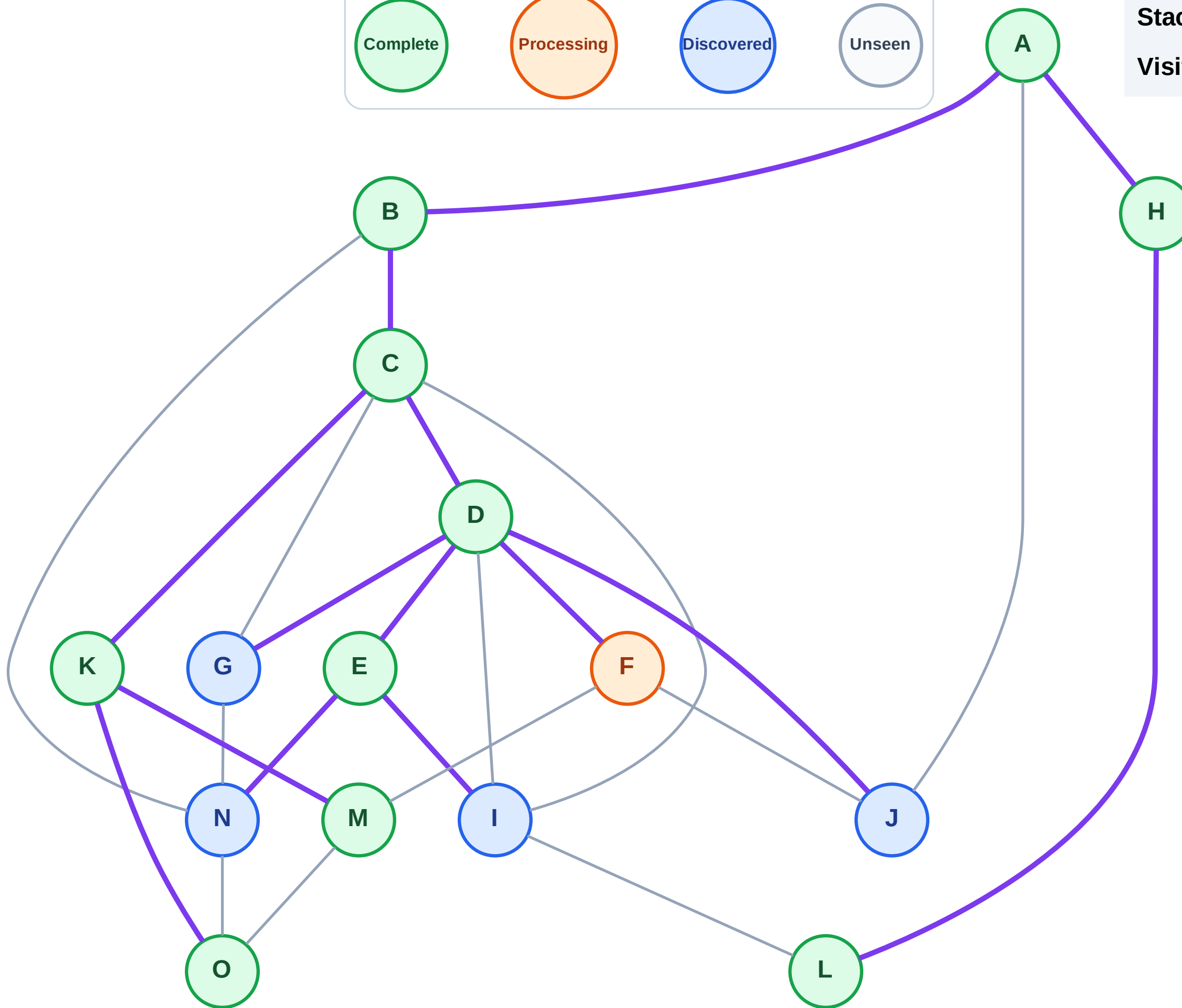
Step 22: Process node F

Legend



Stack (top at right): N | I | J | G

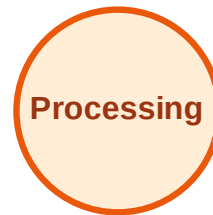
Visited: A, B, C, D, E, H, K, L, M, O



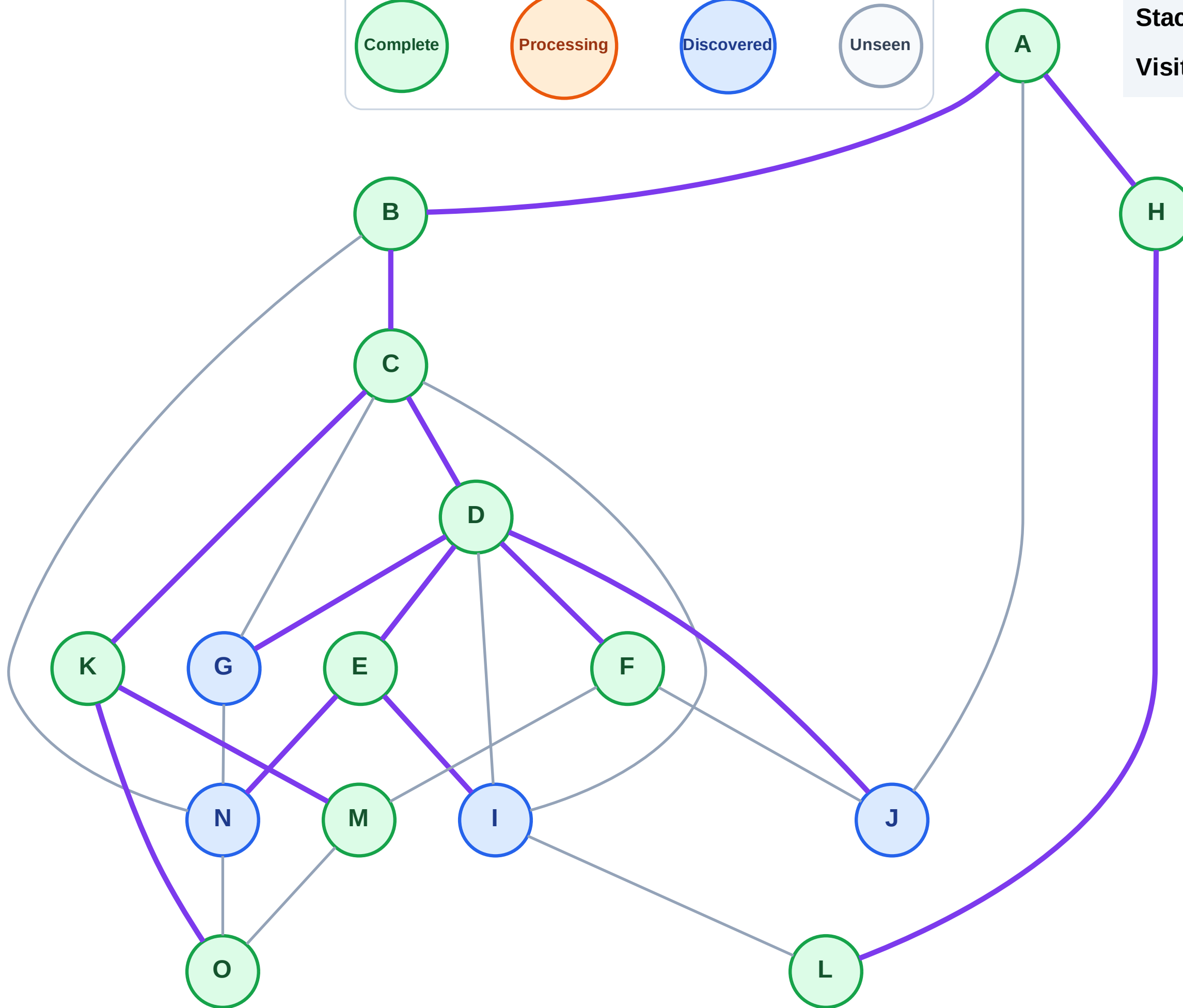
DFS Traversal

Step 23: No new neighbors from F

Legend



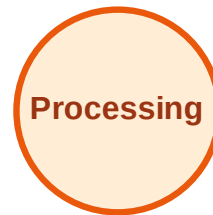
Stack (top at right): N | I | J | G
Visited: A, B, C, D, E, F, H, K, L, M, O



DFS Traversal

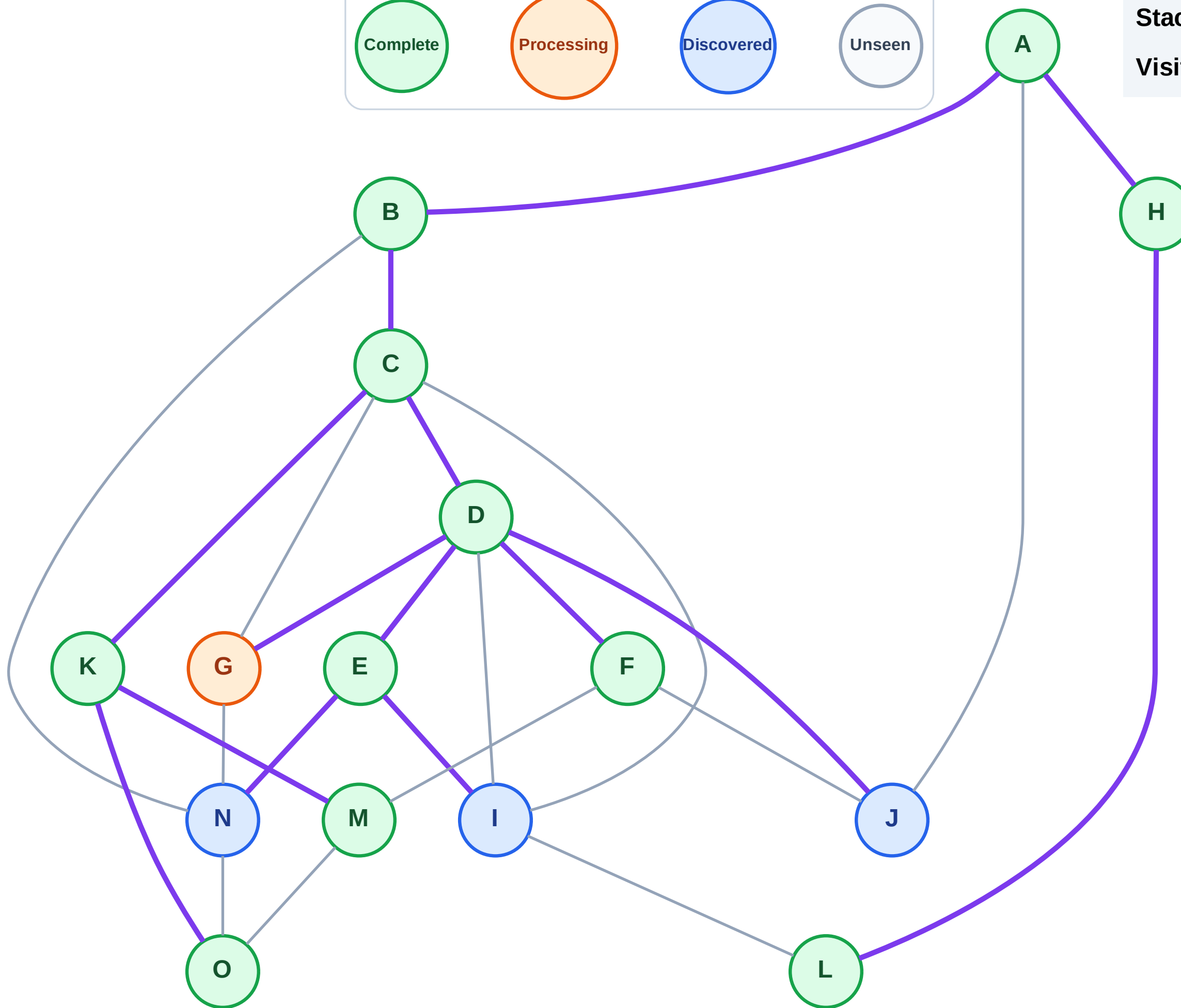
Step 24: Process node G

Legend



Stack (top at right): N | I | J

Visited: A, B, C, D, E, F, H, K, L, M, O



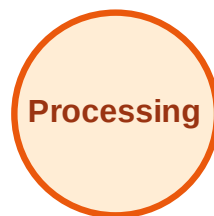
DFS Traversal

Step 25: No new neighbors from G

Legend



Complete



Processing



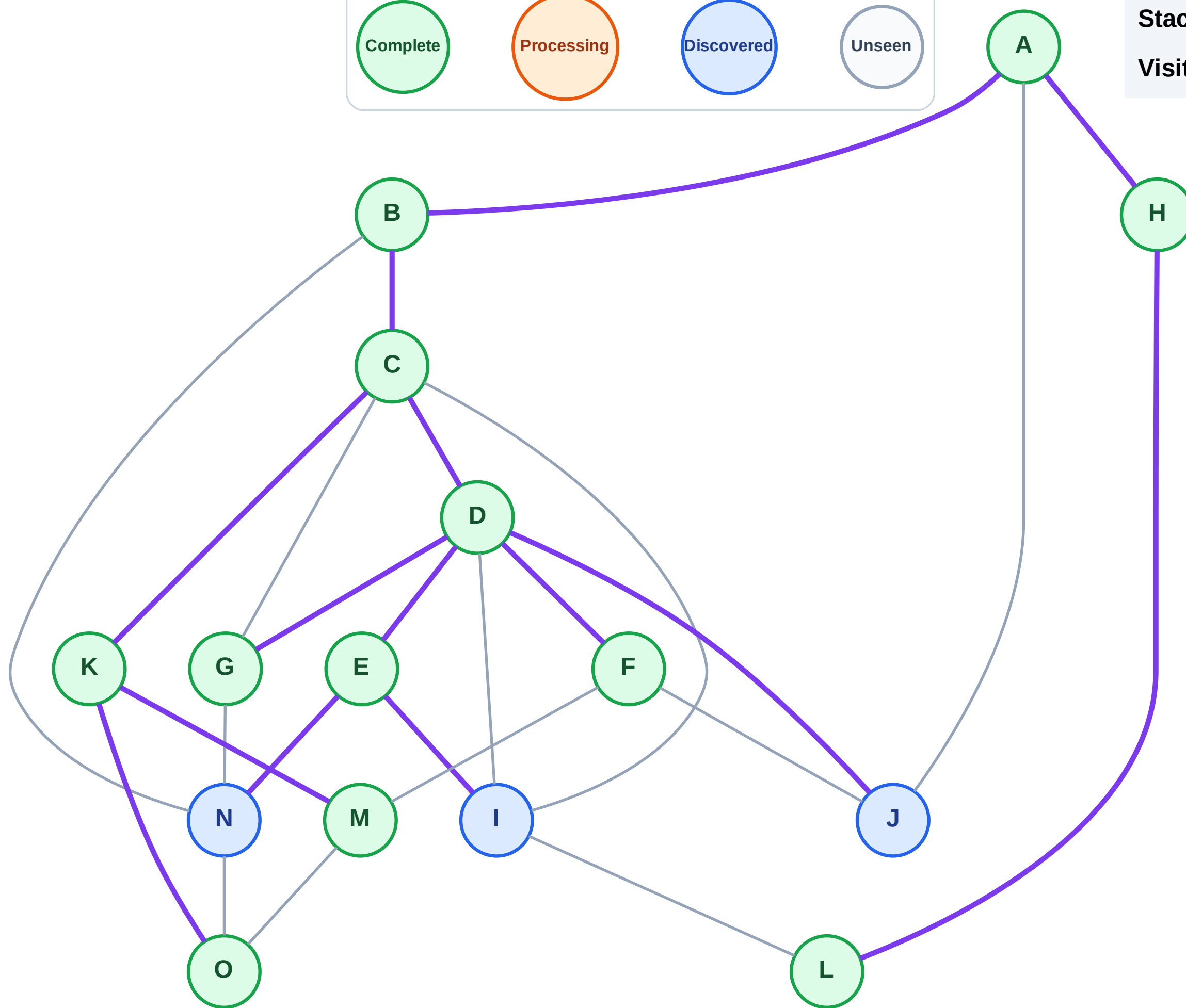
Discovered



Unseen

Stack (top at right): N | I | J

Visited: A, B, C, D, E, F, G, H, K, L, M, O



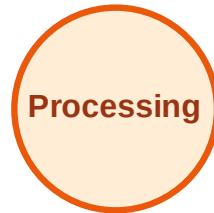
DFS Traversal

Step 26: Process node J

Legend



Complete



Processing



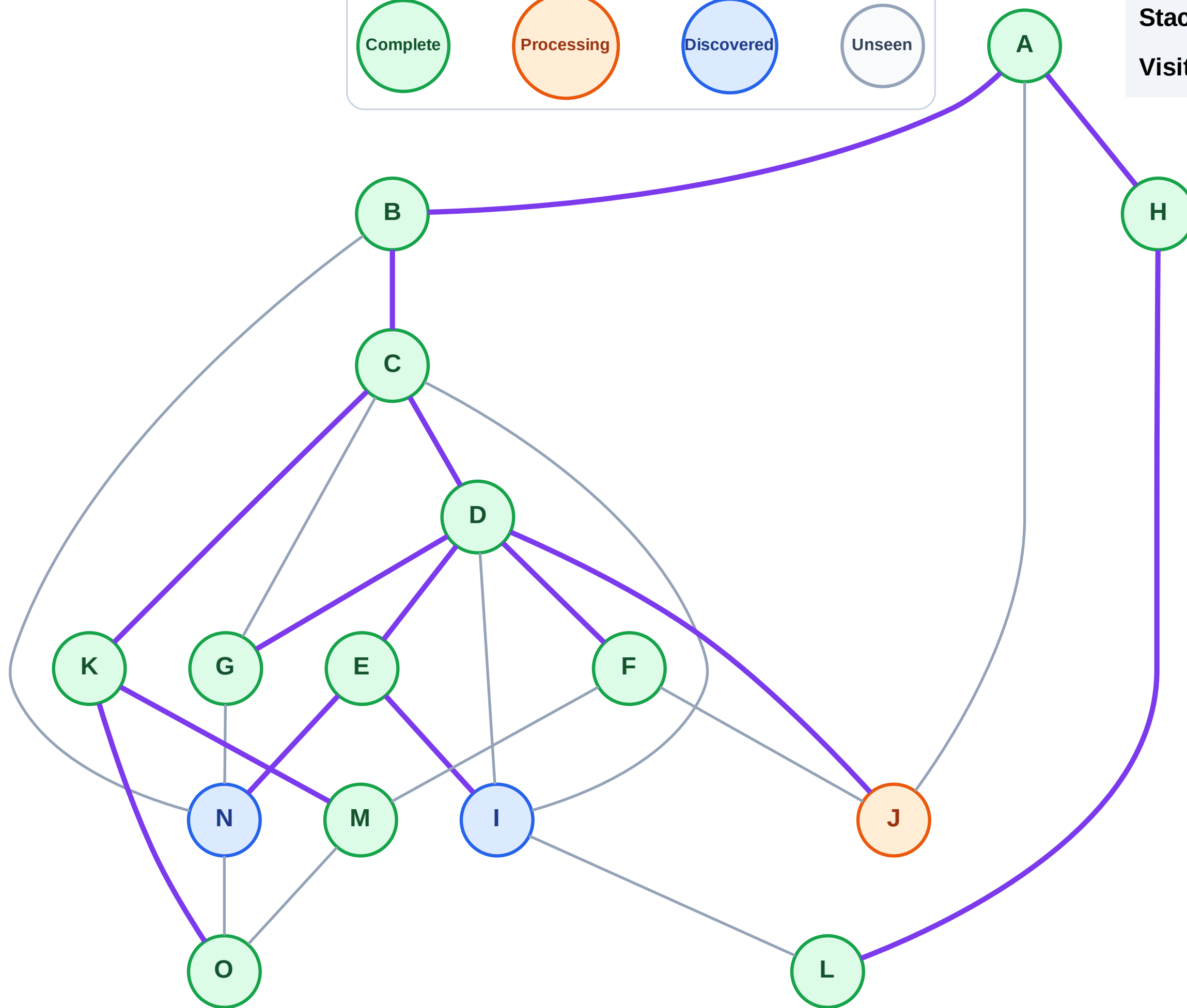
Discovered



Unseen

Stack (top at right): N | I

Visited: A, B, C, D, E, F, G, H, K, L, M, O



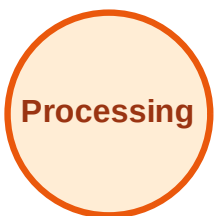
DFS Traversal

Step 27: No new neighbors from J

Legend



Complete



Processing



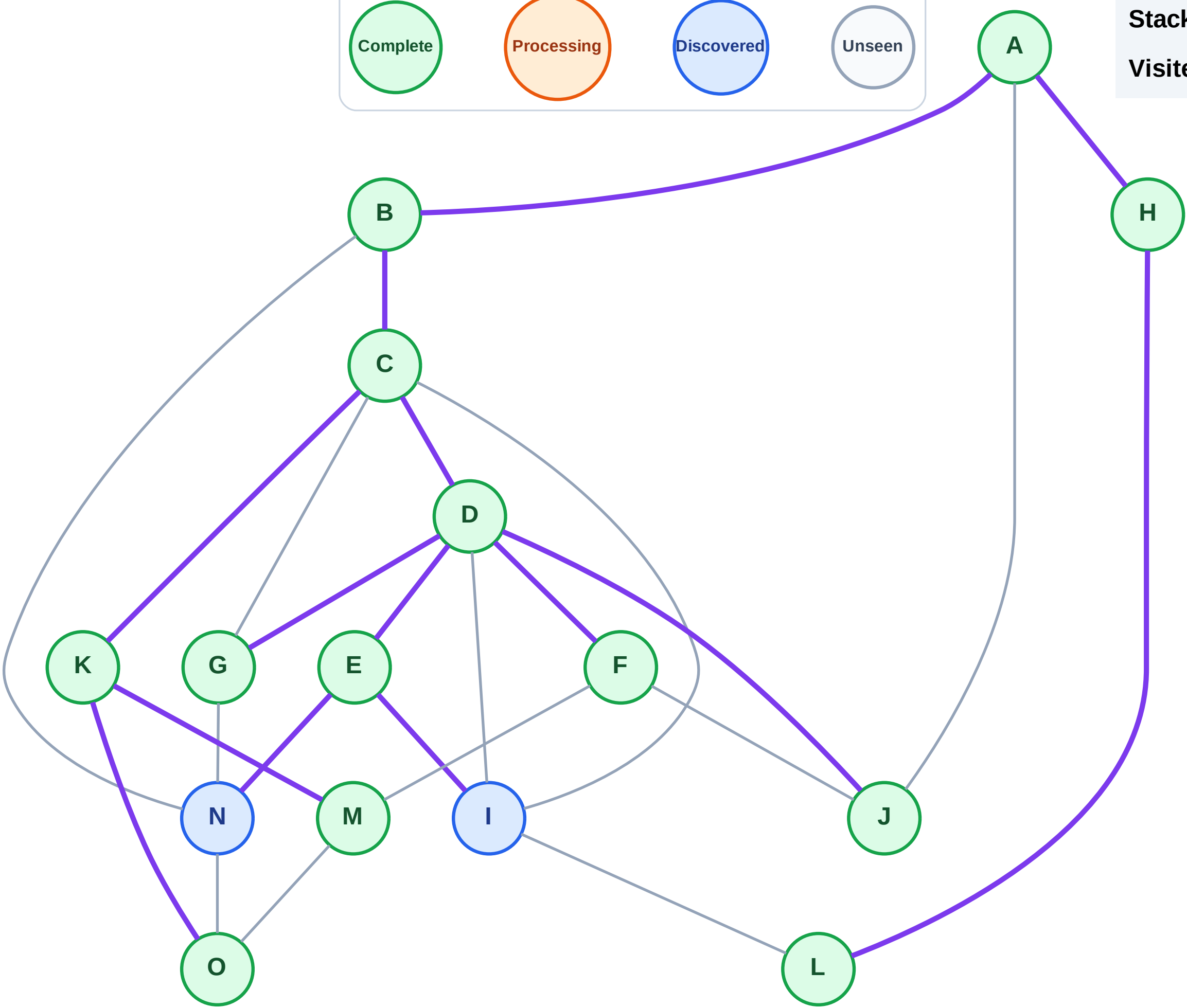
Discovered



Unseen

Stack (top at right): N | I

Visited: A, B, C, D, E, F, G, H, J, K, L, M, O



DFS Traversal

Step 28: Process node I

Legend

Complete

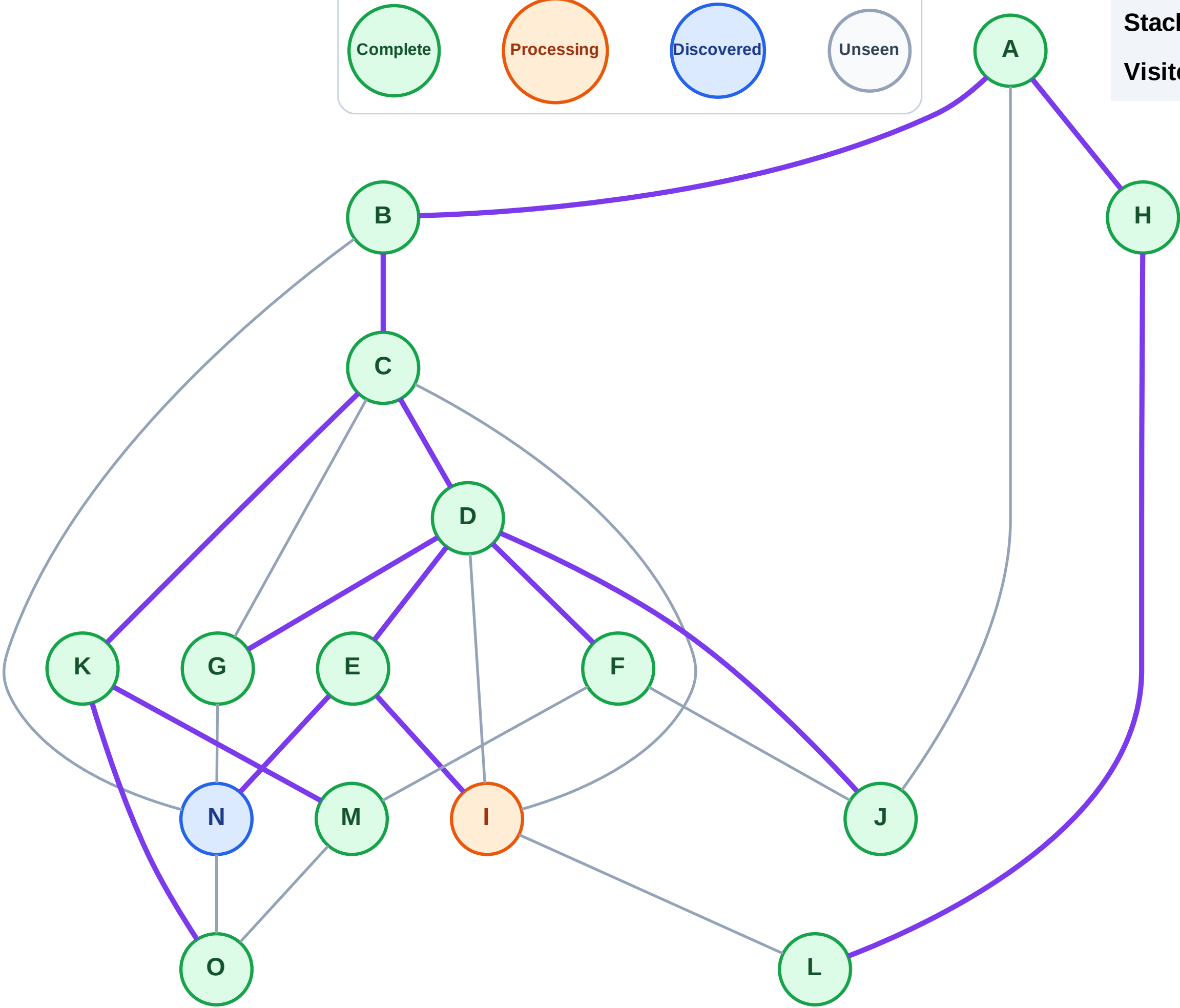
Processing

Discovered

Unseen

Stack (top at right): N

Visited: A, B, C, D, E, F, G, H, J, K, L, M, O



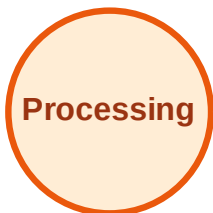
DFS Traversal

Step 29: No new neighbors from I

Legend



Complete



Processing



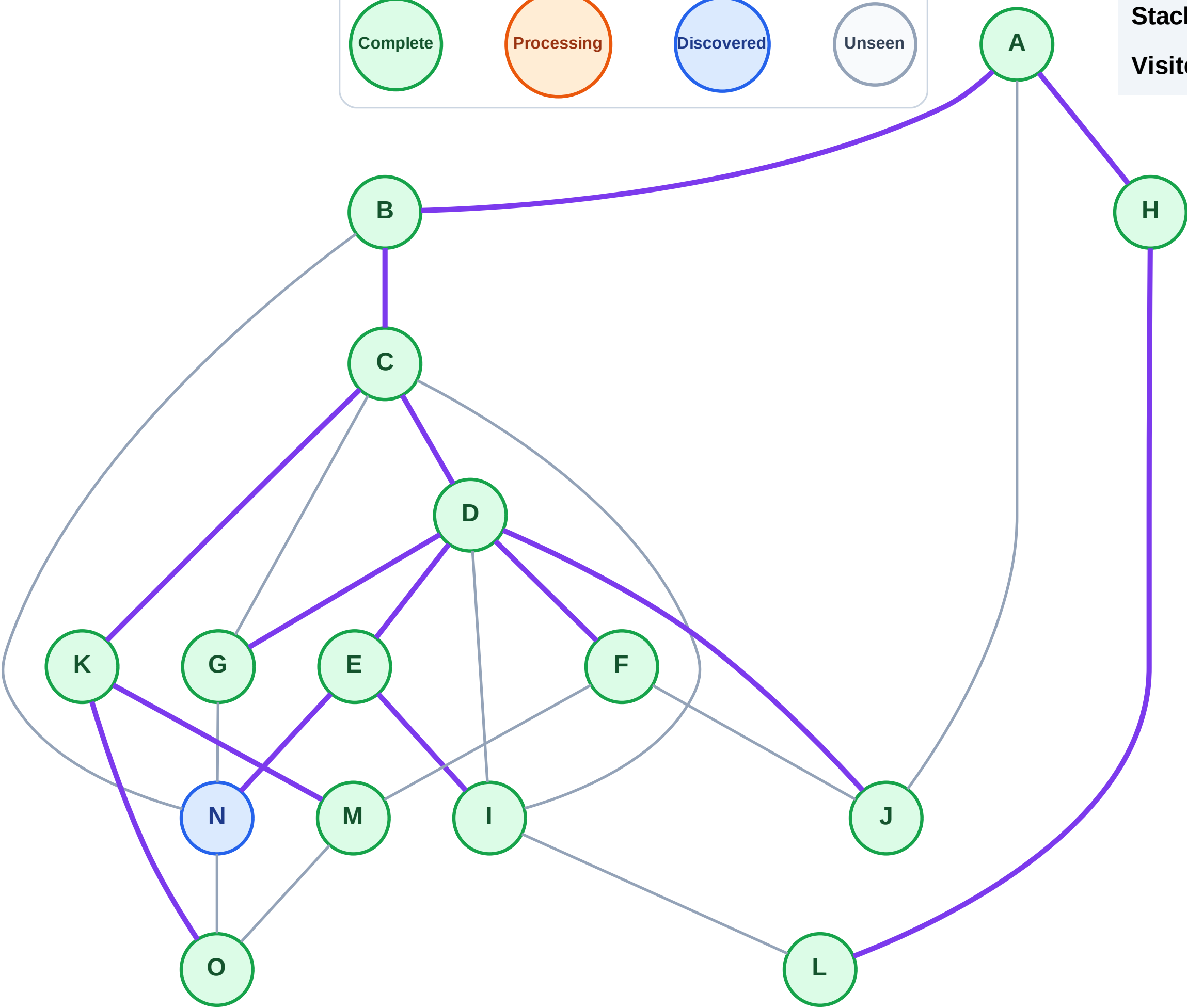
Discovered



Unseen

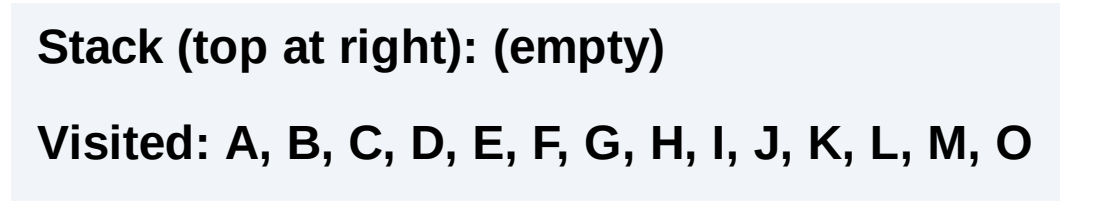
Stack (top at right): N

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, O



Step 30: Process node N

Step 30: Process node N



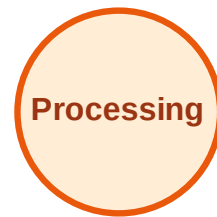
DFS Traversal

Step 31: No new neighbors from N

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O

